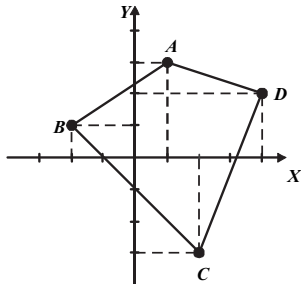


5.



EJERCICIO 75

1. $m = 1$

2. $m = -12$

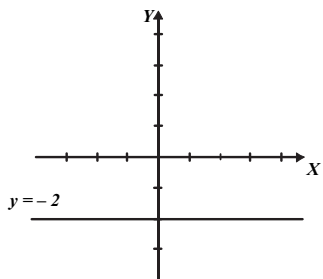
3. $m = \frac{8}{9}$

4. $m = -\frac{22}{27}$

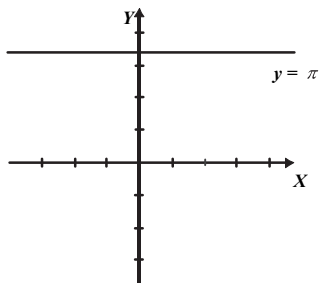
5. $m = \frac{5}{14}$

EJERCICIO 76

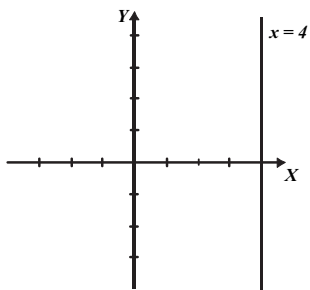
1.



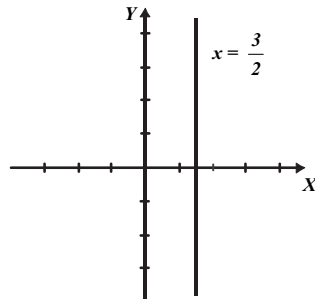
2.



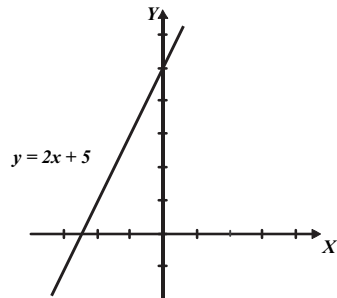
3.



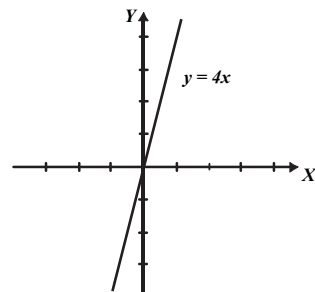
4.



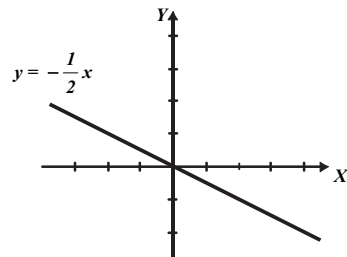
5.



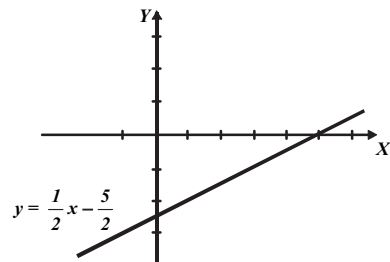
6.



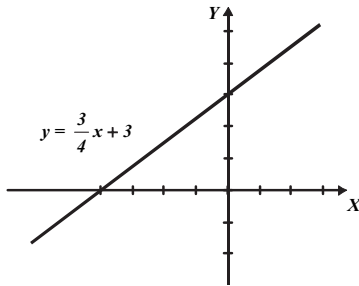
7.



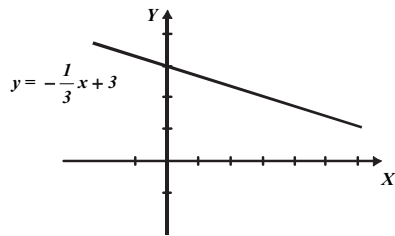
8.



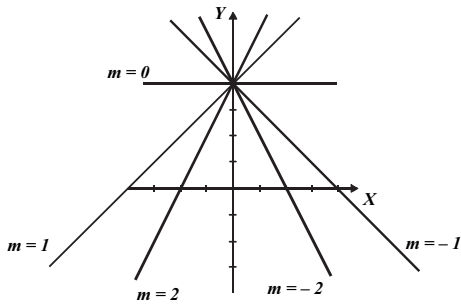
9.



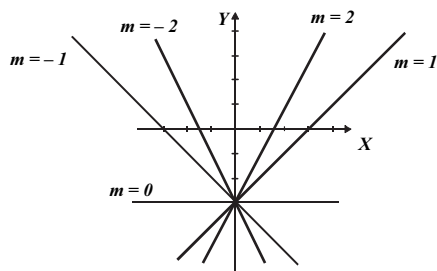
10.


EJERCICIO 77

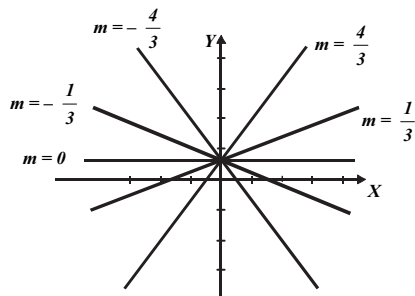
1.



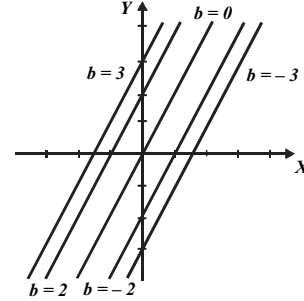
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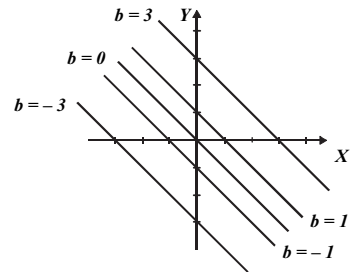
3.



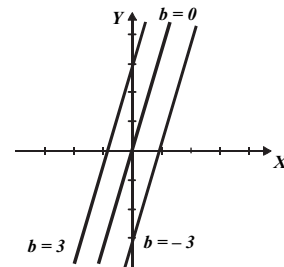
4.



5.



6.


EJERCICIO 78

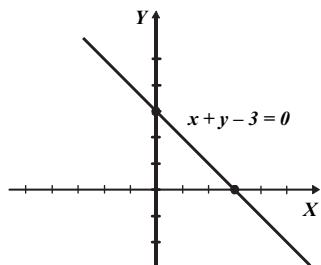
1. $S = 40t$
2. a) $P = \frac{7}{3}t + 3.5$
b) $P = 24.5$ kg
c) $t = 10$ años 6 meses
3. $C = \frac{19}{30}T + \frac{1681}{3}$
4. a) $G = \frac{11}{20}I + 1800$
b) $U = \frac{9}{20}I - 1800$
c) $R = \$4000$
5. a) $C = F = -40^\circ$
b) $C = 160^\circ$ y $F = 320^\circ$

CAPÍTULO 8
EJERCICIO 79

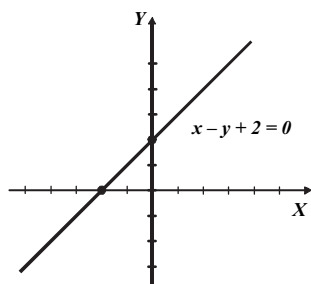
1. $(2, -3), (7, 0)$ son solución
2. $\left(\frac{1}{2}, -\frac{3}{4}\right)$ es solución
3. $(3, -4), (-3, -12)$ son solución
4. $\left(\frac{1}{5}, \frac{2}{3}\right)$ es solución
5. $\left(-\frac{1}{2}, \frac{3}{10}\right)$ es solución

EJERCICIO 80

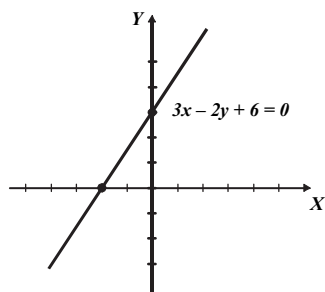
1.



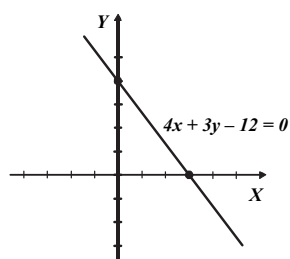
2.



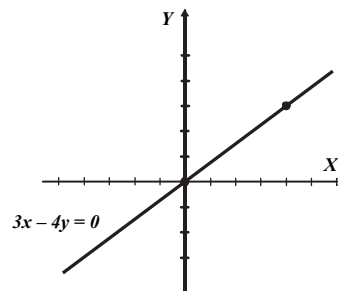
3.



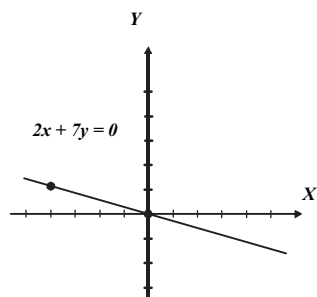
4.



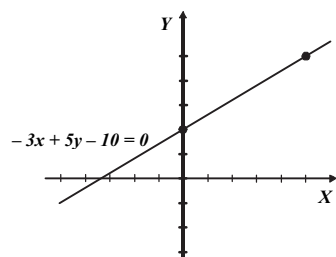
5.



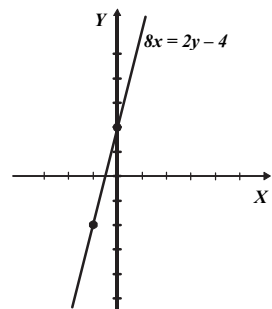
6.



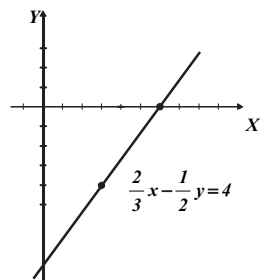
7.



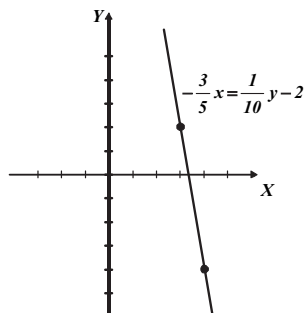
8.

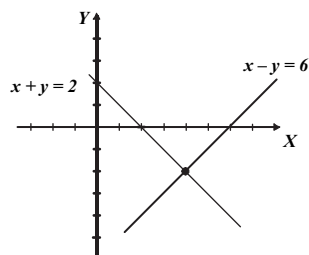
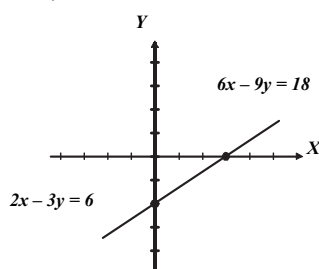


9.

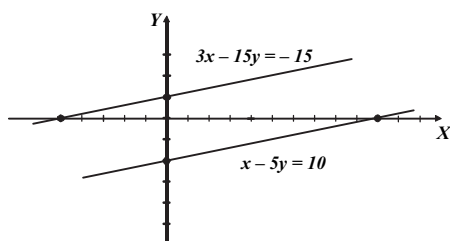
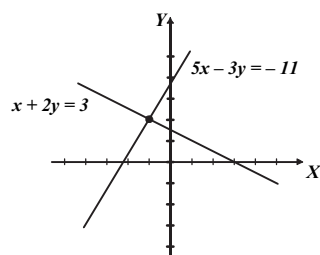
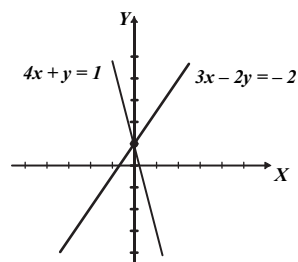


10.

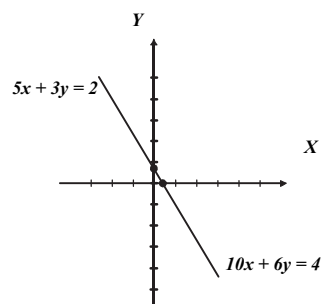


EJERCICIO 811. $(4, -2)$ 2. Conjunto infinito de soluciones
(rectas coincidentes)

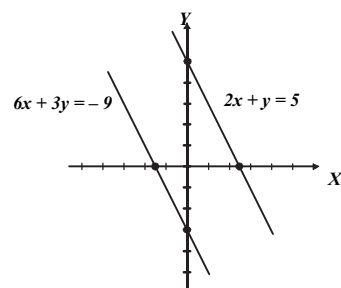
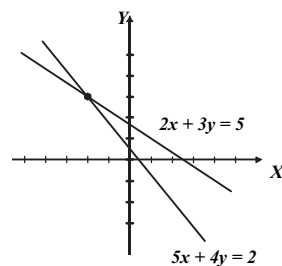
3. Conjunto vacío (rectas paralelas)

4. $(-1, 2)$ 5. $(0, 1)$ 

6. Conjunto infinito de soluciones (rectas coincidentes)



7. Conjunto vacío (rectas paralelas)

8. $(-2, 3)$ **EJERCICIO 82**

1. $\begin{cases} x=3 \\ y=1 \end{cases}$

7. $\begin{cases} m=-1 \\ n=4 \end{cases}$

2. $\begin{cases} x=\frac{1}{3} \\ y=-\frac{1}{2} \end{cases}$

8. $\begin{cases} x=-1 \\ y=2 \end{cases}$

3. $\begin{cases} x=-2 \\ y=5 \end{cases}$

9. $\begin{cases} u=\frac{2}{3} \\ v=\frac{1}{4} \end{cases}$

4. $\begin{cases} x=2 \\ y=3 \end{cases}$

10. Conjunto infinito de soluciones

5. $\begin{cases} x=2 \\ y=4 \end{cases}$

11. No hay solución

6. $\begin{cases} a=3 \\ b=2 \end{cases}$

12. No hay solución

EJERCICIO 83

- | | | |
|---|---|-------------------------------------|
| 1. $\begin{cases} x = -4 \\ y = -2 \end{cases}$ | 5. $\begin{cases} p = \frac{3}{2} \\ q = \frac{4}{3} \end{cases}$ | 9. Conjunto infinito de soluciones |
| 2. $\begin{cases} m = -3 \\ n = -4 \end{cases}$ | 6. $\begin{cases} x = 8 \\ y = -2 \end{cases}$ | 10. No hay solución |
| 3. $\begin{cases} r = -1 \\ t = 1 \end{cases}$ | 7. $\begin{cases} p = -4 \\ q = 0 \end{cases}$ | 11. Conjunto infinito de soluciones |
| 4. $\begin{cases} x = \frac{1}{3} \\ y = 3 \end{cases}$ | 8. $\begin{cases} x = 12 \\ y = 9 \end{cases}$ | 12. No hay solución |

EJERCICIO 84

- | | | |
|---|---|--|
| 1. $\begin{cases} x = 3 \\ y = -4 \end{cases}$ | 5. $\begin{cases} p = -1 \\ q = 1 \end{cases}$ | 9. $\begin{cases} u = \frac{5}{6} \\ v = -\frac{2}{3} \end{cases}$ |
| 2. $\begin{cases} m = \frac{3}{2} \\ n = \frac{1}{2} \end{cases}$ | 6. $\begin{cases} x = -4 \\ y = 0 \end{cases}$ | 10. Conjunto infinito de soluciones |
| 3. $\begin{cases} a = -2 \\ b = 1 \end{cases}$ | 7. $\begin{cases} a = -1 \\ b = 3 \end{cases}$ | 11. No hay solución |
| 4. $\begin{cases} x = -3 \\ y = 4 \end{cases}$ | 8. $\begin{cases} m = \frac{2}{3} \\ n = \frac{1}{5} \end{cases}$ | 12. No hay solución |

EJERCICIO 85

- | | | |
|-------|--------------------|-----------------------|
| 1. 23 | 5. $\frac{9}{4}$ | 9. $-\frac{7}{9}$ |
| 2. 62 | 6. $\frac{73}{30}$ | 10. $\frac{a+b}{a}$ |
| 3. 0 | 7. $2ab - a^2$ | 11. $\frac{x-2}{x+5}$ |
| 4. 39 | 8. $n^2 - 3mn$ | |

EJERCICIO 86

- | | |
|---|--|
| 1. $\begin{cases} x = -3 \\ y = -6 \end{cases}$ | 7. $\begin{cases} a = 2 \\ b = 0 \end{cases}$ |
| 2. $\begin{cases} m = -2 \\ n = -3 \end{cases}$ | 8. $\begin{cases} m = \frac{9}{5} \\ n = -\frac{1}{3} \end{cases}$ |
| 3. $\begin{cases} a = -2 \\ b = 5 \end{cases}$ | 9. Conjunto infinito de soluciones |
| 4. $\begin{cases} x = -7 \\ y = -1 \end{cases}$ | 10. No hay solución |
| 5. $\begin{cases} p = 2 \\ q = 3 \end{cases}$ | 11. No hay solución |
| 6. $\begin{cases} x = -\frac{2}{3} \\ y = -\frac{7}{2} \end{cases}$ | 12. Conjunto infinito de soluciones |

EJERCICIO 87

- | | | |
|---|--|--|
| 1. $\begin{cases} x = -1 \\ y = 2 \end{cases}$ | 9. $\begin{cases} x = 6 \\ y = 5 \end{cases}$ | 17. $\begin{cases} x = \frac{1}{3} \\ y = -\frac{1}{2} \end{cases}$ |
| 2. $\begin{cases} a = -3 \\ b = 4 \end{cases}$ | 10. $\begin{cases} x = -1 \\ y = 2 \end{cases}$ | 18. $\begin{cases} x = 2 \\ y = 5 \end{cases}$ |
| 3. $\begin{cases} m = -2 \\ n = -2 \end{cases}$ | 11. $\begin{cases} x = 4 \\ y = -1 \end{cases}$ | 19. $\begin{cases} x = -\frac{1}{2} \\ y = -\frac{1}{4} \end{cases}$ |
| 4. $\begin{cases} x = 1 \\ y = 2 \end{cases}$ | 12. $\begin{cases} p = 5 \\ q = -1 \end{cases}$ | 20. $\begin{cases} x = \frac{1}{10} \\ y = \frac{1}{5} \end{cases}$ |
| 5. $\begin{cases} m = 3 \\ n = 5 \end{cases}$ | 13. $\begin{cases} x = 2 \\ y = -3 \end{cases}$ | 21. $\begin{cases} x = b \\ y = a \end{cases}$ |
| 6. $\begin{cases} x = \sqrt{3} \\ y = \sqrt{2} \end{cases}$ | 14. $\begin{cases} x = 3 \\ y = -7 \end{cases}$ | 22. $\begin{cases} x = a^2 \\ y = b^2 \end{cases}$ |
| 7. $\begin{cases} x = -5 \\ y = 4 \end{cases}$ | 15. $\begin{cases} a = 1 \\ b = -2 \end{cases}$ | 23. $\begin{cases} x = a + b \\ y = a - b \end{cases}$ |
| 8. $\begin{cases} x = \frac{2}{5} \\ y = \frac{1}{2} \end{cases}$ | 16. $\begin{cases} m = \frac{1}{3} \\ n = \frac{1}{2} \end{cases}$ | 24. $\begin{cases} x = \sqrt{a} \\ y = \sqrt{b} \end{cases}$ |

EJERCICIO 88

- | | |
|--|---|
| 1. $\begin{cases} 180 \\ 45 \end{cases}$ | 12. $\begin{cases} \text{Primera parte} = 350 \\ \text{Segunda parte} = 200 \end{cases}$ |
| 2. $\begin{cases} 140^\circ \\ 40^\circ \end{cases}$ | 13. $\begin{cases} 14 \\ 70 \end{cases}$ |
| 3. $\begin{cases} 80 \\ 50 \end{cases}$ | 14. $\begin{cases} 7 \\ 45 \end{cases}$ |
| 4. $\begin{cases} \frac{1}{8} \\ \frac{1}{6} \end{cases}$ | 15. $\begin{cases} \text{Alejandra tiene} = \$120 \\ \text{Beatriz tiene} = \$50 \end{cases}$ |
| 5. $\begin{cases} \$80 \text{ por adulto} \\ \$50 \text{ por niño} \end{cases}$ | 16. Lancha: 10 km/h
Corriente: 1 km/h |
| 6. 5 monedas de \$10 | 17. $\begin{cases} 25 \text{ gallinas} \\ 19 \text{ borregos} \end{cases}$ |
| 7. $\begin{cases} \text{Lados iguales} = 19 \text{ cm} \\ \text{Base} = 10 \text{ cm} \end{cases}$ | 18. $\begin{cases} \text{Gallinas} = \$30 \\ \text{Borregos} = \$300 \end{cases}$ |
| 8. $\begin{cases} \text{Agenda} = \$750 \\ \text{Traductor} = \$550 \end{cases}$ | 19. $\begin{cases} \text{Álgebra L.} = \$120 \\ \text{Geometría A.} = \$90 \end{cases}$ |
| 9. $\begin{cases} \text{Hermano} = 15 \text{ años} \\ \text{Antonio} = 5 \text{ años} \end{cases}$ | 20. $\begin{cases} 12.5 \text{ lt de la de } 30\% \\ 37.5 \text{ lt de la de } 6\% \end{cases}$ |
| 10. $\begin{cases} 73 \\ 65 \end{cases}$ | 21. $\begin{cases} \text{Veracruz} = 0.75 \text{ kg} \\ \text{Chiapas} = 0.25 \text{ kg} \end{cases}$ |
| 11. $\begin{cases} \text{Carlos tenía } \$300 \\ \text{Gabriel tenía } \$200 \end{cases}$ | |

EJERCICIO 89

$$1. \begin{cases} x=7 \\ y=3 \\ z=1 \end{cases}$$

$$2. \begin{cases} d=6 \\ e=-2 \\ f=3 \end{cases}$$

$$3. \begin{cases} x=3 \\ y=-3 \\ z=\frac{1}{3} \end{cases}$$

$$4. \begin{cases} x=\frac{2}{3} \\ y=\frac{1}{5} \\ z=-1 \end{cases}$$

$$5. \begin{cases} m=-3 \\ n=-2 \\ r=-1 \end{cases}$$

$$6. \begin{cases} a=\frac{1}{4} \\ b=\frac{1}{3} \\ c=\frac{1}{2} \end{cases}$$

$$7. \begin{cases} x=8 \\ y=6 \\ z=4 \end{cases}$$

$$8. \begin{cases} a=5 \\ b=-2 \\ c=-3 \end{cases}$$

$$9. \begin{cases} m=7 \\ n=3 \\ r=1 \end{cases}$$

$$10. \begin{cases} x=-4 \\ y=3 \\ z=2 \end{cases}$$

$$11. \begin{cases} x=3 \\ y=4 \\ z=5 \end{cases}$$

$$12. \begin{cases} a=\frac{1}{3} \\ b=\frac{1}{2} \\ c=1 \end{cases}$$

EJERCICIO 90

$$1. \begin{cases} \text{Paleta}=\$2 \\ \text{Helado}=\$4 \\ \text{Dulce}=\$1 \end{cases}$$

$$2. \begin{cases} \text{Camisa}=\$300 \\ \text{Pantalón}=\$500 \\ \text{Playera}=\$400 \end{cases}$$

$$3. \begin{cases} \text{Pares de calcetas}=\$50 \\ \text{Pantalón}=\$550 \\ \text{Playera}=\$120 \end{cases}$$

$$4. \begin{cases} \text{Centenas}=8 \\ \text{Decenas}=6 \\ \text{Unidades}=2 \\ \text{Número}=862 \end{cases}$$

EJERCICIO 91

$$1. \frac{2}{x+1} + \frac{3}{x-1}$$

$$2. \frac{7}{3x-7} + \frac{5}{2x-3}$$

$$3. \frac{1}{5x-4} - \frac{1}{5x+4}$$

$$4. \frac{2}{x+2} - \frac{1}{x-5}$$

$$5. -\frac{3}{x-7} - \frac{1}{x-4}$$

$$6. \frac{8}{2x-1} - \frac{6}{2x+1}$$

$$7. \frac{1}{x+3} + \frac{1}{x+2}$$

$$8. \frac{3}{2x+5} - \frac{2}{3x-1}$$

$$9. \frac{3}{4-x} - \frac{2}{x+3}$$

$$10. \frac{1}{2x+7} + \frac{5}{x-2}$$

$$11. \frac{1}{3x-2} - \frac{3}{(3x-2)^2}$$

$$12. \frac{2}{x} - \frac{1}{x+2} + \frac{3}{x-3}$$

$$13. \frac{2}{x-2} - \frac{1}{x+1} + \frac{1}{x+4}$$

$$14. \frac{1}{x+3} + \frac{2}{x-2} - \frac{3}{(x-2)^2}$$

$$15. \frac{5}{x} + \frac{4}{2x-1} + \frac{1}{x-3}$$

$$16. \frac{2}{x} + \frac{3}{x-1} + \frac{4}{x+2}$$

$$17. \frac{1}{2x+3} - \frac{1}{3x-2} - \frac{5}{x}$$

$$18. \frac{3}{x+1} + \frac{1}{x+3} - \frac{2}{x-2}$$

$$19. \frac{2}{x+1} - \frac{6}{2x+1} + \frac{3}{x-2}$$

$$20. \frac{2}{(x+1)^2} - \frac{1}{x+1} - \frac{1}{(x+1)^3}$$

$$21. \frac{1}{x} + \frac{1}{(x-1)^3} - \frac{3}{(x-1)^2} - \frac{2}{x-1}$$

$$22. \frac{1}{(x+3)^2} + \frac{1}{x+3} - \frac{1}{(x-3)^2} + \frac{1}{x-3}$$

EJERCICIO 92

$$1. \frac{3}{x} + \frac{x+1}{x^2-3}$$

$$2. \frac{1}{x+1} + \frac{x-2}{3x^2+1}$$

$$3. \frac{1}{x-2} + \frac{x-1}{x^2+1}$$

$$4. \frac{2}{x^2+5} - \frac{1}{x^2-7}$$

$$5. \frac{1}{x-1} + \frac{2x+3}{x^2+x+1}$$

$$6. \frac{x+1}{x^2+5} - \frac{7x}{x^2+3}$$

$$7. \frac{1}{x} - \frac{x+3}{x^2-2} + \frac{x+1}{x^2+3}$$

$$8. \frac{5}{x-3} + \frac{1}{x^2-2x-1}$$

$$9. \frac{x-3}{2x^2-7} + \frac{5x-1}{x^2+5}$$

$$10. \frac{x-2}{x^2+5x-3} - \frac{8}{x}$$

$$11. \frac{3x-1}{x^2+4x+5} - \frac{1}{2x-4}$$

12. $\frac{1}{x} + \frac{x+1}{(x^2-3)^2} + \frac{1}{x^2-3}$
13. $\frac{x}{x^2+x-1} + \frac{2x+1}{(x^2+x-1)^2}$
14. $\frac{x-3}{(x^2+1)^3} + \frac{1}{(x^2+1)^2} - \frac{5}{x^2+1}$
15. $\frac{2}{x+1} + \frac{1}{(x^2-2)^2} - \frac{1}{x^2-2}$
16. $\frac{x+1}{x^2+3x+4} - \frac{1}{(x^2+3x+4)^2} + \frac{1}{x}$
17. $\frac{2}{x^2} - \frac{1}{x} + \frac{1}{(x^2+1)^2} - \frac{1}{x^2+1}$
18. $\frac{1}{x^4} - \frac{1}{(x^2+2)^2}$
19. $\frac{1}{(x^2-2)^2} + \frac{3x+1}{x^2-2} - \frac{1}{x^2+1}$
20. $\frac{5x+17}{3(x^2-x+1)^2} + \frac{14-34x}{9(x^2-x+1)} + \frac{16}{9(x+1)} + \frac{4}{x-1}$

CAPÍTULO 9

EJERCICIO 93

- | | | |
|---------------------------|----------------------------|-------------------------|
| 1. $27x^6$ | 13. a^8 | 25. $\frac{a^3}{b^2}$ |
| 2. $16x^2y^2$ | 14. $\frac{9}{m^6}$ | 26. $\frac{1}{b^{12}}$ |
| 3. $\frac{16}{625}a^{16}$ | 15. $\frac{3a^2}{b}$ | 27. $\frac{1}{z^3}$ |
| 4. $-216x^6y^9$ | 16. $\frac{m^5}{n^3}$ | 28. $(x+2y)^6$ |
| 5. $-32a^{30}$ | 17. $\frac{3}{17a^4b}$ | 29. $108a^9$ |
| 6. $\frac{49}{16}m^{-4}$ | 18. $\frac{x^4}{y}$ | 30. $\frac{y^9}{12x^4}$ |
| 7. $\frac{32a^5}{b^5}$ | 19. $-\frac{1}{243}m^{10}$ | 31. $-\frac{2}{x^9}$ |
| 8. $16a^4x^4$ | 20. $16x^4$ | 32. 1 |
| 9. $-15y^3$ | 21. -9 | 33. a^2b^2 |
| 10. xy^7 | 22. 2 | 34. $72a^{13}$ |
| 11. x | 23. $\frac{5}{x^3}$ | |
| 12. $-mn$ | 24. $-\frac{1}{36x^2}$ | |

EJERCICIO 94

- | | | |
|-------------------------------------|-----------------------------|-------------------------------|
| 1. $x^9y^8z^6$ | 10. 1 | 19. $16y^{20}$ |
| $\frac{y}{x^{\frac{3}{4}}}$ | 11. $(x+3y)^{\frac{7}{6}}$ | 20. x^2+y^2 |
| 3. $x^{\frac{1}{4}}y^{\frac{1}{2}}$ | 12. $\frac{y^{12}}{64x^3}$ | 21. $(x-2y)^2$ |
| 4. $\frac{1}{4z^4}$ | 13. $\frac{a^3b^2}{c}$ | 22. $\frac{b^3-a^3}{a^3+b^3}$ |
| 5. $\frac{2x^2}{y^4}$ | 14. $\frac{y^6}{x^7}$ | 23. $\frac{1}{y-x}$ |
| 6. c | 15. $\frac{y^2}{x^{10}z^5}$ | 24. $\frac{y}{y+1}$ |
| 7. $6x^2y^5$ | 16. $\frac{c^4}{a^2b^4}$ | 25. $\frac{y^6-x^4}{x^4y^6}$ |
| 8. $16ab^2$ | 17. $\frac{9ab^{12}}{8}$ | 26. $x+y$ |
| 9. $\frac{y^{12}}{4x^{10}z^{14}}$ | 18. m^3n^7 | 27. $\frac{x^2-xy+y^2}{xy}$ |

EJERCICIO 95

1. $27-54x+36x^2-8x^3$
2. $1+4x+6x^2+4x^3+x^4$
3. $x^3-6x^2y+12xy^2-8y^3$
4. $1+\frac{3x}{2}+\frac{3x^2}{4}+\frac{x^3}{8}$
5. $x^6-6x^5+15x^4-20x^3+15x^2-6x+1$
6. $16-32x+24x^2-8x^3+x^4$
7. $x^{10}+5x^8y^2+10x^6y^4+10x^4y^6+5x^2y^8+y^{10}$
8. $\frac{x^5}{32}-\frac{5x^4}{16}+\frac{5x^3}{4}-\frac{5x^2}{2}+\frac{5x}{2}-1$
9. $\frac{1}{81}-\frac{2x}{27}+\frac{x^2}{6}-\frac{x^3}{6}+\frac{x^4}{16}$
10. $x^9+15x^6y^3+75x^3y^6+125y^9$
11. $\frac{1}{x^2}+\frac{1}{x^4}+\frac{1}{x^6}+\frac{1}{x^8}+\frac{1}{x^{10}}+\dots$
12. $\frac{1}{8x^3}+\frac{3}{16x^4}+\frac{3}{16x^5}+\frac{5}{32x^6}+\frac{15}{128x^7}+\dots$
13. $\frac{1}{x^4}+\frac{4}{x^5}+\frac{10}{x^6}+\frac{20}{x^7}+\frac{35}{x^8}+\dots$
14. $(3x)^{\frac{1}{3}}+\frac{1}{3(3x)^{\frac{2}{3}}}-\frac{1}{9(3x)^{\frac{5}{3}}}+\frac{5}{81(3x)^{\frac{8}{3}}}-\frac{10}{243(3x)^{\frac{11}{3}}}+\dots$
15. $x^{\frac{4}{3}}+\frac{8x^{\frac{1}{3}}}{3}+\frac{8}{9x^{\frac{2}{3}}}-\frac{32}{81x^{\frac{5}{3}}}+\frac{80}{243x^{\frac{8}{3}}}+\dots$
16. $\frac{1}{x^{\frac{3}{4}}}+\frac{3}{2x^{\frac{1}{4}}}+\frac{21}{8x^{\frac{1}{4}}}+\frac{77}{16x^{\frac{1}{4}}}+\frac{1155}{128x^{\frac{1}{4}}}+\dots$

EJERCICIO 96

- $127575x^5$
- $\frac{35}{8}x^4$
- $-439040x^3y^3$
- $\frac{22}{729(8x)^{\frac{14}{3}}}$
- $-253125000x^3$
- $\frac{1792}{x^9}$
- $\frac{1}{x^5}$
- $\frac{729}{512x^{\frac{3}{2}}}$

EJERCICIO 97

- $16x^4 + 32x^3 + 24x^2 + 8x + 1$
- $2187 - 10206y + 20412y^2 - 22680y^3 + 15120y^4 - 6048y^5 + 1344y^6 - 128y^7$
- $x^8 + 8x^7 + 28x^6 + 56x^5 + 70x^4 + 56x^3 + 28x^2 + 8x + 1$
- $x^6 - 6x^5 + 15x^4 - 20x^3 + 15x^2 - 6x + 1$
- $3125m^5 - 6250m^4n + 5000m^3n^2 - 2000m^2n^3 + 400mn^4 - 32n^5$
- $a^8 + 16a^7b + 112a^6b^2 + 448a^5b^3 + 1120a^4b^4 + 1792a^3b^5 + 1792a^2b^6 + 1024ab^7 + 256b^8$
- $x^{12} + 30x^{10}y + 375x^8y^2 + 2500x^6y^3 + 9375x^4y^4 + 18750x^2y^5 + 15625y^6$
- $\frac{x^7}{128} + \frac{7x^5}{64} + \frac{21x^3}{32} + \frac{35x}{16} + \frac{35}{8x} + \frac{21}{4x^3} + \frac{7}{2x^5} + \frac{1}{x^7}$
- $x^3 + 3x^2y - 6x^2 + 3xy^2 - 12xy + 12x + y^3 - 6y^2 + 12y - 8$
- $\frac{x^{10}}{y^5} - \frac{5x^7}{y^2} + 10x^4y - 10xy^4 + \frac{5y^7}{x^2} - \frac{y^{10}}{x^5}$
- $x^{12} - 12x^{11} + 66x^{10} - 220x^9 + 495x^8 - 792x^7 + 924x^6 - 792x^5 + 495x^4 - 220x^3 + 66x^2 - 12x + 1$
- $\frac{32}{x^5} + \frac{40}{x^4} + \frac{20}{x^3} + \frac{5}{x^2} + \frac{1}{8x} + \frac{1}{32}$

EJERCICIO 99

- $\sqrt[3]{2}$
- $\sqrt[7]{5^4}$
- $\sqrt[3]{m^2}$
- $\sqrt{(3y)^{11}}$
- $\sqrt[4]{(2xy^2)^3}$
- $\sqrt{x^3y}$
- $7\sqrt[5]{y^2}$
- $3\sqrt[5]{a^3}\sqrt[7]{b^8}$
- $\frac{3}{4}\sqrt[5]{z^2}\sqrt[4]{y}$
- $\sqrt[5]{m^2} - \sqrt[3]{n}$
- $\sqrt[7]{a} + \sqrt[7]{b}$
- $\sqrt[3]{x} - \sqrt[4]{y}$
- $\sqrt[5]{(2x+y)^4}$
- $\sqrt{m+n}$
- $\sqrt[3]{(a^3+b^3)^2}$
- $\sqrt[7]{(m^{-1}-n^{-2})^3}$

EJERCICIO 100

- 27
- 2
- 3
- 14
- 4
- $\frac{9}{4}$
- 6
- 8
- 4
- $3m^2n^3$
- $2x^4$
- $2x^3y^5$
- $2m^3n$
- $\frac{5}{y^{2n}}$
- $5m^2 - x_n^{4y-3}$
- $2(1+x)$
- $|x-y|$
- $\frac{3x}{2}, -\frac{3x}{2}$
- $\frac{|x+2y|}{|xy|}$

CAPÍTULO 10**EJERCICIO 98**

- $m^{\frac{5}{2}}$
- $x^{\frac{2}{7}}$
- $y^{\frac{4}{3}}$
- $a^{\frac{2}{5}}$
- $b^{\frac{11}{2}}$
- $(5x)^{\frac{1}{2}}$
- $(2x)^{\frac{5}{6}}$
- $(3y^2)^{\frac{3}{4}}$
- $(2xy)^{\frac{9}{2}}$
- $(x^2y)^{\frac{2}{9}}$
- $(x^4y^4)^{\frac{1}{8}}$
- $(x^6+y^6)^{\frac{1}{3}}$
- $(x^7-y^7)^{\frac{1}{2}}$
- $(x^2+y^2)^{\frac{4}{3}}$
- $(x+2y)^{\frac{11}{5}}$
- $a^{\frac{9}{5}} - b^{\frac{3}{7}}$
- $\left(\frac{1}{x^2} + \frac{1}{y^2}\right)^{\frac{5}{2}}$
- $m^{\frac{7}{4}}n^{\frac{3}{5}}$
- $m^{\frac{1}{2}}(n+p)^{\frac{3}{2}}$
- $a^{\frac{2}{3}}m^{\frac{13}{3}}n^{\frac{7}{4}}$

EJERCICIO 101

- $x\sqrt{x}$
- $3xy^3\sqrt{3y}$
- $8mnz^2\sqrt{m}$
- $3mn^5\sqrt[3]{m^2}$
- $xyz^2\sqrt[3]{x^2y}$
- $5xy^2\sqrt[4]{x}$
- $15a^2b\sqrt{2b}$
- $15p^2q^3\sqrt{q}$
- $6xy^4\sqrt{3xz}$
- $10b\sqrt[4]{5a^3b^3}$
- $6mn^2\sqrt[5]{3m^3n^2}$
- $2x^2yz^3\sqrt[3]{xy^2}$
- $-6m^3n^3\sqrt[4]{8mn^2}$
- $a^2\sqrt{2a}$
- $5a\sqrt[5]{a^4}$
- $\frac{4}{3}mn^3\sqrt[3]{20mp^2}$
- $xy^3\sqrt{y}$
- $\frac{2}{x}\sqrt[4]{x}$
- $\frac{3x}{y}\sqrt{x}$
- $\frac{2ab^2}{m}\sqrt[3]{\frac{2a}{3m^2}}$
- $\frac{xy}{2z^3}\sqrt[4]{x^3y}$
- $2x^2yz^3\sqrt[3]{2a^2}$
- $\frac{3}{4}\sqrt[4]{\frac{5}{3}}$
- $\frac{2}{3}\sqrt[4]{\frac{5}{x^2}}$
- $3m\sqrt{m-2n}$
- $|4x^2+5y^3|\sqrt{x}$
- $3ab\sqrt[3]{a^4-2ab}$
- $|m-n|\sqrt{m-n}$
- $3(x^2-y^2)\sqrt[5]{(x^2-y^2)^2}$
- $\frac{1}{\sqrt[3]{(2-m)^2}}; -\frac{1}{\sqrt[3]{(2-m)^2}}$

EJERCICIO 102

1. $\sqrt{45}$
2. $\sqrt{175}$
3. $\sqrt[3]{128}$
4. $\sqrt{\frac{25}{8}}$
5. $\sqrt[5]{\frac{32}{81}}$
6. $\sqrt{x^3}$
7. $\sqrt[3]{8x^5}$
8. $\sqrt[4]{m^{13}n^5}$
9. $\sqrt[4]{x^5y^9}$
10. $\sqrt{50a^5b^6c^3}$

EJERCICIO 103

1. $5\sqrt{5}$
2. $-6\sqrt[3]{3}$
3. 0
4. $8\sqrt{7} - \sqrt{5}$
5. $-3\sqrt{3} - 7\sqrt{2}$
6. $\frac{5}{4}\sqrt{10} + \frac{1}{2}\sqrt{13}$
7. $-\frac{5}{8}\sqrt{6}$
8. $-4\sqrt[3]{m}$
9. $3\sqrt{x}$
10. $\frac{5}{3}\sqrt[4]{xy}$
11. $4\sqrt{7}$
12. $27\sqrt{2}$
13. $-17\sqrt{3}$
14. $8\sqrt{5} + 7\sqrt{2}$
15. $-2\sqrt{3}$
16. $\sqrt{2} - 10\sqrt{5}$
17. $47\sqrt{5} - 50\sqrt{11}$
18. $-2\sqrt{2} + \frac{4}{5}\sqrt{5}$
19. $9\sqrt{7a^2}$

11. $\frac{1}{\sqrt[3]{4a}}$
12. $\sqrt[3]{\frac{4a^2}{5b^2}}$
13. $\sqrt[4]{\frac{27y^7}{128x^2}}$
14. $\sqrt{3ax^2}$
15. $\sqrt[3]{4x}$
16. $\sqrt{4a^3b + 4a^2b^2 + ab^3}$
17. $\sqrt{\frac{9}{a+b}}$
18. $\sqrt{\frac{x+1}{(x-1)^3}}$
19. $\sqrt[3]{\frac{x^2+2x+4}{x^2-4x+4}}$
20. $\sqrt{2a^3x}$

20. $8a\sqrt{b}$
21. $11x\sqrt[3]{3x}$
22. $-14x^2\sqrt[4]{2}$
23. $3ay\sqrt{x}$
24. $\frac{11}{6}ab\sqrt{2b} + \frac{13}{4}ab\sqrt{3b}$
25. $\frac{17}{12}a^2b\sqrt{c}$
26. $-\frac{23}{24}a^2b^4\sqrt{2ab}$
27. $10x\sqrt{y} - 7x^2\sqrt{2y}$
28. $-xy^2\sqrt{xy} - x^2y\sqrt{3x}$
29. $x\sqrt{2y} + 4y\sqrt{3x}$
30. $-2ab\sqrt{2c} + 16ac\sqrt{3b}$
31. $-2x\sqrt[3]{y^2} - 3y\sqrt[3]{4x}$
32. $10ab\sqrt[4]{5a^2b^3} - 6a^2b^3\sqrt[4]{3ab^2}$
33. $\frac{1}{3}a\sqrt{5a} - \frac{1}{3}b\sqrt{3ab}$
34. $\frac{19}{20}x\sqrt[3]{y} + \frac{5}{12}xy\sqrt[3]{xy^2}$
35. $-\frac{1}{9}a^3b\sqrt{\frac{2a}{b}} + \frac{2}{3}ab^3\sqrt{\frac{5a}{3}}$
36. $6\sqrt{a-2}$
37. $-x\sqrt{x+2}$
38. $10xy\sqrt{x-3y}$

EJERCICIO 104

1. $3\sqrt{2}$
2. $5\sqrt{6}$
3. $2\sqrt[3]{9}$
4. $27\sqrt{10}$
5. xy^2
6. $xy\sqrt[3]{xy^2}$
7. $\frac{2}{15}x^3$
8. $18a^2b\sqrt{2b}$
9. $2xy^3\sqrt{3xy}$
10. $a^4\sqrt{a}$
11. $2a^2\sqrt[4]{a}$
12. $-8ab^2\sqrt[3]{6b}$
13. $6a\sqrt{a}$
14. $8x\sqrt[4]{x^3}$
15. $-6a^3b^3\sqrt{2b}$
16. $\frac{3a}{5x}\sqrt{\frac{2a}{x}}$
17. $\frac{2x}{am^2}\sqrt{3}$
18. $6 - 4\sqrt{6}$
19. $5 - \sqrt[3]{25}$
20. $\frac{8}{3}x^2\sqrt{2} - x$
21. $19 - 8\sqrt{3}$
22. 95
23. $2m - 3n\sqrt{2m} - 4n^2$
24. $x - y$
25. $|x+y|\sqrt{x-y}$
26. $\sqrt{x^2 - y}$
27. $|1 - x|$
28. $\sqrt[3]{3}(x - y)$
29. $|x+y|\sqrt{x-y}$
30. 1

EJERCICIO 105

1. $\sqrt[6]{18}$
2. $x\sqrt{x}$
3. $\sqrt[16]{x^{15}}$
4. $xy\sqrt[6]{108x^3y^2}$
5. $3x^2y\sqrt[4]{x^3}$
6. $a\sqrt[4]{4ab^3}$
7. $x\sqrt[6]{72x}$
8. $\sqrt[6]{2x^5y^5}$
9. $a^{12}\sqrt{a}$
10. $x\sqrt[6]{x^5}$
11. $y^6\sqrt{y}$
12. $\frac{1}{y}\sqrt[3]{2xy}$
13. $\frac{3^9\sqrt{8mn^2}}{2}$
14. $\frac{3}{2}x^{2/10}\sqrt[10]{16xy^4}$
15. $\sqrt[12]{x^4y^3z^2}$

EJERCICIO 106

1. $n\sqrt{m}$
2. $xy^2\sqrt[6]{xy}$
3. $3a^3b^2c$
4. $4y\sqrt{x}$
5. $3x^2y\sqrt[3]{3y^2}$
6. $6ab$
7. $\frac{2a^3}{5b^4}$
8. $9m^2x^2$
9. $\frac{3n^6}{4m^2}$
10. $\frac{2}{3zw^2}$
11. $\frac{5z^2}{3}$
12. $\frac{2u^3v^2}{5}$
13. $\frac{4a^2}{3b^2}$
14. $\frac{3x^3}{2y^4}$
15. $\frac{2p^4}{3m^3n^2}$
16. $\frac{2n^2}{p^2}$
17. $\frac{3}{2x}\sqrt[18]{y^{13}}$
18. $\frac{5}{2}y^2z$
19. $-\frac{5n}{4m^2}\sqrt[3]{n^2}$
20. $\frac{y^4x^4}{2}$

EJERCICIO 107

$$\begin{array}{lll}
1. \frac{1}{\sqrt[6]{6}} & 6. 2\sqrt[6]{\frac{9}{8a}} & 11. \sqrt[6]{\frac{4x}{y^4}} \\
2. \sqrt[3]{\frac{4}{y}} & 7. a\sqrt[9]{a} & 12. \sqrt[n^2+n]{x} \\
3. \sqrt[6]{\frac{2y^2}{3}} & 8. a\sqrt[6]{108ab^4} & 13. \sqrt[n^2+n]{x+1} \\
4. 8\sqrt[3]{\frac{b}{a^2}} & 9. \frac{1}{\sqrt[6]{xy}} & 14. \sqrt[6]{x-1} \\
5. \sqrt[15]{\frac{32a}{b}} & 10. \frac{1}{x\sqrt[4]{x}} & 15. \sqrt[12]{(a-b)^5}
\end{array}$$

EJERCICIO 108

$$\begin{array}{lll}
1. \frac{\sqrt{3}}{3} & 8. \frac{1}{b}\sqrt[3]{3a^2b^2} & 15. \frac{1}{10}(4-\sqrt{6}) \\
2. \frac{2\sqrt{5}}{5} & 9. \frac{1}{4x^2}\sqrt{6xy} & 16. \sqrt{3a} + \sqrt{2b} \\
3. \frac{\sqrt[3]{300}}{6} & 10. \frac{2}{5ab}\sqrt[4]{25a^3b^2} & 17. (1+x)(1-\sqrt{x}) \\
4. 3\sqrt[3]{x} & 11. \frac{1}{11}(1+2\sqrt{3}) & 18. \frac{2xy(\sqrt[3]{x^2}-\sqrt[3]{xy}+\sqrt[3]{y^2})}{x+y} \\
5. 2x\sqrt{3xy} & 12. \frac{2}{7}(3+\sqrt{2}) & 19. -(1+\sqrt[3]{x}+\sqrt[3]{x^2}) \\
6. \frac{1}{xy}\sqrt[4]{8x^3y^3} & 13. \frac{5x+\sqrt{5x}}{1-5x} & 20. \sqrt[3]{9a^2}-\sqrt[3]{3ab}+\sqrt[3]{b^2} \\
7. \frac{3}{2}\sqrt[5]{16a^4} & 14. \sqrt{3}-2 &
\end{array}$$

EJERCICIO 109

$$\begin{array}{ll}
1. \frac{3}{\sqrt{3}} & 11. \frac{7}{7-2\sqrt{7}} \\
2. \frac{10}{\sqrt{2}} & 12. \frac{1}{5+\sqrt{2}} \\
3. \frac{2}{\sqrt[4]{8}} & 13. \frac{1}{5+2\sqrt{6}} \\
4. \frac{x}{\sqrt[6]{x^5}} & 14. -\frac{1}{(x+3)(\sqrt{x}+\sqrt{3})} \\
5. \frac{1}{2y\sqrt[3]{x^2y^2}} & 15. \frac{x-y}{x+3\sqrt{xy}+2y} \\
6. \frac{1}{3\sqrt[4]{x^3}} & 16. \frac{1}{2(\sqrt{5x}+\sqrt{6y})} \\
7. \frac{3}{2\sqrt{6x}} & 17. \frac{x-5}{\sqrt{x}+\sqrt{5}} \\
8. \frac{1}{2\sqrt[3]{4x}} & 18. \frac{1}{\sqrt[3]{x^2}+3\sqrt[3]{x}+9} \\
9. \frac{2}{x\sqrt[5]{2x^2}} & 19. \frac{1}{\sqrt[3]{a^2}+2\sqrt[3]{ab}+4\sqrt[3]{b^2}} \\
10. \frac{x}{2y\sqrt{3x}} & 20. -\frac{1}{(y+2)(\sqrt[3]{y^2}+\sqrt[3]{2y}+\sqrt[3]{4})}
\end{array}$$

CAPÍTULO 11**EJERCICIO 110**

$$\begin{array}{llll}
1. 4i & 5. 25i & 9. 5\sqrt{5}i & 13. 3+6i \\
2. 6i & 6. 2\sqrt{2}i & 10. 9\sqrt{2}i & 14. 2-4\sqrt{7}i \\
3. 7i & 7. 5\sqrt{2}i & 11. \frac{2}{7}\sqrt{3}i & 15. \frac{2}{3}+\frac{1}{2}\sqrt{5}i \\
4. 11i & 8. 3\sqrt{6}i & 12. \frac{5}{2}\sqrt{3}i & 16. \frac{4}{5}-2\sqrt{2}i
\end{array}$$

EJERCICIO 111

$$\begin{array}{ll}
1. 9i & 8. \frac{1}{2}\sqrt{3}i + \frac{5}{2}\sqrt{2}i \\
2. 3+i & 9. 11-i \\
3. -9i & 10. 7 \\
4. 11i & 11. 0 \\
5. 6\sqrt{6}i & 12. 0 \\
6. 0 & 13. 5\sqrt[4]{i} \\
7. -4\sqrt{2}i + 3\sqrt{3}i & 14. \frac{23}{12}x\sqrt{\sqrt{x}i}
\end{array}$$

EJERCICIO 112

$$\begin{array}{lll}
1. -1 & 6. i & 11. 3i-2 \\
2. -i & 7. -i & 12. 0 \\
3. -3i & 8. 3i-2 & 13. -1 \\
4. -1 & 9. 4i & 14. \text{Si } n \text{ es par: } 0 \\
& & \text{Si } n \text{ es impar: } -i \\
5. i & 10. 1 & 15. 0
\end{array}$$

EJERCICIO 113

$$\begin{array}{llll}
1. -9 & 6. -\frac{18}{5} & 11. \frac{2}{5} & 16. 0 \\
2. -12\sqrt{3}i & 7. -60 & 12. \sqrt{2}-4 & 17. \frac{1}{4} \\
3. -4\sqrt{3}i & 8. -6-3\sqrt{6} & 13. \frac{9}{10}i & 18. -i^n \\
4. -2 & 9. 4i & 14. \frac{6}{5} & 19. 2i \\
5. -\frac{3}{2}-\frac{5}{4}i & 10. 3 & 15. 0 & 20. -i
\end{array}$$

EJERCICIO 114

$$\begin{array}{lll}
1. (2, 3) & 5. (5, -2) & 9. 3 \\
2. -1+5i & 6. \frac{1}{2}-\frac{6}{7}i & 10. \left(5, -\frac{2}{11}\right) \\
3. (0, 7) & 7. -2i & 11. \frac{5}{2}-8i \\
4. \left(\frac{2}{3}, -\frac{5}{4}\right) & 8. \left(-\frac{1}{3}, 0\right) & 12. (1, -1)
\end{array}$$

EJERCICIO 115

$$\begin{array}{lll}
1. (10, 1) & 4. (5, -6) & 7. \left(\frac{4}{5}, -\frac{1}{2}\right) \\
2. (1, 0) & 5. \left(\frac{31}{20}, -\frac{1}{3}\right) & 8. (\sqrt{2}, -5) \\
3. (-2, -5) & 6. (0, 1) & 9. (\sqrt{3}, \sqrt{2})
\end{array}$$

10. $7 - i$ 15. $11 - \frac{7}{2}i$ 20. $16 - 4i$
 11. 6 16. $4 - 10i$ 21. $2 + 3i$
 12. 0 17. 1 22. $4 + 5i$
 13. $-1 + 11i$ 18. $-\frac{1}{2} + \frac{19}{6}i$
 14. $3i$ 19. $4 - 3i$

EJERCICIO 116

1. $-17 + 6i$ 7. $\frac{13}{6}i$ 13. $-8 - 6i$
 2. $5 + i$ 8. $-2 + \frac{\sqrt{2}}{2}i$ 14. $\frac{39}{400} - \frac{1}{5}i$
 3. $6 + 4i$ 9. $6 + 18i$ 15. $32 - 126i$
 4. $1 - 3i$ 10. $-2 + 10i$ 16. 4
 5. $-3 + 4i$ 11. $-\frac{14}{3} + 4i$ 17. $-5 + 13i$
 6. $-1 + 2\sqrt{6}i$ 12. i 18. $-3 + 4i$

$$19. (a + bi)(a - bi) = a^2 - b^2i^2 = a^2 + b^2$$

$$= \operatorname{Re}(z)^2 + \operatorname{Im}(z)^2$$

$$20. (1 + i)^n \cdot (1 - i)^n = (1 - i^2)^n = (1 + 1)^n$$

$$= [\operatorname{Re}(z_1) + \operatorname{Re}(z_2)]^n$$

$$21. w^{2n} = (1 + i)^{2n} = [(1 + i)^2]^n = (2i)^n \text{ si } n \text{ es par,}$$

entonces $n = 2k$ con $k \in \mathbb{Z}$, sustituyendo:

$$(2i)^n = (2i)^{2k} = (4i^2)^k = (-4)^k = (-1 \cdot 4)^k$$

$$= (-1)^k \cdot (4)^k$$

$$= (-1)^{\frac{n}{2}} (2)^{2k}$$

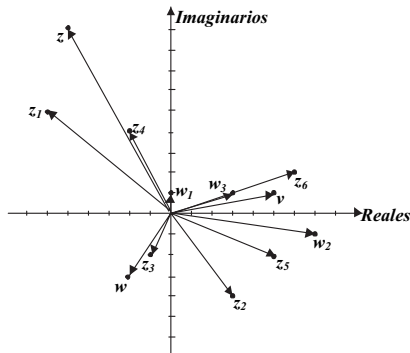
$$= (-1)^{\frac{n}{2}} (2, 0)^n$$

$$22. w^{2n} = (1 + i)^{2n} = [(1 + i)^2]^n = (2i)^n = (0, 2)^n$$

EJERCICIO 117

1. $\frac{i - 2}{5}$ 8. $\frac{-1 + 8i}{5}$
 2. $\frac{5 - 12i}{13}$ 9. $\frac{-2 + 3i}{2}$
 3. $-3 - i$ 10. $\frac{-13 - 9i}{5}$
 4. $\frac{-1 - 2\sqrt{6}i}{5}$ 11. $\frac{-2 + 11i}{25}$
 5. $\frac{-4 - \sqrt{2}i}{2}$ 12. $\frac{-129 - 107i}{10}$
 6. $1 + i$ 13. $\frac{4\sqrt{2} - 4i}{3}$
 7. $\frac{3 + i}{2}$ 14. $-i$

EJERCICIO 118



EJERCICIO 119

1. $\sqrt{13}$ 16. $-5i$ 31. $3 + 2i$
 2. $\sqrt{41}$ 17. $-\frac{1}{2}i$ 32. $2 - 4i$
 3. $\sqrt{41}$ 18. $(2, -1)$ 33. $1 + 8i$
 4. 3 19. $(0, 3)$ 34. $2 - 9i$
 5. $\sqrt{5}$ 20. $-\frac{3}{7} + \frac{2}{5}i$ 35. 4
 6. $\sqrt{85}$ 21. $-2 - 6i$ 36. 11
 7. $\frac{3}{2}$ 22. $(-1, 1)$ 37. $-4 + 6i$
 8. $\sqrt{5}$ 23. $-2 - \frac{11}{4}i$ 38. $7 - 6i$
 9. $\sqrt{2}$ 24. $\left(-\frac{1}{2}, -\frac{1}{3}\right)$ 39. $1 + \frac{2}{3}i$
 10. $\sqrt{\frac{79}{3}}$ 25 a 30. 40. $\frac{5}{6}i$
 11. $\frac{2}{3}\sqrt{13}$ No se incluye la solución 41. $\frac{17 - 7i}{13}$
 12. $\sqrt{11}$ 42. $\frac{2 + i}{5}$
 13. $5 - 4i$ 43. 1
 14. -5 44. $\frac{5}{12}$
 15. $1 - i$ 45. $\frac{i}{4}$

CAPÍTULO 12

EJERCICIO 120

1. $x_1 = -4, x_2 = -1$ 6. $z_1 = 15, z_2 = -2$
 2. $x_1 = -9, x_2 = 3$ 7. $x_1 = 6, x_2 = 4$
 3. $x_1 = -6, x_2 = -5$ 8. $x_1 = -20, x_2 = 12$
 4. $y_1 = 3 + i, y_2 = 3 - i$ 9. $x_1 = -1 + 2i, x_2 = -1 - 2i$
 5. $w_1 = -5, w_2 = 8$ 10. $x_1 = 1, x_2 = -\frac{2}{3}$

11. $x_1 = -\frac{1}{2}, x_2 = -2$

12. $w_1 = \frac{3}{2}, w_2 = -\frac{1}{5}$

13. $x_1 = 3, x_2 = -\frac{2}{3}$

14. $x_1 = \frac{1}{2} - \frac{1}{3}i, x_2 = \frac{1}{2} + \frac{1}{3}i$

15. $x_1 = -b, x_2 = -\frac{1}{4}b$

16. $w_1 = -\frac{3}{7}a, w_2 = 5a$

17. $x_1 = -5b, x_2 = 2b$

18. $x_1 = -\frac{5}{b}, x_2 = \frac{6}{b}$

19. $y_1 = -\frac{2b}{a}, y_2 = -\frac{b}{a}$

20. $y_1 = \frac{10}{7}a, y_2 = \frac{a}{2}$

10. $\begin{cases} x_1 = -\frac{1}{7} \\ x_2 = \frac{5}{2} \end{cases}$

11. $\begin{cases} x_1 = -\frac{2}{5} \\ x_2 = \frac{1}{4} \end{cases}$

12. $\begin{cases} z_1 = \frac{7}{5} \\ z_2 = 2 \end{cases}$

13. $\begin{cases} w_1 = -\frac{1}{2} \\ w_2 = \frac{6}{5} \end{cases}$

14. $\begin{cases} x_1 = -\frac{3}{2} \\ x_2 = \frac{2}{7} \end{cases}$

15. $\begin{cases} x_1 = -5 \\ x_2 = \frac{3}{2} \end{cases}$

16. $\begin{cases} x_1 = -\frac{5}{2}b \\ x_2 = \frac{2}{3}b \end{cases}$

17. $\begin{cases} x_1 = \frac{ab}{2} \\ x_2 = 2ab \end{cases}$

18. $\begin{cases} x_1 = -\frac{1}{a} \\ x_2 = \frac{3}{a} \end{cases}$

19. $\begin{cases} x_1 = -\frac{3b}{a} \\ x_2 = \frac{2b}{a} \end{cases}$

20. $\begin{cases} z_1 = 2\sqrt{3} \\ z_2 = -\sqrt{3} \end{cases}$

21. $\begin{cases} x_1 = -3\sqrt{3} \\ x_2 = 5\sqrt{3} \end{cases}$

22. $\begin{cases} x_1 = 2\sqrt{7} \\ x_2 = 5\sqrt{7} \end{cases}$

23. $\begin{cases} y_1 = -\frac{1}{2} \\ y_2 = -\frac{1}{15} \end{cases}$

24. $\begin{cases} x_1 = \frac{2}{3} \\ x_2 = -\frac{1}{4} \end{cases}$

25. $\begin{cases} w_1 = -\frac{1}{3} \\ w_2 = \frac{2}{5} \end{cases}$

EJERCICIO 121

1. $x_1 = 3, x_2 = 5$

2. $x_1 = 3, x_2 = -2$

3. $x_1 = -4, x_2 = -2$

4. $x_1 = 5, x_2 = -3$

5. $x_1 = \frac{5}{2}, x_2 = \frac{5}{2}$

6. $x_1 = -\frac{5}{2}, x_2 = \frac{1}{3}$

7. $y_1 = -\frac{3}{5}, y_2 = 1$

8. $x_1 = 3 - \sqrt{7}, x_2 = 3 + \sqrt{7}$

9. $x_1 = -1 - \sqrt{6}, x_2 = -1 + \sqrt{6}$

10. $x_1 = 2 - i, x_2 = 2 + i$

11. $x_1 = -\frac{1}{2} - 2i, x_2 = -\frac{1}{2} + 2i$

12. $x_1 = \frac{1}{3} + \frac{3}{2}i, x_2 = \frac{1}{3} - \frac{3}{2}i$

13. $w_1 = 0, w_2 = 5$

14. $z_1 = 0, z_2 = -\frac{5}{2}$

15. $y_1 = 0, y_2 = \frac{a}{3}$

16. $x_1 = 0, x_2 = \frac{b}{a}$

17. $x_1 = -5, x_2 = 5$

18. $x_1 = -\frac{1}{2}, x_2 = \frac{1}{2}$

19. $x_1 = -\frac{b}{a}i, x_2 = \frac{b}{a}i$

20. $w_1 = -\frac{4}{a}, w_2 = \frac{4}{a}$

EJERCICIO 122

1. Reales y diferentes
2. Complejas
3. Complejas
4. Reales e iguales
5. Reales y diferentes
6. Complejas
7. Reales y diferentes
8. Complejas
9. Reales y diferentes
10. Reales e iguales
11. Complejas
12. Reales e iguales

EJERCICIO 123

1. $\begin{cases} x_1 = -1 \\ x_2 = 6 \end{cases}$

2. $\begin{cases} x_1 = -8 \\ x_2 = -3 \end{cases}$

3. $\begin{cases} y_1 = -4 \\ y_2 = 5 \end{cases}$

4. $\begin{cases} x_1 = -9 \\ x_2 = 10 \end{cases}$

5. $\begin{cases} w_1 = 1 \\ w_2 = 4 \end{cases}$

6. $\begin{cases} y_1 = \frac{5}{3} \\ y_2 = 2 \end{cases}$

7. $\begin{cases} x_1 = 1 \\ x_2 = -\frac{2}{3} \end{cases}$

8. $\begin{cases} y_1 = -4 \\ y_2 = \frac{1}{2} \end{cases}$

9. $\begin{cases} x_1 = -\frac{2}{3} \\ x_2 = 3 \end{cases}$

EJERCICIO 124

1. $\begin{cases} x_1 = 0 \\ x_2 = -6 \end{cases}$

2. $\begin{cases} x_1 = 0 \\ x_2 = 2 \end{cases}$

3. $\begin{cases} x_1 = 0 \\ x_2 = 5 \end{cases}$

4. $\begin{cases} x_1 = 0 \\ x_2 = -\frac{2}{3} \end{cases}$

5. $\begin{cases} x_1 = 0 \\ x_2 = 1 \end{cases}$

6. $\begin{cases} x_1 = 0 \\ x_2 = \frac{5}{7} \end{cases}$

7. $\begin{cases} x_1 = 0 \\ x_2 = \frac{1}{2} \end{cases}$

8. $\begin{cases} x_1 = 0 \\ x_2 = -4 \end{cases}$

9. $\begin{cases} x_1 = 0 \\ x_2 = -8 \end{cases}$

10. $\begin{cases} x_1 = 0 \\ x_2 = 2 \end{cases}$

EJERCICIO 125

1. $\begin{cases} x_1 = -2 \\ x_2 = 2 \end{cases}$

2. $\begin{cases} x_1 = -1 \\ x_2 = 1 \end{cases}$

3. $\begin{cases} w_1 = -10 \\ w_2 = 10 \end{cases}$

4. $\begin{cases} x_1 = -8 \\ x_2 = 8 \end{cases}$

5. $\begin{cases} y_1 = -\sqrt{3} \\ y_2 = \sqrt{3} \end{cases}$

6. $\begin{cases} x_1 = -\frac{a}{4} \\ x_2 = \frac{a}{4} \end{cases}$

7. $\begin{cases} z_1 = -\frac{6}{5} \\ z_2 = \frac{6}{5} \end{cases}$

8. $\begin{cases} y_1 = -6 \\ y_2 = 6 \end{cases}$

9. $\begin{cases} w_1 = -\sqrt{7} \\ w_2 = \sqrt{7} \end{cases}$

10. $\begin{cases} x_1 = -\sqrt{3} \\ x_2 = \sqrt{3} \end{cases}$

11. $\begin{cases} x_1 = -\frac{2}{3} \\ x_2 = \frac{2}{3} \end{cases}$

12. $\begin{cases} x_1 = -1 \\ x_2 = 1 \end{cases}$

13. $\begin{cases} y_1 = -4i \\ y_2 = 4i \end{cases}$

14. $\begin{cases} w_1 = -5i \\ w_2 = 5i \end{cases}$

15. $\begin{cases} x_1 = -i \\ x_2 = i \end{cases}$

EJERCICIO 126

1. 27 y 15
2. 30 y 12
3. 3, 5 y 7
4. 12, 14 y 16
5. $\frac{1}{5}$ y 5
6. 5 y 20
7. $\begin{cases} \text{largo}=200 \text{ m} \\ \text{base}=125 \text{ m} \end{cases}$
8. $\begin{cases} \text{largo}=250 \text{ m} \\ \text{base}=100 \text{ m} \end{cases}$
9. 3, 4 y 5
10. 96 m^3
11. $\begin{cases} \text{altura}=18 \text{ m} \\ \text{base}=54 \text{ m} \end{cases}$
12. $\begin{cases} \text{Alejandro}=8 \text{ años} \\ \text{Alfredo}=4 \text{ años} \end{cases}$
13. 7
14. $\begin{cases} \text{árboles}=15 \\ \text{filas}=13 \end{cases}$
15. $r=4 \text{ cm}$
16. $\begin{cases} \text{largo}=17 \text{ cm} \\ \text{ancho}=16 \text{ cm} \end{cases}$
17. 39 años
18. 5 segundos
19. 7.5 segundos
20. $\begin{cases} \text{primera llave}=8 \text{ h} \\ \text{segunda llave}=24 \text{ h} \end{cases}$
21. \$20
22. 8, 6 y 10 unidades

EJERCICIO 127

1. $V(2, -2)$ $x_1=1, x_2=3$
2. $V\left(\frac{1}{2}, \frac{25}{2}\right)$ $x_1=-2, x_2=3$
3. $V\left(\frac{1}{2}, -\frac{81}{4}\right)$ $x_1=-4, x_2=5$
4. $V(-2, -7)$ $x_1=-\sqrt{7}-2, x_2=\sqrt{7}-2$
5. $V(-1, 4)$ $x_1=-1+2i, x_2=-1-2i$
6. $V(1, 0)$ $x_1=1, x_2=1$
7. $V(2, 9)$ $x_1=2+3i, x_2=2-3i$
8. $V(5, 0)$ $x_1=5, x_2=5$
9. $V(0, -9)$ $x_1=3i, x_2=-3i$
10. $V\left(\frac{3}{2}, -\frac{9}{2}\right)$ $x_1=0, x_2=3$

EJERCICIO 128

1. 50 y 50
2. -10 y 10
3. 20 y 20
4. 55 y 55
5. 72 pies
6. 19 cm y 19 cm
7. 500 ejemplares
8. 20 pelotas
9. 35 cajas
10. $\begin{cases} 56.5 \text{ cm} \\ 43.5 \text{ cm} \end{cases}$

EJERCICIO 129

1. $\begin{cases} x_1+x_2=0 \\ x_1 \cdot x_2=-\frac{9}{4} \end{cases}$
2. $\begin{cases} x_1+x_2=0 \\ x_1 \cdot x_2=-25 \end{cases}$
3. $\begin{cases} x_1+x_2=1 \\ x_1 \cdot x_2=0 \end{cases}$
4. $\begin{cases} x_1+x_2=-\frac{8}{3} \\ x_1 \cdot x_2=0 \end{cases}$
5. $\begin{cases} x_1+x_2=5 \\ x_1 \cdot x_2=6 \end{cases}$
6. $\begin{cases} x_1+x_2=-4 \\ x_1 \cdot x_2=3 \end{cases}$

7. $\begin{cases} x_1+x_2=1 \\ x_1 \cdot x_2=-12 \end{cases}$
8. $\begin{cases} x_1+x_2=-\frac{1}{2} \\ x_1 \cdot x_2=-\frac{1}{2} \end{cases}$
9. $\begin{cases} x_1+x_2=-3 \\ x_1 \cdot x_2=\frac{14}{9} \end{cases}$
10. $\begin{cases} x_1+x_2=-7a \\ x_1 \cdot x_2=12a^2 \end{cases}$

EJERCICIO 130

1. $x^2-9=0$
2. $x^2+7x=0$
3. $x^2+16=0$
4. $x^2-5x+4=0$
5. $x^2+8x+15=0$
6. $x^2+4x+29=0$
7. $2x^2-5x+2=0$
8. $20x^2+19x+3=0$
9. $x^2+2bx-3b^2=0$
10. $x^2-7ax+10a^2=0$

EJERCICIO 131

1. $x=49$
2. $x=-8$
3. $x=\frac{13}{2}$
4. $x=5$
5. $x=5$
6. $x=2$
7. $x=3$
8. $x=7$
9. $x=-1$
10. $x=1$
11. $x=4$
12. $x=1$
13. $x=\frac{1}{256}$
14. $x=1$
15. $x=3$
16. $x=-2, -7$
17. $x=-1, 1$
18. $x=9$

EJERCICIO 132

1. (0, 0), (4, 4)
2. (0, 3), (3, 0)
3. (3, -3), (-3, 3)
4. (2, 4), (-2, -4)
5. (-3, -5), (5, 3)
6. (3, 2), (-3, -1)
7. (4, 3), (4, -3), (-4, 3), (-4, -3)
8. $\left(\frac{1}{3}, 0\right), \left(-\frac{1}{3}, 0\right)$
9. $\left(2, 4\sqrt{2}\right), \left(2, -4\sqrt{2}\right), \left(-2, 4\sqrt{2}\right), \left(-2, -4\sqrt{2}\right)$
10. $(7, -7), (-7, 7), (2\sqrt{7}, \sqrt{7}), (-2\sqrt{7}, -\sqrt{7})$
11. (4, 2), (-4, -2), (5, 1), (-5, -1)
12. $(5, 1), (-5, -1), (-\sqrt{3}, 2\sqrt{3}), (\sqrt{3}, -2\sqrt{3})$
13. (1, 1), (-1, -1), (-2, 0), (2, 0)
14. (3, 2), (-3, -2), $(4i, i), (-4i, -i)$
15. $\left(\frac{\sqrt{30}}{5}, \frac{2\sqrt{30}}{5}\right), \left(-\frac{\sqrt{30}}{5}, -\frac{2\sqrt{30}}{5}\right), (2, -3), (-2, 3)$
16. (3, 6), (-3, -6)
17. $(2, 1), (-2, -1), \left(-\frac{\sqrt{2}}{2}, \sqrt{2}\right), \left(\frac{\sqrt{2}}{2}, -\sqrt{2}\right)$
18. (-2, 1), (2, -1)
19. $(1, 0), (-1, 0), \left(\frac{5\sqrt{17}}{17}, \frac{\sqrt{17}}{17}\right), \left(-\frac{5\sqrt{17}}{17}, -\frac{\sqrt{17}}{17}\right)$
20. (-1, -4), (2, -7), (1, 4), (-2, 7)

CAPÍTULO 13

EJERCICIO 133

1. $(3, \infty)$
2. $(-\infty, 3)$
3. $(-\infty, -4)$
4. $(-\infty, -1]$
5. $(-\infty, -7)$
6. $\left[-\frac{8}{7}, \infty\right)$
7. $\left(\frac{3}{2}, \infty\right)$
8. $(-\infty, 2)$
9. $(-\infty, 10]$
10. $(-\infty, 5]$
11. $\left(\frac{5}{3}, \infty\right)$
12. $\left(-\infty, \frac{8}{53}\right]$
13. $\left(\frac{4}{5}, \infty\right)$
14. $\left(-\infty, -\frac{1}{23}\right)$
15. $(2, \infty)$
16. $(6, \infty)$
17. $[-21, \infty)$
18. $[6, \infty)$
19. $(-\infty, -6]$
20. $\left[-\frac{19}{28}, \infty\right)$
21. $\left(\frac{18}{5}, \infty\right)$
22. $(-\infty, 13]$
23. $(-2, 3)$
24. $\left(-\frac{3}{2}, \frac{7}{2}\right)$
25. $[-3, -1]$
26. $[-9, -1)$
27. $(-3, 3)$
28. $[-23, -10]$
29. $(-2, 4)$
30. $[-4, 1)$
31. $(-1000, 100)$
32. $\left[\frac{2}{5}, \frac{3}{5}\right]$
33. $[-16, 8)$
34. $\left(-\frac{21}{2}, \frac{11}{2}\right)$
35. $\left(-\frac{1}{2}, \frac{19}{2}\right]$
36. $\left[-\frac{10}{3}, \frac{32}{3}\right]$
37. $(-14, -2)$
38. $[-2, 4]$
39. $\left[\frac{9}{2}, \frac{19}{2}\right]$
40. $\left(\frac{14}{9}, \frac{8}{3}\right)$

EJERCICIO 134

1. $(-3, 3)$
2. $[-4, 4]$
3. $(-\infty, -5] \cup [5, \infty)$
4. $(-\infty, -6) \cup (6, \infty)$
5. $\left[0, \frac{1}{3}\right]$
6. $(-\infty, 0) \cup (5, \infty)$
7. $(-\infty, 0) \cup (4, \infty)$
8. $(-\infty, -4) \cup (5, \infty)$
9. $\left(-\frac{1}{2}, 3\right)$
10. $\left[-\frac{1}{3}, \frac{3}{2}\right]$
11. $(-\infty, -4) \cup (-1, \infty)$
12. $(-\infty, -1] \cup \left[\frac{3}{4}, \infty\right)$

13. $\left(-\frac{2}{3}, 1\right)$

EJERCICIO 135

1. $\left(\frac{3}{4}, \infty\right)$
2. $\left(-\infty, \frac{5}{2}\right)$
3. $\left(2, \frac{5}{2}\right]$
4. $(-\infty, 2) \cup (2, \infty)$
5. $(-\infty, 3)$
6. $[-3, 2)$
7. $(-\infty, -1] \cup (3, \infty)$
8. $(-1, 3) \cup (11, \infty)$
9. $\left[-9, -\frac{1}{3}\right) \cup (4, \infty)$
10. $(-\infty, -2) \cup (2, 4]$
11. $(-4, -2) \cup (1, \infty)$
12. $(-\infty, -2) \cup \left[\frac{3}{2}, 4\right)$
13. $(-\infty, -6) \cup (1, 4]$

EJERCICIO 136

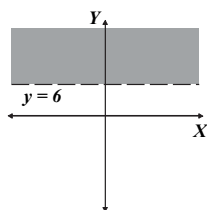
1. $[-2, -1] \cup [2, 4]$
2. $[2, \infty) \cup \{-2\}$
3. $(-\infty, -2) \cup (-1, 1)$
4. $(-\infty, -4)$
5. $(-3, 0) \cup (3, \infty)$
6. $(-\infty, -2) \cup (4, \infty)$

EJERCICIO 137

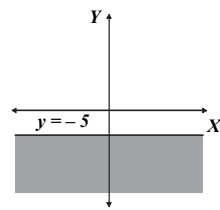
1. $(-\infty, -7] \cup [7, \infty)$
2. $(-7, 7)$
3. $(-\infty, 1) \cup (9, \infty)$
4. $\left[-\frac{9}{5}, 3\right]$
5. $(-\infty, 3) \cup (5, \infty)$
6. $\emptyset$
7. $[-9, 10]$
8. $(-\infty, -4) \cup (20, \infty)$
9. $[2, 18]$
10. $\left[\frac{1}{2}, \frac{5}{6}\right]$
11. $\left(\frac{1}{3}, \infty\right)$
12. $(-\infty, -2] \cup [0, \infty)$
13. $\emptyset$
14. $\left(-\infty, \frac{1}{2}\right)$
15. $\left(-\frac{4}{3}, 0\right) \cup (0, 4)$
16. $\left(-\infty, \frac{1}{2}\right]$
17. $(-\infty, 0) \cup (4, \infty)$

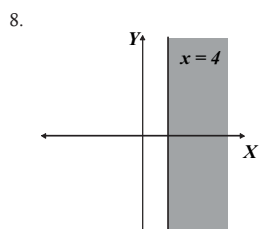
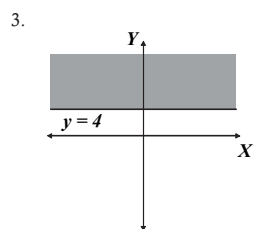
EJERCICIO 138

1.

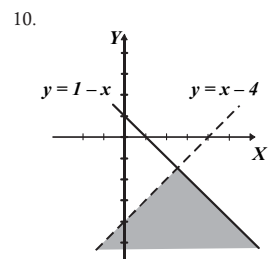
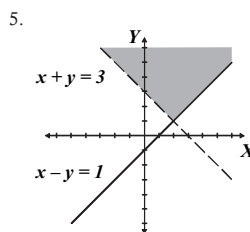
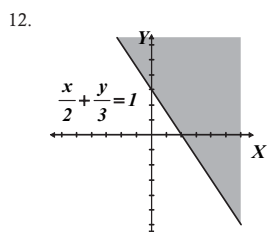
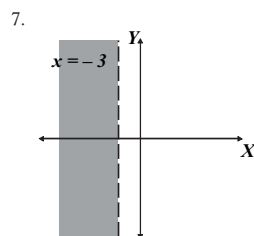
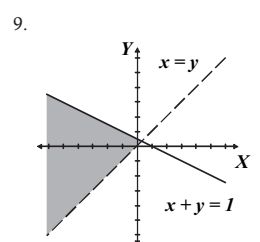
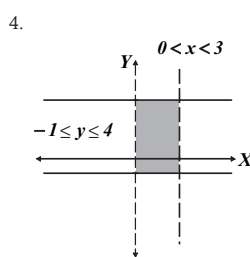
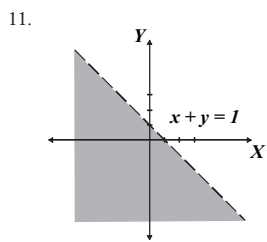
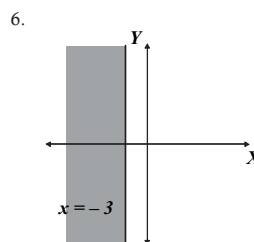
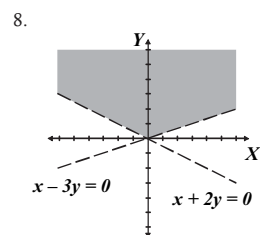
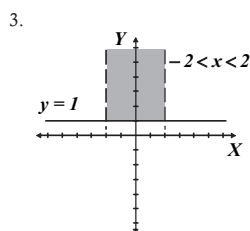
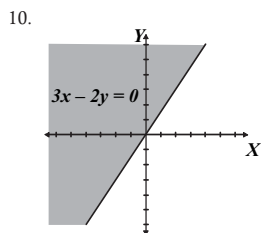
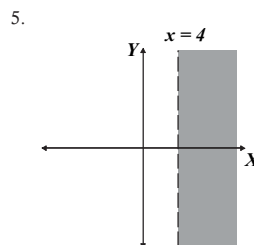
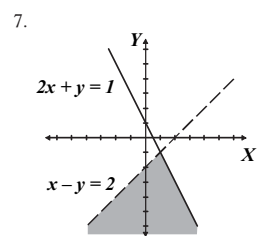
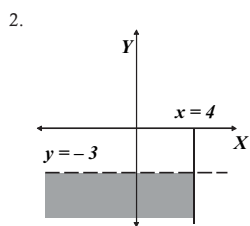
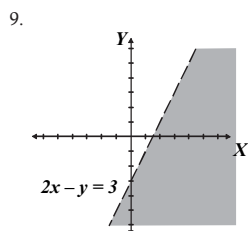
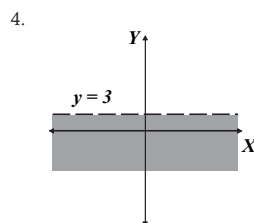
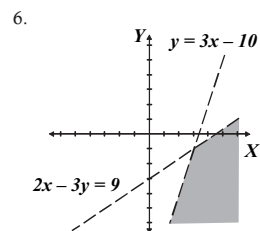
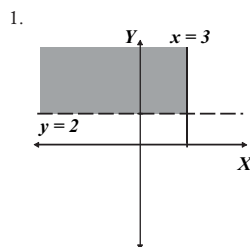


2.





EJERCICIO 139



CAPÍTULO 14

EJERCICIO 140

1. $8 = 2^3$
2. $16 = x^4$
3. $81 = 3^4$
4. $\frac{1}{36} = 6^{-2}$
5. $9 = (\sqrt{3})^4$
6. $343 = 7^x$
7. $\sqrt{6} = (a)^{\frac{1}{2}}$
8. $x - 1 = 3^2$
9. $625 = w^4$
10. $128 = (x - 1)^7$
11. $243 = (3x)^5$
12. $256 = (2x - 1)^8$
13. $\log_{17} a = 2$
14. $\log_5 625 = 4$
15. $\log_{64} 4 = \frac{1}{3}$
16. $\log_N \frac{1}{16} = 2$
17. $\log_{\frac{2}{3}} \left(\frac{4}{9} \right) = 2$
18. $\log_2 (x + 3) = 4$
19. $\log_2 256 = x$
20. $\log_{(x-2)} 8 = 3$
21. $\log_x z = w$
22. $\log_3 \frac{1}{81} = -4$
23. $\log_5 125 = -3x$
24. $\log_{(3x+2)} 441 = 2$

EJERCICIO 141

1. $x = 5$
2. $x = 4$
3. $y = 3$
4. $b = \frac{1}{5}$
5. $x = 4$
6. $a = 343$
7. $x = 81$
8. $m = 8$
9. $y = \frac{1}{32}$
10. $N = 8$
11. $w = 3$
12. $x = \frac{4}{9}$
13. $b = 2$
14. $x = 2$
15. $x = 3$
16. $a = -\frac{2}{5}$
17. $x = -6$
18. $y = -\frac{1}{4}$
19. $x = -3$

EJERCICIO 142

1. $4 \log_a 7$
2. $-\frac{3}{2} \log_6 3$
3. $\frac{7}{3} + \frac{1}{3} \log_e x$
4. $\log 5 + \log x + 2 \log y$
5. $3 \log_3 x + 2 \log_3 y + \log_3 z$
6. $8 + 2 \ln 3 + 4 \ln x$
7. $3 \log (x + y) + \log (x - z)$
8. $\log_{\frac{1}{2}} 7 - 2 \log_{\frac{1}{2}} x$
9. $\ln x + 2 \ln y - 3 - 4 \ln z$
10. $\log_5 3 + 3 \log_5 x + 6 \log_5 (1 - 2x) - \log_5 2 - y \log_5 x - \log_5 (x^2 - y^2)$

11. $\frac{1}{2} \log_4 3 + \log_4 x + 2 \log_4 y$
12. $2 \log (x + y) + \frac{5}{2} \log z$
13. $\frac{1}{3} \log x - \frac{1}{2} \log y$
14. $\frac{3}{2} \log a + \frac{1}{2} \log b - \frac{2}{3} \log c - \frac{1}{3} \log d$
15. $\frac{1}{2} \log_2 (x + y) - 4 \log_2 (x - y)$
16. $2 \log x - \frac{1}{3} \log (x - 3) - 2 \log (x + z)$
17. $\frac{1}{2} \log (x + 3) + \frac{1}{2} \log (y - 5) - 2 \log (x + 6) - \frac{1}{4} \log (y - 2)$
18. $\frac{2}{3} - \frac{x}{3} + \frac{2}{5} \ln (x + 1) + \frac{7}{30} \ln (x - 1)$
19. $\ln 25x^2$
20. $\log \frac{m^3}{n^2}$
21. $\log_7 \sqrt[6]{x^3 y^2}$
22. $\ln 8e^{4x}$
23. $\log n^4 \sqrt[5]{m^2}$
24. $\log_2 3 \cdot 4^x$
25. $\log_b \frac{1}{\sqrt[12]{(x+1)^8 (x+2)^3}}$
26. $\log \frac{3y}{x}$
27. $\log_2 \frac{x}{yz}$
28. $\log_4 \frac{4}{m^2 - 1}$
29. $\log \frac{x^{\frac{1}{8}} y^{\frac{1}{3}}}{\frac{1}{z^4}}$
30. $\ln \frac{5ey}{x^7}$
31. $\ln \frac{e^{2-x} (x+y)^3}{(x-y)^3}$
32. $\log \frac{(x-2)^{\frac{2}{3}} (x+1)^2}{(x+2)^{\frac{4}{5}}}$
33. $\log_2 \sqrt[2]{\frac{2x^{14}}{y^3}}$
34. $\log \frac{(x+1)^{\frac{1}{3}} (x-1)^{\frac{1}{2}}}{10x^{\frac{1}{6}}}$
35. $\log \frac{10^{x^2+x+1} (x+1)^3}{x^2}$
36. $\ln \left(\frac{9m^2 p}{7xy^3} \right)^2$

EJERCICIO 143

1. $x = 1$
2. $x = -20$
3. $x = 9, x = -\frac{27}{5}$
4. $x = 17$
5. $x = 6, x = -6$
6. $x = 13$
7. $x = 40$
8. $x = 25$
9. $x = 17, x = 7$
10. $x = \frac{9}{2}$
11. $x = 8, x = \frac{22}{9}$
12. $x = -1$
13. $x = 0, x = -35$
14. $x = 6$
15. $x = 3$
16. $x = 12, x = \frac{7}{11}$

17. $x = 5$

18. $x = 6$

19. $x = 7$

20. $x = 4$

EJERCICIO 144

1. $x = 4$

2. $x = \frac{3 \log 2}{\log 3}$

3. $x = 0$

4. $x = \frac{1}{2}$

5. $x = 1.20557$

6. $x = 2$

7. $x = 3$

8. $x = 2$

9. $x = -1$

10. $x = 3$

11. $x = \frac{2 \log 2 + 3 \log 5}{2 \log 5}$

12. $x = -1.72683$

13. $x = -\frac{7}{3}$

14. $x = \frac{7}{3}$

15. $x = -4$

16. $x = \frac{2}{5}$

17. $x = \frac{5}{3}$

18. $x = \frac{5}{2}$

EJERCICIO 145

1. pH = 4.7212

2. pH = 3.2218

3. 1×10^{-9}

4. 4.3010

5. 0.9 segundos

6. 3500 micrómetros

7. 59.46%

8. 6.4321 años

21. $x = \frac{e+1}{e-1}$

22. $x = 4e$

23. $x = 4$

24. $x = \frac{2\sqrt{e}-1}{\sqrt{e}-1}$

19. $x = -\frac{1}{2}$

20. $x = \sqrt{6}, x = -\sqrt{6}$

21. $x = 3, x = -1$

22. $x = 2$

23. $x = -\frac{3}{2}$

24. $x = -1, x = -2$

25. $x = -1$

26. $x = 2$

27. $x = \frac{\log 2}{2 \log 2 - \log 3}$

28. $x = 0, x = 2$

29. $x = \frac{2 \log 7 + \log 5}{2 \log 7 - \log 5}$

30. $x = 0$

31. $y = \ln 11 - \ln 13$

32. $x = 2, x = 1$

33. $x = \ln \sqrt[3]{2}$

34. $x = 0$

35. $x = \ln(1 - \sqrt{e})$

36. $x = \ln \sqrt{5}$

9. 138.62 años

10. 18321 habitantes

11. 3.5 horas

12. 29.15 años

13. $T = 64.762^\circ\text{C}$

14. $T = 94.84^\circ\text{C}$

15. $t = 133.9$ min

EJERCICIO 146

1. $1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}$

2. 9.9, 9.99, 9.999, 9.9999, 9.99999

3. $2, \frac{5}{4}, \frac{10}{9}, \frac{17}{16}, \frac{26}{25}$

4. $\frac{1}{4}, \frac{2}{5}, \frac{2}{3}, \frac{8}{7}, 2$

5. $1, \frac{3}{2}, \frac{5}{6}, \frac{7}{24}, \frac{3}{40}$

6. -1, 4, -9, 16, -25

7. 0, 0, 2, 6, 12

8. $-\frac{1}{2}, -\frac{2}{3}, -\frac{3}{4}, -\frac{4}{5}, -\frac{5}{6}$

9. 1, 2, 3, 4, 5

10. $1, -\frac{4}{3}, \frac{3}{2}, -\frac{8}{5}, \frac{5}{3}$

11. 2, 5, 11, 23, 47

12. $\frac{1}{2}, 1, \frac{1}{2}, 1, \frac{1}{2}$

13. $\frac{2}{3}, -\frac{1}{3}, -\frac{4}{3}, -\frac{7}{3}, -\frac{10}{3}$

14. 27, -9, 3, -1, $\frac{1}{3}$

15. -1, -1, -2, -6, -24

16. -2, 4, 16, 256, 65536

17. 4, 2, 1, $\frac{1}{\sqrt{3}}, \frac{1}{2\sqrt{3}}$

18. 3, 1, -1, 1, -1

EJERCICIO 147

1. 48

2. 165

3. $\frac{617}{140}$

4. 126

5. $\sqrt{7} - 1$

6. 18

7. $-\frac{11}{2}$

8. 21

9. n^2

10. $\frac{n(n+1)}{2}$

11. $c = 3$

12. $c = 1$

13. $c = 3$

14. $c = 2, -\frac{20}{13}$

EJERCICIO 148

1. Sí es

2. No es

3. Sí es

4. No es

5. Sí es

6. Sí es

7. $a_8 = 23$

8. $a_{11} = \frac{7}{2}$

9. $a_{15} = \frac{103}{12}$

10. $a_{10} = 55$

11. $a_{16} = \frac{27}{4}$

12. $a_7 = 48$

13. $a_{12} = -5$

14. $a_{18} = 454$

15. $a_{13} = -27$

16. $a_{17} = \frac{11}{4}$

17. $a_1 = 7$

18. $r = -2$

19. $n = 9$

20. $r = -\frac{1}{4}$

21. $a_{11} = -28$

22. $a_1 = -15$

23. $n = 10$

24. $a_1 = 7$

25. $a_1 = -\frac{17}{6}$

27. $r = \frac{1}{4}$

29. $a_7 = 8n - 5$

19. $a_1 = 2$

23. $n = 5$

27. $a_4 = \frac{1}{m^4}$

26. $n = 10$

28. $a_5 = -\frac{15}{4}$

30. $a_4 = \frac{20n-7}{6}$

20. $a_2 = 4$

24. $n = 8$

28. $a_{11} = 2^{\frac{23}{6}x-9}$

EJERCICIO 149

1. $S_8 = 176$

6. $S_{12} = 450$

11. $S = n^2$

2. $S_9 = 9$

7. $S_{11} = 0$

12. $n = 12$

3. $S_8 = 31$

8. $S = 40600$

13. $n = 10$

4. $S_9 = 648$

9. $S = \frac{n(n+1)}{2}$

14. $a_1 = -9$

5. $S_{13} = -78$

10. $S = n(n+1)$

15. $a_1 = 2, a_n = 100$

EJERCICIO 150

1. 365 lugares

2. 518 ladrillos

3. \$1375

4. 9 rollos

5. 18 filas

EJERCICIO 151

1. $27\frac{1}{2}, 34, 40\frac{1}{2}, 47, 53\frac{1}{2}$

2. $6\frac{1}{2}, 8, 9\frac{1}{2}, 11, 12\frac{1}{2}, 14, 15\frac{1}{2}$

3. $1, 1\frac{1}{3}, 1\frac{2}{3}, 2, 2\frac{1}{3}, 2\frac{2}{3}$

4. $1\frac{1}{2}, 2\frac{1}{2}, 3\frac{1}{2}, 4\frac{1}{2}, 5\frac{1}{2}, 6\frac{1}{2}, 7\frac{1}{2}$

5. $-2.5, -2, -1.5, -1, -0.5, 0$

6. $\frac{5}{6}, \frac{4}{3}, \frac{11}{6}$

7. Promedio = 8.24

EJERCICIO 152

1. 8 años

2. 9.8 de calificación

3. Promedio = 9

4. Promedio = $\frac{a_1 + a_n}{2}$

5. 7 hileras y constan de 80, 76, 72, 68, 64, 60, 56 tejas

6. 8 hileras de 58, 62, 66, 70, 74, 78, 82 y 86 tejas

EJERCICIO 153

1. Sí es

7. $a_6 = -81$

13. $a_{12} = \frac{32}{243}$

2. Sí es

8. $a_9 = \frac{128}{2187}$

14. $a_9 = m^{24}$

3. No es

9. $a_5 = -80$

15. $a_{10} = n^{14}$

4. No es

10. $a_7 = \frac{5}{128}$

16. $a_7 = \frac{n^3}{n+1}$

5. No es

11. $a_{10} = -\frac{1}{2187}$

17. $a_{13} = 2^{27x-16}$

6. Sí es

12. $a_8 = \frac{1}{16}$

18. $a_9 = a_1 r^{16}$

EJERCICIO 154

1. $9, \frac{3}{2}, \frac{1}{4}, -1$

9, 3, 1, -1

2. 4096 células

3. 3, 5, 7

4. 6 células

5. 30388 bacterias

6. $a_t = 25000(1.05)^{3t}$

7. \$339814.7

8. 67392 bebés

9. 4 cm^2 **EJERCICIO 155**

1. $S_6 = -\frac{364}{27}$

11. $S_n = \frac{a_1(r^{2n}-1)}{r^2-1}$

2. $S_7 = \frac{2059}{486}$

12. $S_n = \frac{2^n-1}{2^n}$

3. $S_9 = -855$

13. $n = 8$

4. $S_{10} = \frac{989527}{2187}$

14. $r = \frac{1}{2}$

5. $S_{15} = \frac{32767}{8}$

15. $a_1 = 27$

6. $S_{18} = 524286$

16. $a_n = \frac{1}{64}$

7. $S_{12} = 1092 + 364\sqrt{3}$

17. $a_1 = -2$

8. $S_{10} = 31 - 31\sqrt{2}$

18. $r = \frac{3}{2}$

9. $S_{12} = \frac{n^{21}-n}{n-1}$

19. $n = 7$

10. $S_9 = 511 \cdot 2^{x-2}$

EJERCICIO 156

1. 5461 triángulos

2. 127 personas

3. 65761.7 ton.

4. 34316.76 partos

5. 1.01 % por año, 110.4 millones

EJERCICIO 157

1. $S = -4$

6. $r = \frac{3}{5}$

2. $S = \frac{9}{4}$

7. $r = \frac{1}{3}$

3. $S = 9$

8. $S = \frac{4}{3} \text{ cm}^2$

4. $S = \frac{27}{4}$

9. $S = 2048 \text{ cm}^2$

5. $S = \frac{23}{8}$

EJERCICIO 158

- 1, 2, 4, 8, 16
- $4, \frac{4}{3}, \frac{4}{9}$
- 6, -12, -24, -48
- 6, 24, 96, 384, 1536
- $6, 6\sqrt{3}, 18$
- $\frac{2}{3}, \frac{8}{9}, \frac{32}{27}, \frac{128}{81}$
- 64, -32, -16, -8, -4, -2
- $\frac{(x-1)^3}{3}, \frac{(x-1)^4}{9}, \frac{(x-1)^5}{27}$
- $a, 2, \frac{4}{a}$

- $1, \sqrt{2}, 2, 2\sqrt{2}$
- $3\sqrt{6}$
- $-4\sqrt{2}$
- $5\sqrt{5}$
- 12
- $\sqrt[3]{36}$
- $8\sqrt[3]{2}$
- $3\sqrt{3}$
- $\frac{\sqrt{2}}{8}$

EJERCICIO 159

- \$25 937.4
- \$64 390.28
- \$49 783.2
- \$43 346.6
- \$13 324.4
- \$18 824.8
- \$1 292.2
- \$8 723.2
- \$8 682.5
- \$188 542
- \$17 483
- $\begin{cases} 2.5\% \text{ trimestral} \\ 10\% \text{ anual} \end{cases}$
- 14.86%
- 7%
- 11.1 años
- 9 955 habitantes
- 3 años
- \$655 446.5
- 3%
- \$12 244.5

EJERCICIO 160

- \$55 700.19
- \$3 652.26
- 25%
- 8%
- \$156 738.56
- 20%
- 10 años de vida útil

CAPÍTULO 16

EJERCICIO 161

- $a = 2, b = -1$
- $x = -2, y = 4, z = 0$
- $q = 2, r = 1, t = 1,$
- $x = 7, y = 1, z = -2$

EJERCICIO 162

- $A+B=\begin{bmatrix} -6 & 2 \\ 0 & 4 \end{bmatrix}, A-B=\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}, A-A=\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$
 $4A-3B=\begin{bmatrix} -3 & 1 \\ 0 & 2 \end{bmatrix}, 2A-0B=\begin{bmatrix} -6 & 2 \\ 0 & 4 \end{bmatrix}$
- $A+B=\begin{bmatrix} -4 & 7 & 4 \end{bmatrix}, A-B=\begin{bmatrix} 8 & -7 & -2 \end{bmatrix}, A-A=\begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$
 $4A-3B=\begin{bmatrix} 26 & -21 & -5 \end{bmatrix}, 2A-0B=\begin{bmatrix} 4 & 0 & 2 \end{bmatrix}$
- $A+B=\begin{bmatrix} -2 & -2 \\ 3 & -6 \\ 3 & 4 \end{bmatrix}, A-B=\begin{bmatrix} 6 & -12 \\ -1 & 6 \\ 1 & -10 \end{bmatrix}, A-A=\begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$
 $4A-3B=\begin{bmatrix} 20 & -43 \\ -2 & 18 \\ 5 & -33 \end{bmatrix}, 2A-0B=\begin{bmatrix} 4 & -14 \\ 2 & 0 \\ 4 & -6 \end{bmatrix}$

$$4. A+B=\begin{bmatrix} 3 & -9 & 3 \\ 1 & -4 & 8 \end{bmatrix}, A-B=\begin{bmatrix} 1 & 3 & -5 \\ 7 & -8 & -6 \end{bmatrix}, A-A=\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$4A-3B=\begin{bmatrix} 5 & 6 & -16 \\ 25 & -30 & -17 \end{bmatrix}, 2A-0B=\begin{bmatrix} 4 & -6 & -2 \\ 8 & -12 & 2 \end{bmatrix}$$

$$5. A+B=\begin{bmatrix} -\frac{3}{5} & \frac{16}{3} & \frac{1}{8} \\ \frac{1}{3} & -2 & 10 \\ \frac{23}{3} & 1 & -\frac{3}{2} \end{bmatrix}, A-B=\begin{bmatrix} \frac{7}{5} & \frac{14}{3} & \frac{1}{8} \\ -\frac{1}{3} & 8 & -6 \\ \frac{19}{3} & -\frac{3}{5} & \frac{3}{2} \end{bmatrix}$$

$$A-A=\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, 4A-3B=\begin{bmatrix} \frac{23}{5} & 19 & \frac{1}{2} \\ -1 & 27 & -16 \\ 26 & -\frac{8}{5} & \frac{9}{2} \end{bmatrix}$$

$$2A-0B=\begin{bmatrix} \frac{4}{5} & 10 & \frac{1}{4} \\ 0 & 6 & 4 \\ 14 & \frac{2}{5} & 0 \end{bmatrix}$$

- $\begin{cases} a=7, b=2, c=2, d=5, v=-3, w=4 \end{cases}$
- $\begin{cases} n=-3, w=-10, x=3, y=6 \end{cases}$
- $\begin{cases} v=4, w=-2, x=3, y=7, z=2 \end{cases}$

EJERCICIO 163

- $AB=\begin{bmatrix} -2 \end{bmatrix}, BA=\begin{bmatrix} 5 & 7 \\ -5 & -7 \end{bmatrix}$
- $AB=\begin{bmatrix} 5 & -5 \end{bmatrix}$
- $BA=\begin{bmatrix} 5 & -4 \\ -5 & 0 \\ -6 & 1 \end{bmatrix}$
- $AB=\begin{bmatrix} -7 & -7 & -4 \\ -5 & -5 & -8 \end{bmatrix}$
- $AB=\begin{bmatrix} -8 & -8 \\ -2 & -4 \end{bmatrix}, BA=\begin{bmatrix} -4 & -2 \\ -8 & -8 \end{bmatrix}$
- $AB=\begin{bmatrix} 12 & -7 \\ -5 & 2 \end{bmatrix}, BA=\begin{bmatrix} 1 & 3 \\ 8 & 13 \end{bmatrix}$
- $AB=\begin{bmatrix} -1 & 25 \\ -1 & 7 \end{bmatrix}, BA=\begin{bmatrix} 4 & 2 & 0 \\ 1 & -1 & -3 \\ 7 & 5 & 3 \end{bmatrix}, A(BC)=\begin{bmatrix} 74 & 98 \\ 20 & 26 \end{bmatrix}$
- $AB=\begin{bmatrix} 11 & 3 \\ 4 & 2 \\ 2 & 0 \end{bmatrix}, A(B-2C)=\begin{bmatrix} 1 & 5 \\ 4 & 0 \\ -2 & 2 \end{bmatrix}, A(BC)=\begin{bmatrix} 17 & -3 \\ 8 & -2 \\ 2 & 0 \end{bmatrix}$

EJERCICIO 164

- $\det A = 22$
- $\det B = 8$
- $\det C = -50$
- $\det D = 43$
- $\det E = 122$

EJERCICIO 165

$$1. A^{-1} = \begin{bmatrix} \frac{1}{7} & \frac{2}{7} \\ \frac{1}{7} & -\frac{3}{14} \end{bmatrix} = \frac{1}{14} \begin{bmatrix} 2 & 4 \\ 2 & -3 \end{bmatrix}$$

$$2. B^{-1} = \begin{bmatrix} -1 & 0 \\ \frac{5}{2} & \frac{1}{2} \end{bmatrix} = \frac{1}{2} \begin{bmatrix} -2 & 0 \\ 5 & 1 \end{bmatrix}$$

$$3. C^{-1} = \begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix}$$

$$4. D^{-1} = \begin{bmatrix} \frac{6}{7} & \frac{2}{7} \\ -\frac{12}{7} & \frac{3}{7} \end{bmatrix} = \frac{1}{7} \begin{bmatrix} 6 & 2 \\ -12 & 3 \end{bmatrix}$$

$$5. E^{-1} = \begin{bmatrix} \frac{5}{6} & \frac{1}{6} & -\frac{1}{2} \\ -\frac{1}{6} & \frac{1}{6} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \end{bmatrix} = \frac{1}{6} \begin{bmatrix} 5 & 1 & -3 \\ -1 & 1 & 3 \\ 3 & 3 & -3 \end{bmatrix}$$

$$6. F^{-1} = \begin{bmatrix} -\frac{1}{8} & \frac{3}{4} & -\frac{1}{8} \\ \frac{1}{4} & -\frac{1}{2} & \frac{1}{4} \\ \frac{5}{24} & -\frac{7}{12} & -\frac{1}{8} \end{bmatrix} = \frac{1}{24} \begin{bmatrix} -3 & 18 & -3 \\ 6 & -12 & 6 \\ 5 & -14 & -3 \end{bmatrix}$$

$$7. G^{-1} = \begin{bmatrix} -\frac{5}{17} & \frac{2}{17} & \frac{3}{17} \\ \frac{1}{17} & -\frac{11}{34} & -\frac{4}{17} \\ \frac{1}{17} & \frac{3}{17} & -\frac{4}{17} \end{bmatrix} = \frac{1}{34} \begin{bmatrix} -10 & 4 & 6 \\ 2 & -11 & -8 \\ 2 & 6 & -8 \end{bmatrix}$$

$$8. H^{-1} = \begin{bmatrix} \frac{1}{5} & -\frac{1}{10} & -\frac{3}{10} \\ -\frac{1}{5} & \frac{3}{5} & \frac{9}{5} \\ -\frac{1}{5} & \frac{3}{5} & \frac{4}{5} \end{bmatrix} = \frac{1}{10} \begin{bmatrix} 2 & -1 & -3 \\ -2 & 6 & 18 \\ -2 & 6 & 8 \end{bmatrix}$$

$$9. J^{-1} = \begin{bmatrix} \frac{43}{6} & \frac{49}{6} & -\frac{19}{6} & \frac{1}{6} \\ -\frac{1}{6} & -\frac{1}{6} & \frac{1}{6} & -\frac{1}{6} \\ -\frac{67}{6} & -\frac{79}{6} & \frac{31}{6} & -\frac{1}{6} \\ -\frac{16}{3} & -\frac{19}{3} & \frac{7}{3} & -\frac{1}{3} \end{bmatrix} = \frac{1}{6} \begin{bmatrix} 43 & 49 & -19 & 1 \\ -1 & -1 & 1 & -1 \\ -67 & -79 & 31 & -1 \\ -32 & -38 & 14 & -2 \end{bmatrix}$$

EJERCICIO 166

$$1. \begin{cases} x=5 \\ y=-2 \end{cases} \quad 3. \begin{cases} a=11 \\ b=-10 \end{cases} \quad 5. \begin{cases} x=5 \\ y=2 \\ z=-1 \end{cases}$$

$$2. \begin{cases} m=-4 \\ n=2 \end{cases} \quad 4. \begin{cases} a=4 \\ b=-3 \\ c=2 \end{cases} \quad 6. \begin{cases} x=1 \\ y=-1 \\ z=-2 \end{cases}$$

CAPÍTULO 17

EJERCICIO 167

1. $(x-1)y(x-5)$
2. $(x+1)y(2x+3)$
3. $(x+2)(3x-2)y(x-2)$
4. $(x+1)y(x+3i)$
5. $(x+2i)y(x-2i)$
6. $(x)(x+1-i)y(x+2+3i)$
7. Residuo -72

$$8. \text{Residuo } -\frac{9}{2}$$

$$9. \text{Residuo } \frac{273}{2}$$

$$10. \text{Residuo } 12$$

$$11. \text{Residuo } 264$$

$$12. \text{Residuo } -240$$

$$13. k=2$$

$$14. k=3, k=-6$$

$$15. k=6, k=\frac{46}{3}$$

$$16. k=4, k=-\frac{167}{73}$$

$$17. k=4, k=-\frac{1}{3}$$

$$18. \text{Todos son raíces}$$

$$19. \text{Todos son raíces}$$

$$20. \text{Ninguno es raíz}$$

$$21. x=-1, x=\frac{11}{2}$$

$$22. x=2+i, x=-\frac{3}{5}$$

$$23. x=-\frac{1}{2}, x=\frac{5}{3}$$

$$24. x=2i, x=-2i$$

$$25. x=4, x=\frac{3}{5}$$

$$26. f(x) = x^3 + 4x^2 - 5x$$

$$27. f(x) = x^3 + 4x^2 - 9x - 36$$

$$28. f(x) = 3x^3 - x^2 + 48x - 16$$

$$29. f(x) = 8x^3 + 2x^2 - 43x - 30$$

$$30. f(x) = x^4 - 5x^3 - 13x^2 + 133x - 260$$

$$31. f(x) = 6x^4 - 5x^3 + 7x^2 - 5x + 1$$

$$32. f(x) = 3x^3 + 5x^2 + 4x - 2$$

$$33. f(x) = x^3 - x^2 - x + 1$$

$$34. f(x) = x^4 - x^3 - x^2 - x - 2$$

$$35. f(x) = x^4 - 6x^2 + 8x - 3$$

$$36. f(x) = x^4 - 4x^3 + 16x - 16$$

$$37. f(x) = x^5 + 2x^4 - x - 2$$

$$38. x=-1, x=1, x=5$$

$$39. x=5, x=4, x=3$$

$$40. x=1\frac{1}{5}, x=-\frac{2}{3}, x=4$$

$$41. x=-\frac{5}{2}, x=-2+i, x=-2-i$$

$$42. x=2, x=-3, x=7, x=0$$

$$43. x=-4i, x=4i, x=-2, x=3$$

$$44. x=-\frac{2}{3}, x=\frac{5}{2}, x=-1+i, x=-1-i$$

$$45. x=-i, x=i, x=-4, x=-3, x=\frac{1}{2}$$

Solución a los ejercicios de geometría y trigonometría



CAPÍTULO 2

EJERCICIO 1

- | | | |
|-------------|----------------|-----------------|
| 1. 40.1708° | 5. 9.1525° | 9. 18° 15' 18" |
| 2. 61.7058° | 6. 98.3791° | 10. 29° 24' 39" |
| 3. 1.03416° | 7. 40° 19' 12" | 11. 19° 59' 24" |
| 4. 73.6777° | 8. 61° 14' 24" | 12. 44° 00' 36" |

EJERCICIO 2

- | | |
|--|---|
| 1. $\frac{7}{6}\pi \text{ rad} = 3.665 \text{ rads}$ | 8. $\frac{11}{6}\pi \text{ rad} = 5.759 \text{ rads}$ |
| 2. $\frac{5}{3}\pi \text{ rad} = 5.236 \text{ rads}$ | 9. $\frac{2}{3}\pi \text{ rad} = 2.094 \text{ rads}$ |
| 3. $\frac{5}{4}\pi \text{ rad} = 3.927 \text{ rads}$ | 10. $\frac{3}{4}\pi \text{ rad} = 2.356 \text{ rads}$ |
| 4. $\frac{5}{2}\pi \text{ rad} = 7.854 \text{ rads}$ | 11. $\frac{4523}{18000}\pi \text{ rad} = 0.789 \text{ rad}$ |
| 5. $\frac{2}{5}\pi \text{ rad} = 1.256 \text{ rads}$ | 12. $\frac{1283}{1800}\pi \text{ rad} = 2.239 \text{ rads}$ |
| 6. $\frac{5}{9}\pi \text{ rad} = 1.745 \text{ rads}$ | 13. $\frac{2711}{3240}\pi \text{ rad} = 2.628 \text{ rads}$ |
| 7. $\frac{1}{6}\pi \text{ rad} = 0.523 \text{ rad}$ | 14. $\frac{33601}{14400}\pi \text{ rad} = 7.330 \text{ rads}$ |

EJERCICIO 3

- | | | | |
|---------|----------|-----------------|-----------------|
| 1. 120° | 5. 1260° | 9. 90° | 13. 360° |
| 2. 330° | 6. 20° | 10. 270° | 14. 28° 38' 52" |
| 3. 135° | 7. 468° | 11. 9° 38' 34" | |
| 4. 240° | 8. 15° | 12. 64° 10' 37" | |

EJERCICIO 4

- | | | |
|----------------|-----------------|-----------------|
| 1. 55° 46' 50" | 6. 75° 44' 22" | 11. 4° 33' 11" |
| 2. 40° 13' 15" | 7. 246° 34' 15" | 12. 15° 41' 18" |
| 3. 49° 19' 33" | 8. 875° 11' 40" | 13. 3° 21' 41" |
| 4. 59° 19' 45" | 9. 383° 51' 21" | 14. 13° 15' 18" |
| 5. 108° 7' 48" | 10. 227° 3' 18" | |

EJERCICIO 5

- | | | | |
|--------------------|--------------------|---------|----------|
| 1. Suplementarios | 6. Complementarios | | |
| 2. Complementarios | 7. Suplementarios | | |
| 3. Conjugados | 8. Complementarios | | |
| 4. Conjugados | 9. Conjugados | | |
| 5. Conjugados | 10. Suplementarios | | |
| 11. 10° | 13. 80° | 15. 18° | 17. 36° |
| 12. 57° | 14. 30° | 16. 20° | 18. 120° |

19. a) $\angle COB = 30^\circ$, $\angle BOA = 60^\circ$
 b) $\angle AOB = 45^\circ$, $\angle BOC = 30^\circ$, $\angle COD = 15^\circ$
 c) $\angle AOB = 50^\circ$, $\angle DOB = 130^\circ$
 d) $\angle AOB = 65^\circ$, $\angle BOC = 45^\circ$, $\angle COD = 70^\circ$
 e) $\angle AOB = 30^\circ$, $\angle BOC = 90^\circ$, $\angle COD = 60^\circ$
 f) $\angle AOB = \angle COD = 45^\circ$, $\angle BOC = 55^\circ$, $\angle DOE = 35^\circ$
 g) $\angle AOB$ (convexo) = 134° , $\angle AOB$ (cóncavo) = 226°
 h) $\angle AOB$ (convexo) = 50° , $\angle AOB$ (cóncavo) = 310°

EJERCICIO 6

- | | | |
|--|-------------|------------------------|
| 1. 135° | 5. 22° 30' | 9. $48\pi \text{ rad}$ |
| 2. 115° | 6. 115° | 10. 3:40 h |
| 3. $\theta = 25^\circ$, $\alpha = 30^\circ$ | 7. 292° 30' | |
| 4. O63° 18'S, S26° 42'O | 8. 12:30 h | |

CAPÍTULO 3

EJERCICIO 7

- $x = 60^\circ$, $\angle a = 60^\circ$, $\angle b = 120^\circ$
- $x = 46.5^\circ$, $\angle a = \angle b = \angle e = 46.5^\circ$, $\angle c = \angle d = \angle f = 133.5^\circ$
- $x = 40^\circ$, $\angle a = \angle b = \angle e = 80^\circ$, $\angle c = \angle d = \angle f = 100^\circ$
- $\angle a = \angle c = 137^\circ$, $\angle b = 43^\circ$
- $\angle a = \angle c = \angle d = \angle g = 47^\circ$, $\angle b = \angle e = \angle f = 133^\circ$
- $x = 25^\circ$
- $x = 26^\circ$, $\angle a = 128^\circ$, $\angle b = 52^\circ$
- $\angle 10 = \angle 4 = \angle 7 = 70^\circ$, $\angle 1 = \angle 13 = \angle 16 = 110^\circ$
- $x = 115^\circ$, $y = 65^\circ$
- $x = 40^\circ$, $y = 110^\circ$
- $x = 80^\circ$, $y = 60^\circ$
- $R = 120^\circ$
- $\angle a = \angle c = \angle e = \angle f = 126^\circ$, $\angle b = \angle d = 54^\circ$
- $\angle n = \angle z = 50^\circ$, $\angle m = \angle s = \angle y = \angle r = 130^\circ$
- $\angle x = \angle q = \angle p = \angle k = 35^\circ$, $\angle y = \angle r = \angle s = 145^\circ$
- $\angle q = \angle z = \angle y = 60^\circ$, $\angle r = \angle w = \angle p = 120^\circ$
- a), b), d) y f)

CAPÍTULO 4

EJERCICIO 8

- | | |
|------------------|--|
| 1. 105°, 110° | 5. 118°, 38° y 24°; 68°, 70° y 42° |
| 2. 10°, 80° | 6. $\theta = 54^\circ$ y $\beta = 72^\circ$ |
| 3. 80°, 80°, 20° | 7. $\angle A = 35^\circ$, $\angle B = 95^\circ$, $\angle C = 50^\circ$ |
| 4. 55°, 41° | 8. $ABC = 69^\circ$, $BCA = 73^\circ$, $BAC = 38^\circ$,
$ACD = 107^\circ$, $CDA = 35^\circ$, $CAD = 38^\circ$ |

EJERCICIO 9

1. Teorema II (LAL) $x = 85^\circ$ $y = 12$
2. Teorema III (ALA) $x = 13$ $y = 19.8$
3. Teorema I (LLL) $x = 32^\circ$ $y = 62^\circ$

EJERCICIO 10

1 a 8. No se incluye la solución por ser demostraciones.

EJERCICIO 11

1. $a = 36^\circ$, $b = 8^\circ$ 4. $x = 25$, $y = 14$
2. $x = 15$, $y = 45$ 5. $a = 12^\circ$, $b = 25^\circ$
3. $x = 15^\circ$, $y = 20^\circ$

EJERCICIO 12

1. $x = 3$ 4. $x = 7$, $x = 0$ 7. $x = \pm 6y$ 10. $x = 3$
2. $x = 7.2$ 5. $x = \pm 4\sqrt{2}$ 8. $x = \pm 5$
3. $x = \pm 9$ 6. $x = 2$ 9. $x = \pm 4$

EJERCICIO 13

1. $a' = 3$, $c' = 5$
2. $a = 30$, $b' = 16$
3. Lados 12 y 22; $x = 11$, $y = 36$
4. Lados 8 y 4; $x = 7$, $y = 5$
5. Lados 8 y 6; $u = 3$, $t = 10$
6. Lados 10 y 9; $x = 5$, $y = 3$

EJERCICIO 14

1. $x = 10$ 4. $x = \frac{12}{5}$ 7. $x = 4$ 10. $x = 30$
2. $x = \frac{9}{2}$ 5. $x = \frac{25}{3}$ 8. $x = \frac{27}{22}$
3. $x = 6$ 6. $x = 16$ 9. $x = 10$

EJERCICIO 15

1. 68 m 3. 160 m 5. a) 28 m
2. 481.6 m 4. 15 m b) 120 m

EJERCICIO 16

1. $c = 25$ 8. $b = 5\sqrt{2}$ 15. Acutángulo
2. $c = \sqrt{41}$ 9. $c = 3\sqrt{5}$ m 16. Rectángulo
3. $c = 4\sqrt{5}$ 10. $b = 5$ m 17. Rectángulo
4. $c = 7\sqrt{2}$ 11. $c = \sqrt{421}$ cm 18. Obtusángulo
5. $b = 16$ 12. $a = 5\sqrt{7}$ dm 19. Rectángulo
6. $a = 2\sqrt{7}$ 13. Obtusángulo 20. Acutángulo
7. $c = 8$ 14. Rectángulo 21. Rectángulo
22. a) $2\sqrt{15}$, b) $5\sqrt{13}$, c) $2\sqrt{10}$, d) $6\sqrt{21}$ e) $\frac{40}{3}$,
f) $\frac{91\sqrt{218}}{218}$, g) $\frac{169}{60}\sqrt{30}$

EJERCICIO 17

1. $100\sqrt{73}$ m 6. $\sqrt{91}$ m
2. $2\sqrt{5}$ m 7. 5 cm
3. 40 cm 8. $8\sqrt{3}$ cm
4. $5\sqrt{3}$ cm 9. $9\sqrt{2}$ km
5. $4\sqrt{2}$ m 10. $5\sqrt{2}$ cm
11. $\frac{2\sqrt{2}}{3}m$, $\frac{m}{3}$
12. $2\sqrt{\frac{4m^2 - n^2}{15}}$, $2\sqrt{\frac{4n^2 - m^2}{15}}$ y $2\sqrt{\frac{m^2 + n^2}{5}}$

CAPÍTULO 5**EJERCICIO 18**

1. $\angle A = \angle C = 140^\circ$, $\angle B = 40^\circ$
2. $\angle DCA = 40^\circ$, $\angle CAD = 60^\circ$, $\angle DAB = \angle DCB = 100^\circ$, $\angle D = \angle B = 80^\circ$
3. $\angle ADC = \angle B = 110^\circ$, $\angle A = \angle C = 70^\circ$
4. $x = 30^\circ$, $z = 120^\circ$, $y = 60^\circ$
5. $x = 127^\circ$, $y = 53^\circ$
6. $x = 120^\circ$, $y = 55^\circ$, $z = 125^\circ$
7. $x = 60^\circ$, $y = 120^\circ$, $z = 60^\circ$
8. $x = 15^\circ$, $y = 70^\circ$, $z = 110^\circ$

EJERCICIO 19

1 a 6. No se incluye la solución por ser demostraciones.

EJERCICIO 20

1. $x = 4$ cm 4. $\angle NPO = 24^\circ$ 7. $\overline{MN} = 20$ u
2. 4 y 8 u 5. $x = 20^\circ$, $y = 68^\circ$ 8. $\overline{AB} = a$, $\overline{IJ} = b$
3. 41 u 6. $\overline{AB} = 11$ cm 9. $\overline{AE} = 5$

CAPÍTULO 6**EJERCICIO 21**

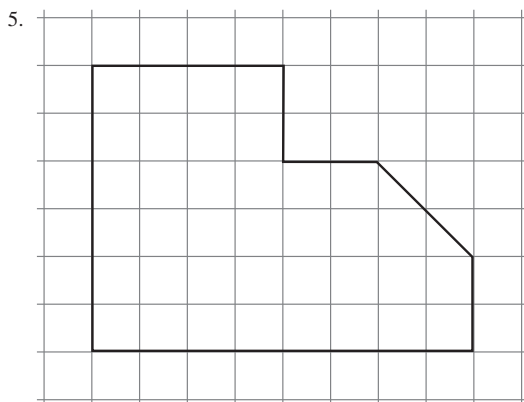
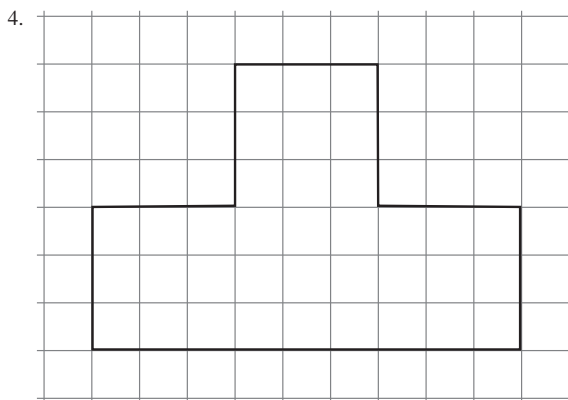
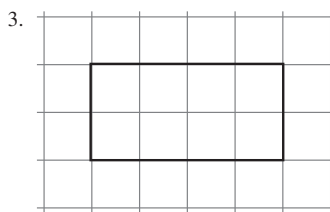
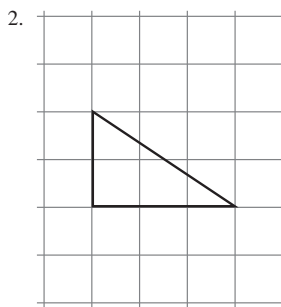
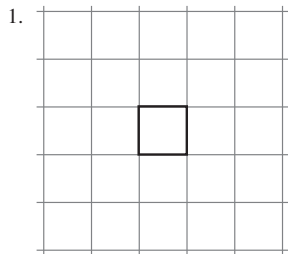
1. $d = 8$
2. Icoságono
3. $d = 7$
4. Dodecágono
5. Nonágono
6. a) 170, b) 54, c) 27, d) 9, e) 90, f) 14, g) 104, h) 135, i) 44
7. Heptágono
8. Hexadecágono
9. Heptadecágono
10. Nonadecágono
11. Heptágono
12. Undecágono
13. Pentágono
14. Tridecágono
15. Dodecágono
16. Octágono
17. Icoságono

EJERCICIO 22

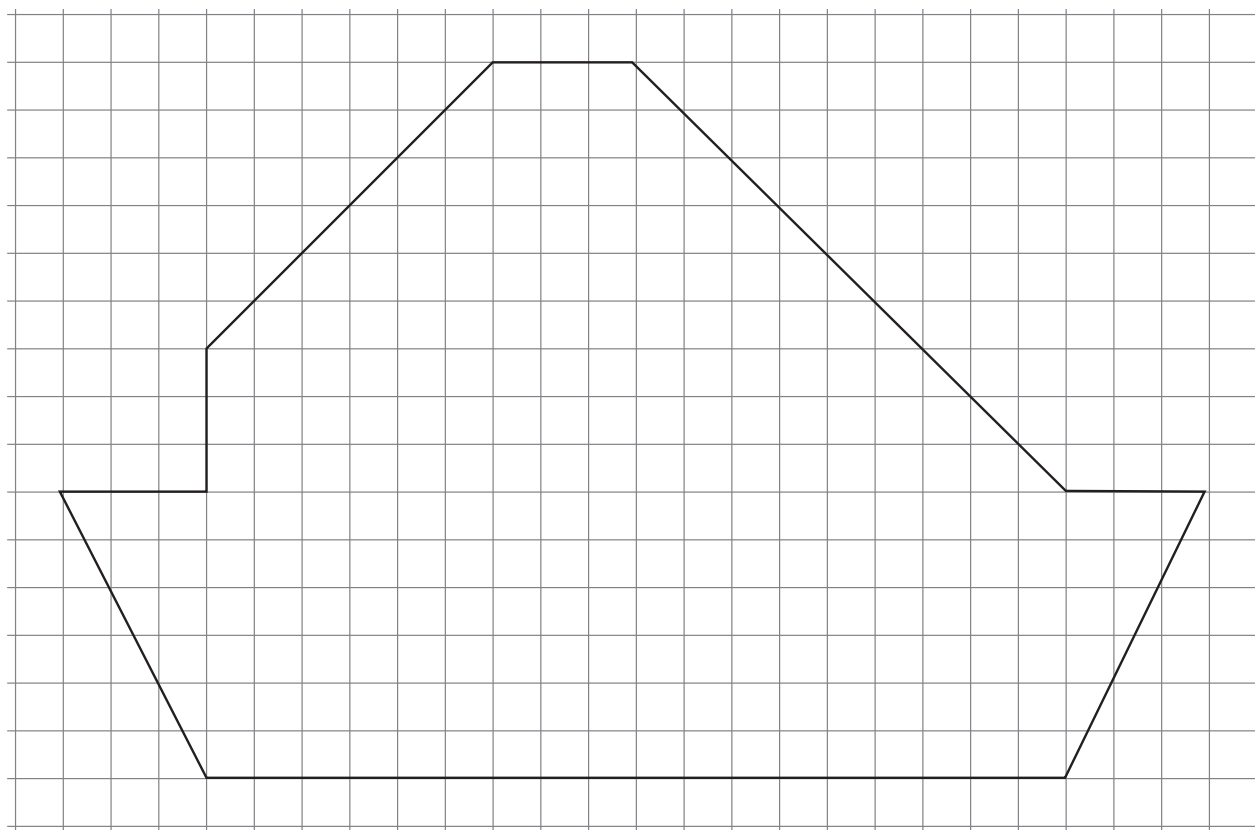
1. a) 120° , b) 135° , c) 150° , d) 162° , e) 160° , f) $171^\circ 25' 42''$
2. a) 540° , b) $1\ 440^\circ$, c) $2\ 340^\circ$, d) $1\ 080^\circ$, e) $1\ 980^\circ$, f) $6\ 300^\circ$
3. Nonágono (nueve lados)
4. Heptágono (siete lados)
5. Hexadecágono (16 lados)
6. Undecágono (11 lados)
7. Hexágono (seis lados)
8. Hexadecágono (16 lados)
9. Nonágono (nueve lados)
10. Dodecágono (12 lados)
11. Octágono (ocho lados)
12. Triángulo
13. Hexágono (seis lados)
14. Pentadecágono (15 lados)
15. Nonágono (nueve lados)
16. Pentágono (cinco lados)
17. 54° , 129.6° , 129.6° , 108° y 118.8°
18. 110° , 100° , 115° , 135° y 80°
19. 30° , 60° , 90° , 120° , 150° , 210° y 240°
20. $\angle A = 70^\circ$, $\angle B = 65^\circ$, $\angle C = 10^\circ$, $\angle D = 110^\circ$ y $\angle E = 105^\circ$
21. $\angle A = 54^\circ$, $\angle B = 64^\circ$, $\angle C = 116^\circ$, $\angle D = 64^\circ$,
 $\angle E = 17^\circ$ y $\angle F = 45^\circ$

CAPÍTULO 7

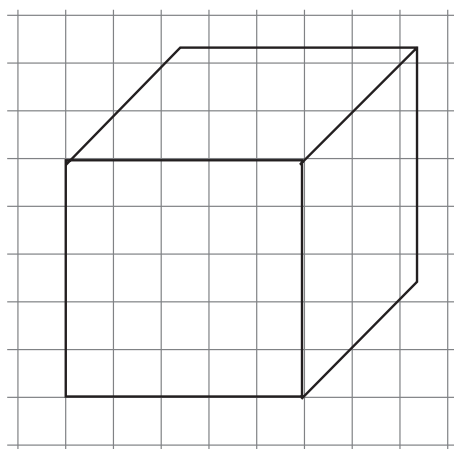
EJERCICIO 23



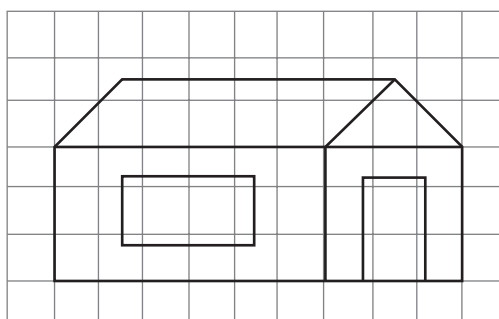
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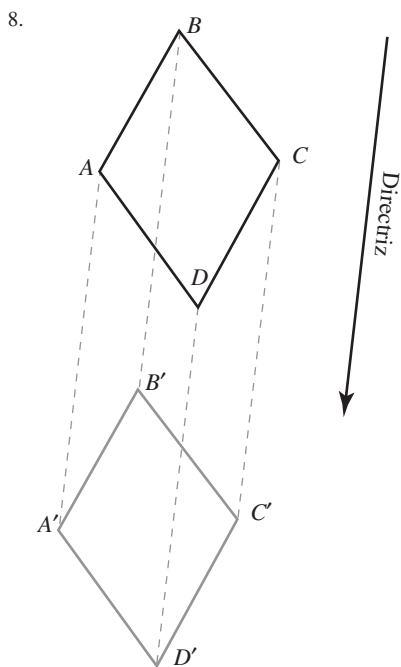
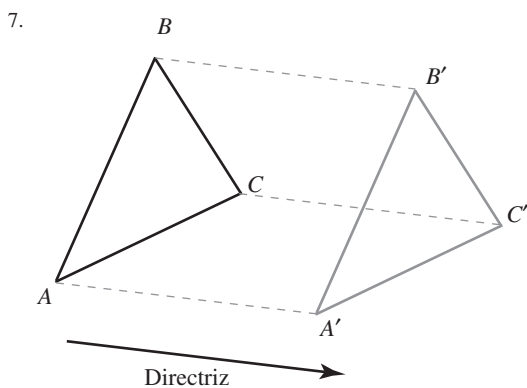
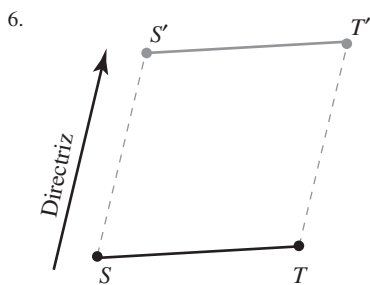
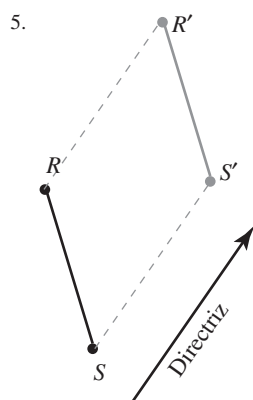
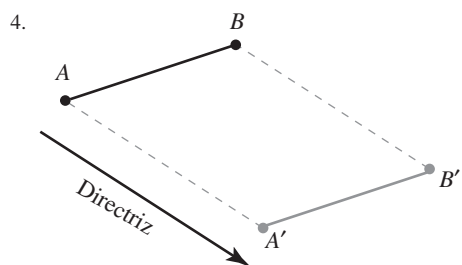
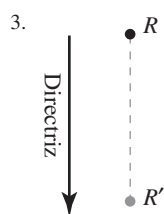
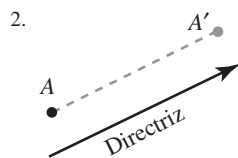
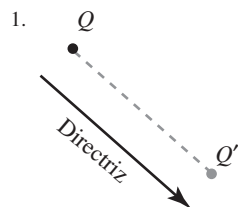
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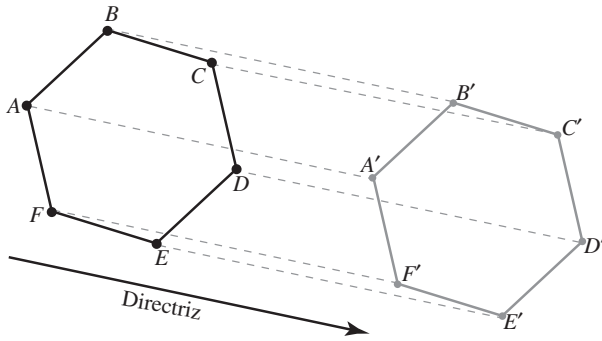
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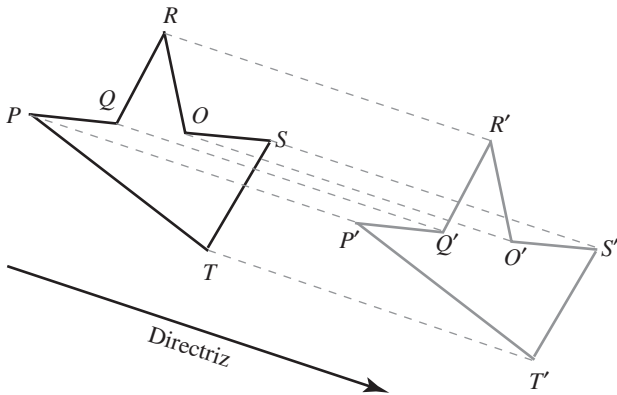
EJERCICIO 24



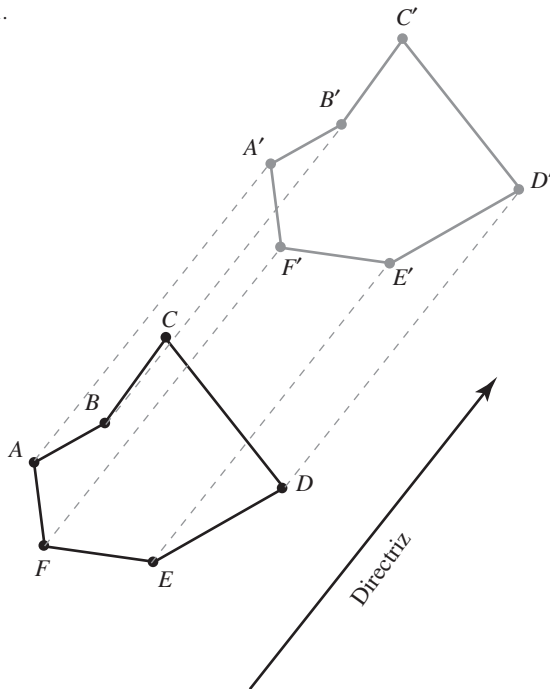
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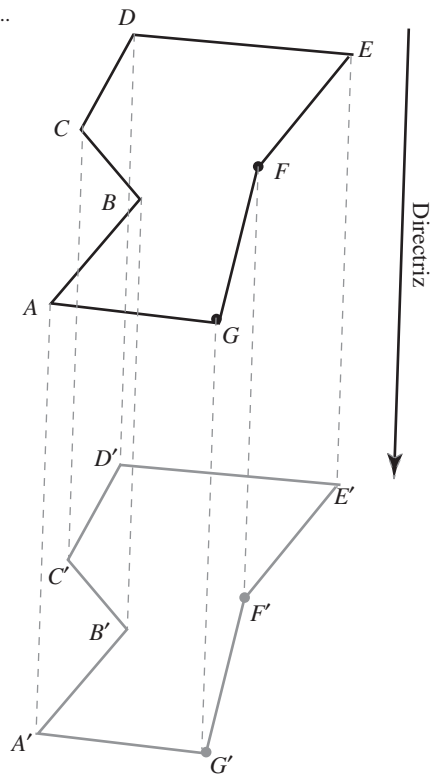
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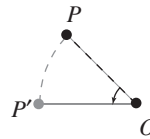


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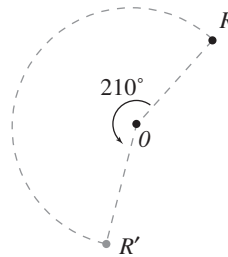


EJERCICIO 25

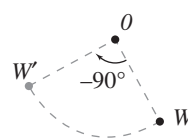
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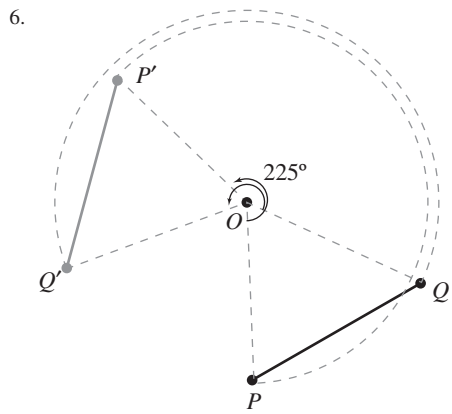
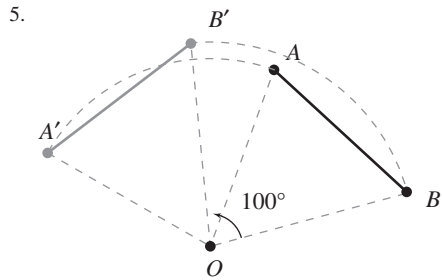
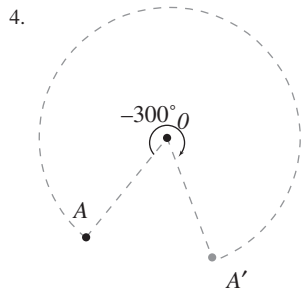


2.

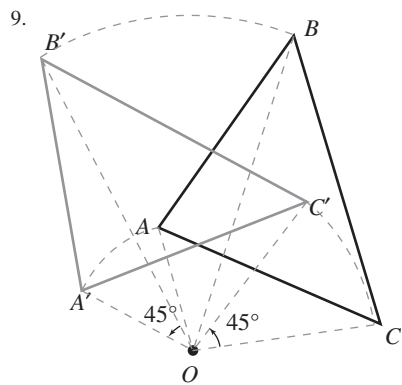
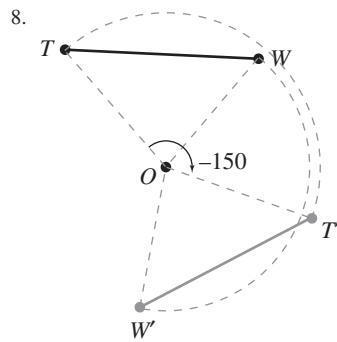
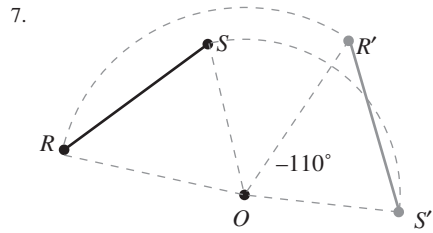
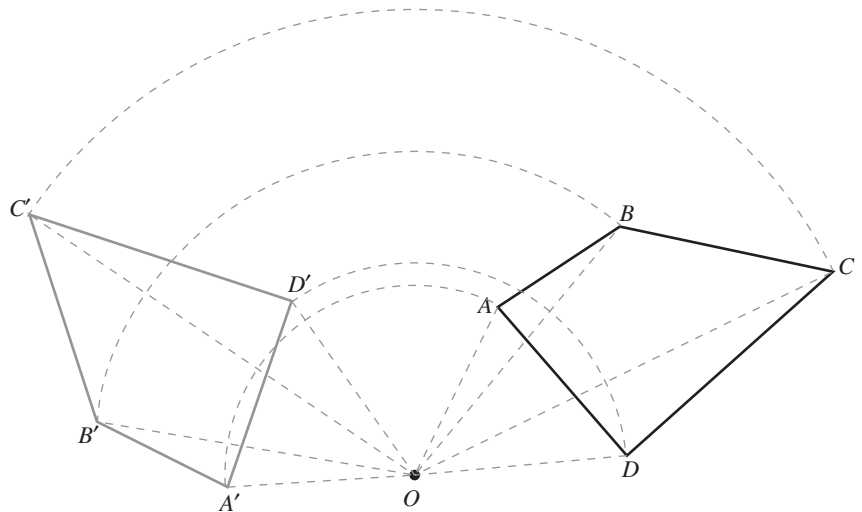


3.

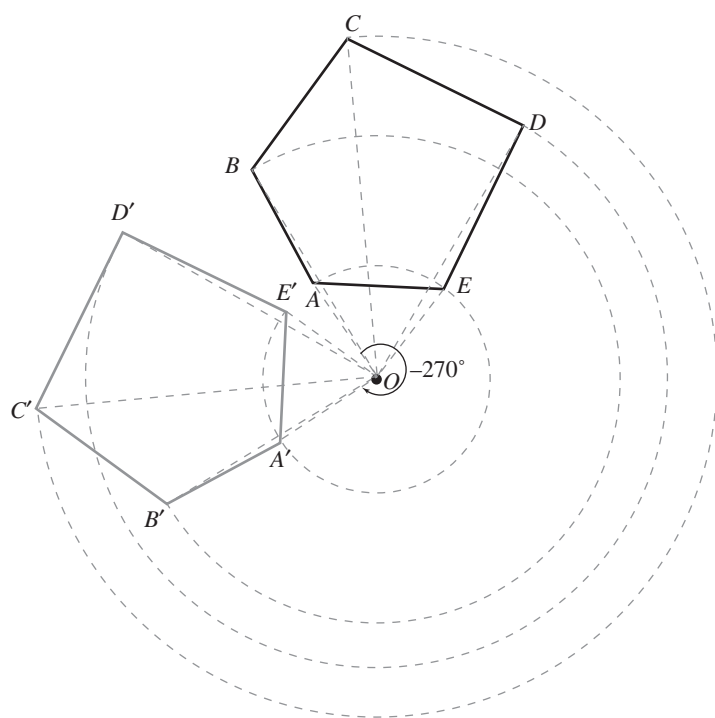




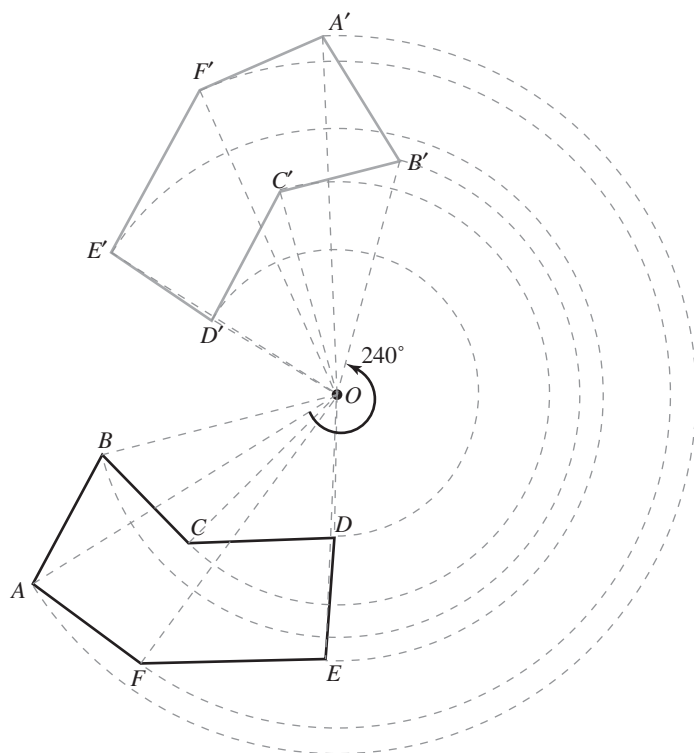
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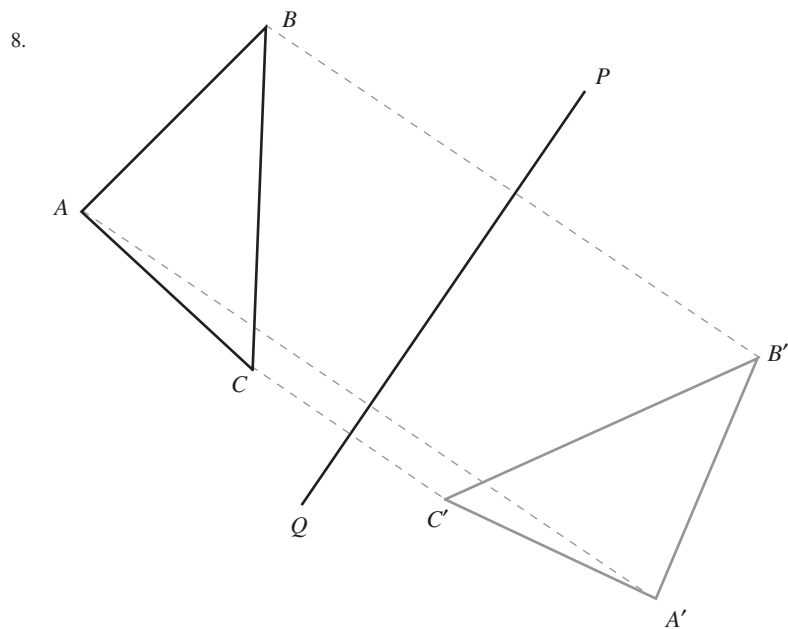
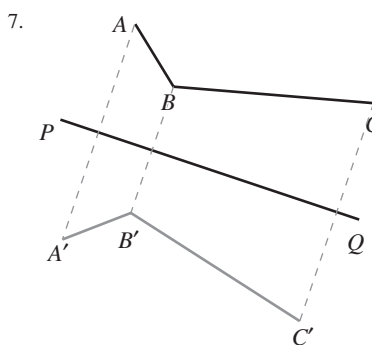
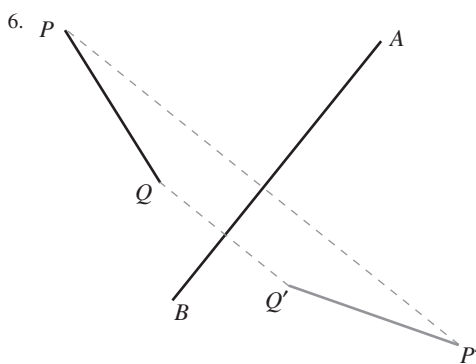
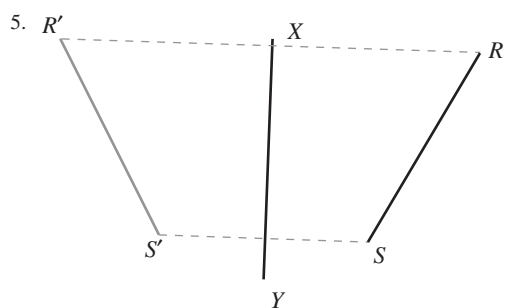
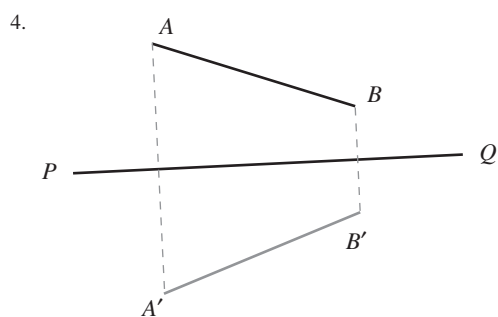
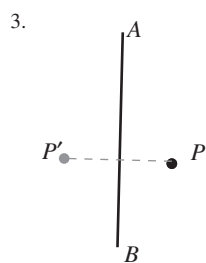
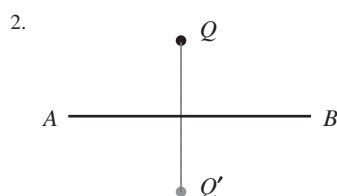
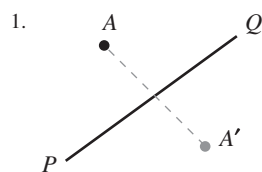


11.

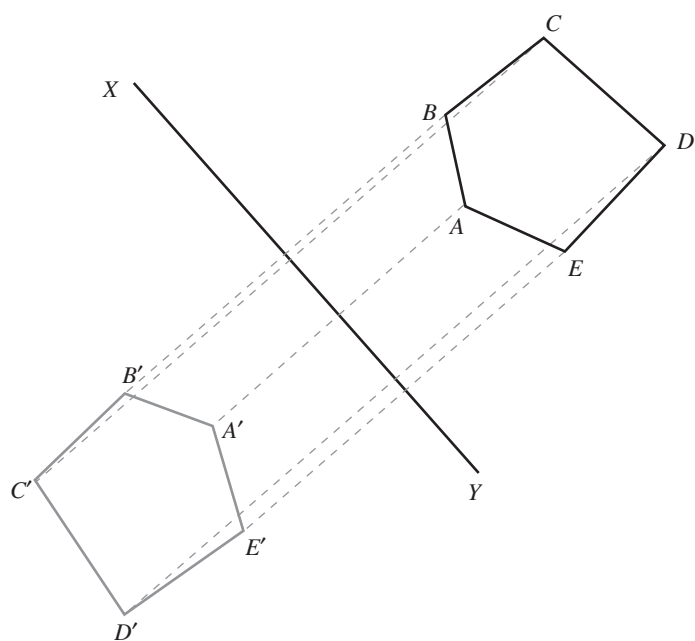


12.

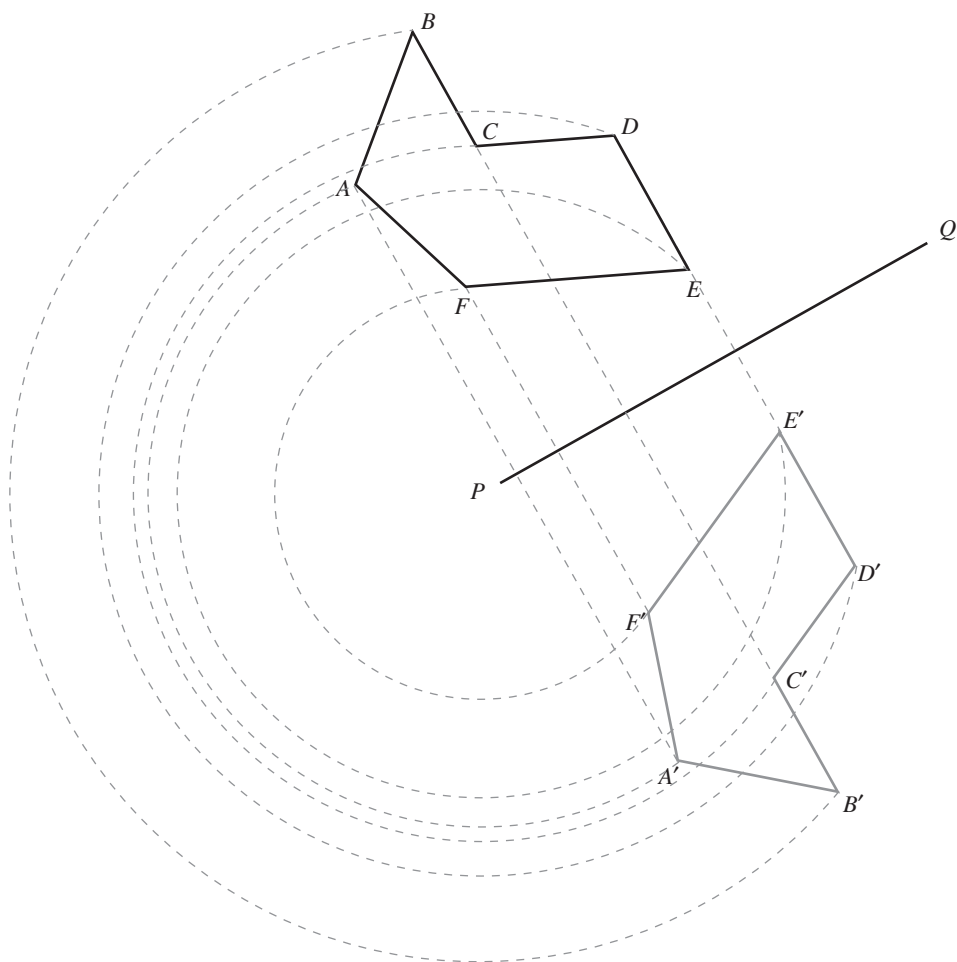


EJERCICIO 26


9.

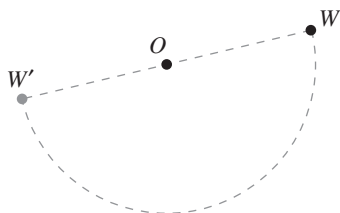


10

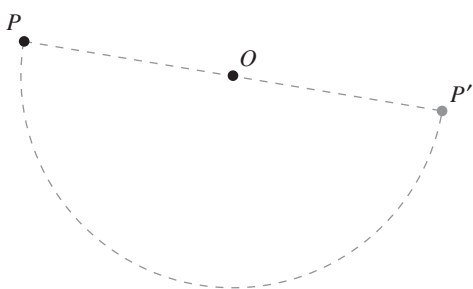


EJERCICIO 27

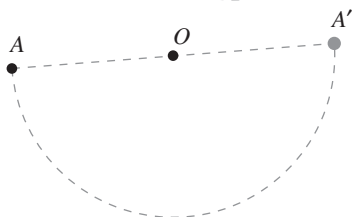
1.



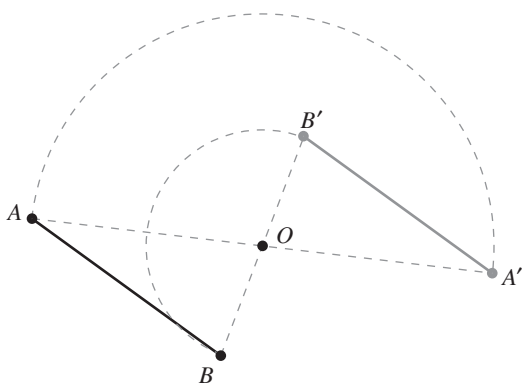
2.



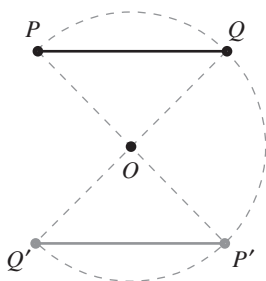
3.



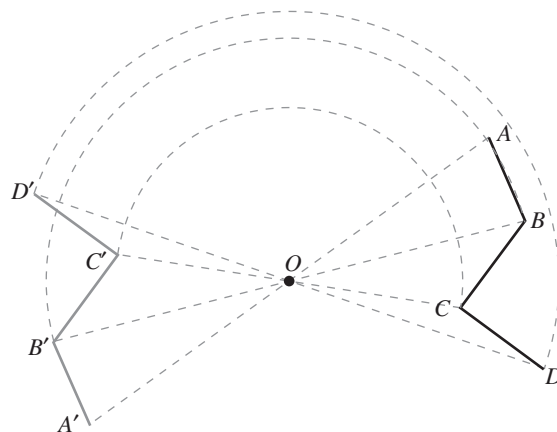
4.



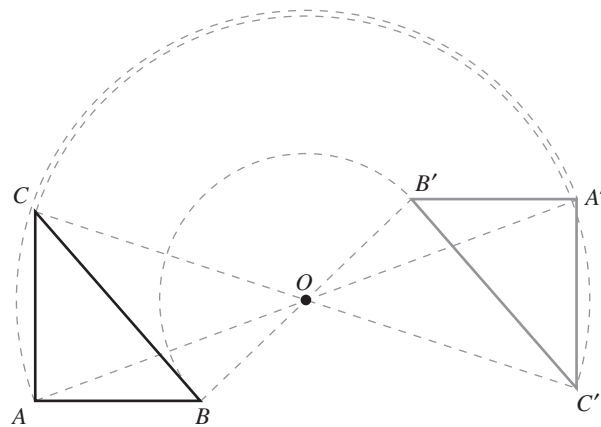
5.



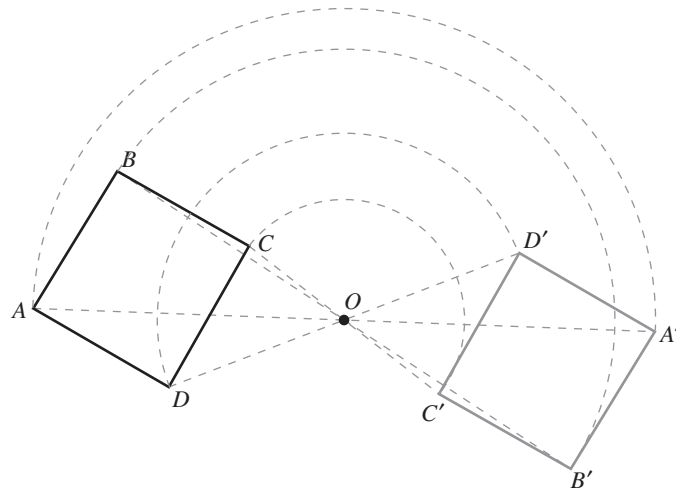
6.



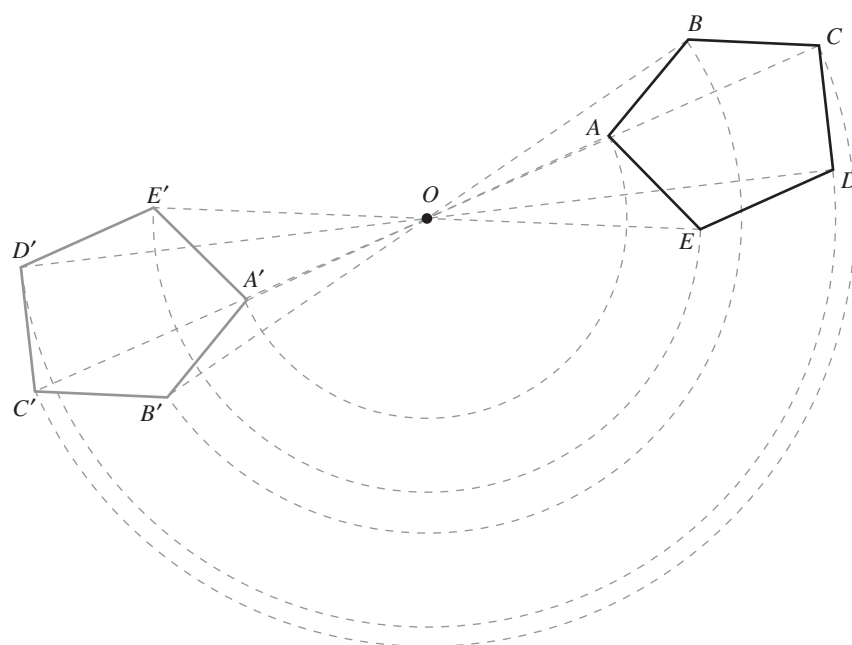
7.



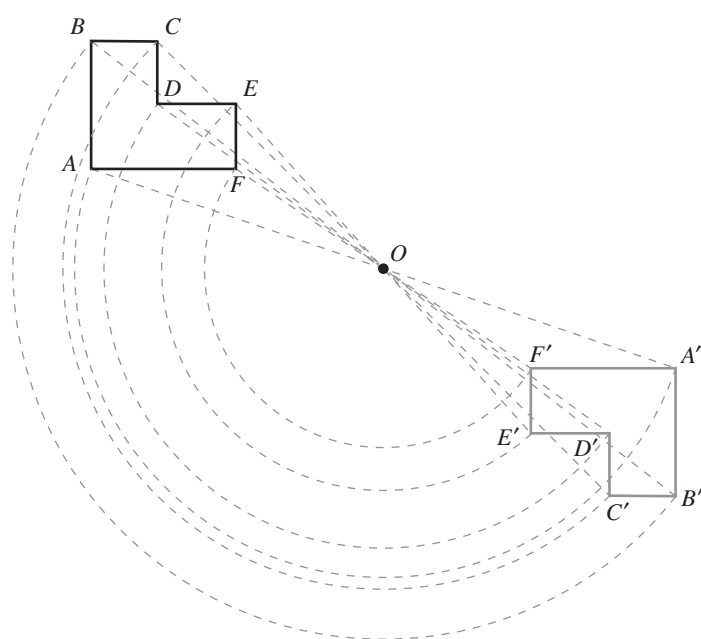
8.



9.



10.



CAPÍTULO 8

EJERCICIO 28

1. $\angle ABC = 30^\circ$, $\angle AOC = 60^\circ$, $\angle BOC = 104^\circ$, $\widehat{AD} = 116^\circ$
2. $\angle a = 75^\circ$, $\angle b = 50^\circ$, $\angle c = 55^\circ$, $\angle d = 55^\circ$, $\angle e = 50^\circ$, $\angle f = 75^\circ$
3. $\angle ABC = 27.5^\circ = 27^\circ 30'$
4. $\angle ABC = 85^\circ$, $\angle DBA = 95^\circ$
5. $\angle A = 105^\circ$, $\angle B = 95^\circ$, $\angle C = 75^\circ$, $\angle D = 85^\circ$
6. a) $\angle A = 30^\circ$, b) $\angle A = 40^\circ$
7. $\angle a = 60^\circ$, $\angle b = 15^\circ$, $\angle c = 25^\circ$, $\angle d = 30^\circ$, $\angle e = 50^\circ$
8. a) $\angle A = 15^\circ$, b) $\angle A = 40^\circ$, c) $\angle A = 30^\circ$, d) $\widehat{a} = 35^\circ$
e) $\widehat{c} = 120^\circ$, f) $\widehat{c} - \widehat{a} = 140^\circ$, g) $\widehat{a} = 70^\circ$, h) $\widehat{a} = 40^\circ$
9. $\angle u = 120^\circ$, $\angle x = 60^\circ$, $\angle y = 30^\circ$, $\angle w = 60^\circ$, $\angle z = 90^\circ$
10. $\angle a = 90^\circ$, $\angle b = 90^\circ$, $\angle c = 90^\circ$, $\angle d = 90^\circ$, $\angle e = 25^\circ$, $\angle f = 25^\circ$,
 $\angle g = 65^\circ$, $\angle h = 65^\circ$, $\angle i = 40^\circ$

EJERCICIO 29

1. a) 10.8, b) 7.8, c) 9.4
2. a) 10.09, b) 16.2, c) 17.29

EJERCICIO 30

1. Exteriores
2. Tangentes exteriores
3. Interiores
4. Secantes
5. Tangentes interiores
6. Tangentes exteriores
7. $3r$
8. $2u$
9. $2\sqrt{3}u$
10. $5cm$
11. $\overline{C_1C_3} = \frac{7}{18}R$, $\overline{C_1C_2} = \frac{1}{6}R$
12. r
13. $\frac{\sqrt{5}}{2}R$

CAPÍTULO 9

EJERCICIO 31

1. $P = 8.4 m$, $A = 4.25 m^2$
2. $P = 24.9 m$, $A = 29.4 m^2$
3. $P = 38.6 m$, $A = 82.5 m^2$
4. $P = 52.5 m$, $A = 118.12 m^2$
5. $P = 40.0 m$, $A = 110 m^2$
6. $P = 65.4 m$, $A = 37.375 m^2$
7. $P = 36 cm$, $A = 81 cm^2$
8. $P = 10 m$, $A = 6 m^2$
9. $A = 150 m^2$
10. $A = (x^2 - 3x + 2)m^2$
11. $A = 63 dm^2$
12. $A = 17.5 dm^2$
13. $A = 900\pi cm^2$
14. $A = 81\pi cm^2$
15. $A = 400 cm^2$
16. $\$ 2.6/m^2$
17. $\$ 725.5$
18. Altura = $36 m$, base = $27 m$
19. Altura = $10 m$
20. 80 círculos, $1280\pi cm^2$
21. a) $12\sqrt{14} u^2$, b) $2\sqrt{255} u^2$,
c) $\frac{15}{4}\sqrt{15} u^2$
22. $x = 9$, $A = 98 m^2$
23. a) $2\pi cm^2$, b) $\frac{1}{6}\pi cm^2$,
c) $\frac{9}{4}\pi cm^2$, d) $\frac{32}{3}\pi cm^2$
24. a) $(\pi - 2)cm^2$
b) $\frac{3}{2}(\pi - \frac{3}{2}\sqrt{3})cm^2$
c) $16(\pi - 2)cm^2$

EJERCICIO 32

1. a) $\overline{TS} = 24 cm$, b) $\overline{BC} = 13 cm$, c) $P = 44 cm$, $A = 14\sqrt{11} cm^2$
2. $A = 84 cm^2$
3. $A = 2r^2(4 - \pi)$
4. $A = 3\pi r^2$
6. $A = 25(2\sqrt{3} - \pi)dm^2$
7. $A_s = 25(4 - \pi)cm^2$
8. $A_s = 100\pi dm^2$
9. $A_s = 64(4 - \pi)mm^2$
10. $A_s = 4(10 + \pi)dm^2$
11. $A_s = 196(4 - \pi)cm^2$
12. $A_s = 1152(\pi - 2)mm^2$
 $P = 96\pi mm$
13. $A_s = 32(6 - \pi)mm^2$
14. $A_s = 128(\pi - 2)mm^2$
15. $A_s = 256(4 - \pi)cm^2$
16. a) $A = 3\sqrt{3} dm^2$
b) $A = 256\sqrt{3} dm^2$
17. $A_s = 36\pi cm^2$
18. $A_s = \frac{1}{8}\pi cm^2$
19. $A_s = 2 cm^2$,
 $P = 2(1 + \pi)cm$
20. $A_s = \frac{5}{2}\pi cm^2$
 $P = (6 + 4\pi)cm$

CAPÍTULO 10

EJERCICIO 33

1. $A_T = 4\sqrt{3} cm^2$, $V_T = \frac{2}{3}\sqrt{2} cm^3$
2. $A_T = 3\sqrt{3} cm^2$, $V_T = \frac{\sqrt{6}}{4} cm^3$
3. $A_T = 72 cm^2$, $V_T = 24\sqrt{3} cm^3$
4. $A_T = 150 cm^2$, $V_T = 125 cm^3$
5. $A_T = 72\sqrt{3} cm^2$, $V_T = 72\sqrt{2} cm^3$

6. $A_T = 6\sqrt{3} \text{ cm}^2$, $V_T = \sqrt{6} \text{ cm}^3$
7. $A_T = \left(60\sqrt{25+10\sqrt{5}}\right) \text{ cm}^2$, $V_T = (350 + 150\sqrt{5}) \text{ cm}^3$
8. $A_T = \left(12\sqrt{25+10\sqrt{5}}\right) \text{ cm}^2$, $V_T = (30 + 14\sqrt{5}) \text{ cm}^3$
9. $A_T = 15\sqrt{3} \text{ cm}^2$, $V_T = \left(\frac{15\sqrt{3} + 5\sqrt{15}}{4}\right) \text{ cm}^3$
10. $A_T = 250\sqrt{3} \text{ dm}^2$, $V_T = \left(\frac{1875\sqrt{2} + 625\sqrt{10}}{6}\right) \text{ dm}^3$
11. $A_T = 9\sqrt{3} \text{ cm}^2$
12. $V_T = \frac{27\sqrt{6}}{4} \text{ cm}^3$
13. $h = \frac{4\sqrt{3}}{3} \text{ cm}$
14. $V_T = 2\sqrt{2} \text{ cm}^3$
15. $L = \sqrt[3]{2} \text{ m}$, $A_T = 6\sqrt[3]{4} \text{ m}^2$
16. $h = 6\sqrt{2} \text{ cm}$, $A_T = 72\sqrt{3} \text{ cm}^2$
17. $V_T = \frac{4}{3} \text{ cm}^3$
18. $V_T = 36 \text{ cm}^3$
19. $V_T = \frac{15\sqrt{2} + 5\sqrt{10}}{6} \text{ cm}^3$
20. $V_T = \frac{(3 + \sqrt{5}) \cdot \sqrt[4]{75} \sqrt{A_i}}{180}$

EJERCICIO 34

1. $A_L = 50 \text{ cm}^2$, $A_T = 62 \text{ cm}^2$, $V_T = 30 \text{ cm}^3$
2. $A_L = 72 \text{ cm}^2$, $A_T = (72 + 8\sqrt{3}) \text{ cm}^2$, $V_T = 24\sqrt{3} \text{ cm}^3$
3. $A_L = 16 \text{ cm}^2$, $A_T = 18 \text{ cm}^2$, $V_T = 4 \text{ cm}^3$
4. $A_L = 97.5 \text{ cm}^2$, $A_T = \frac{195 + 75\sqrt{3}}{2} \text{ cm}^2$, $V_T = \frac{975}{8}\sqrt{3} \text{ cm}^3$
5. $A_L = (16\sqrt{2} + 8) \text{ cm}^2$, $A_T = (24\sqrt{2} + 8) \text{ cm}^2$, $V_T = 16 \text{ cm}^3$
6. $A_L = 16 \text{ cm}^2$, $A_T = 24 \text{ cm}^2$, $V_T = 8 \text{ cm}^3$
7. $A_L = 64\sqrt{3} \text{ cm}^2$, $A_T = (64\sqrt{3} + 24) \text{ cm}^2$, $V_T = 96 \text{ cm}^3$
8. $A_L = 400 \text{ cm}^2$, $A_T = 400(2 + \sqrt{2}) \text{ cm}^2$, $V_T = 1000(1 + \sqrt{2}) \text{ cm}^3$
9. $A_L = 1200 \text{ cm}^2$, $A_T = 300(4 + \sqrt{5}) \text{ cm}^2$, $V_T = 3000\sqrt{3} \text{ cm}^3$
10. $A_L = 16\sqrt{3} \text{ cm}^2$
11. $V_T = 27u^3$
12. $A_L = 48 \text{ cm}^2$
13. $V_T = \frac{21}{4} \text{ cm}^3$

14. $A_T = \frac{180 + 25\sqrt{3}}{2} \text{ cm}^2$
15. $V_T = \frac{81}{2} \text{ cm}^3$
16. $A_L = 16(1 + \sqrt{2}) \text{ cm}^2$
17. $V_T = \frac{\sqrt{6A_i^3}}{36}$
18. $V_T = \frac{\sqrt{A_L^3}}{8}$
19. $V_T = \frac{27\sqrt{2}}{4} \text{ cm}^3$
20. $A_L = 3\sqrt[3]{3V_i^2}$

EJERCICIO 35

1. $A_L = 3\sqrt{55} \text{ cm}^2$, $A_T = (9 + 3\sqrt{55}) \text{ cm}^2$, $V_T = 12 \text{ cm}^3$
2. $A_L = \frac{3\sqrt{3}}{4} \text{ cm}^2$, $A_T = \sqrt{3} \text{ cm}^2$, $V_T = \frac{\sqrt{2}}{12} \text{ cm}^3$
3. $A_L = 12\sqrt{7} \text{ cm}^2$, $A_T = (12\sqrt{7} + 7\sqrt{3}) \text{ cm}^2$,
 $V_T = \frac{35\sqrt{3}}{3} \text{ cm}^3$
4. $A_L = 38.4 \text{ cm}^2$, $A_T = 64 \text{ cm}^2$, $V_T = 81.92 \text{ cm}^3$
5. $A_L = 30\pi \text{ cm}^2$, $A_T = 48\pi \text{ cm}^2$, $V_T = 45\pi \text{ cm}^3$
6. $A_L = 32\pi \text{ cm}^2$, $A_T = 64\pi \text{ cm}^2$, $V_T = 64\pi \text{ cm}^3$
7. $A_L = 7\sqrt{150} \pi \text{ cm}^2$, $A_T = (7\sqrt{150} + 49) \pi \text{ cm}^2$,
 $V_T = 147\pi \text{ cm}^3$
8. $A_L = 4\sqrt{17} \pi \text{ cm}^2$, $A_T = (4 + 4\sqrt{17}) \pi \text{ cm}^2$,
 $V_T = \frac{32}{3} \pi \text{ cm}^3$
9. $A_L = \frac{15\sqrt{3}}{4} \pi \text{ cm}^2$, $A_T = \frac{5}{4}(5 + 3\sqrt{3}) \pi \text{ cm}^2$,
 $V_T = \frac{25\sqrt{3}}{6} \pi \text{ cm}^3$
10. $A_L = 3\pi \text{ cm}^2$, $A_T = 4\pi \text{ cm}^2$, $V_T = \frac{2\sqrt{2}}{3} \pi \text{ cm}^3$
11. $V_T = 12 \text{ cm}^3$
12. $V_T = 8 \text{ cm}^3$
13. $V_T = 12\sqrt{46} \text{ cm}^3$
14. $V_T = \frac{560}{3} \text{ cm}^3$
15. $A_B = 24\sqrt{3} \text{ cm}^2$
16. $V_T = 24\pi \text{ cm}^3$

17. $A_L = 70\pi \text{ cm}^2$
 18. $V_T = 12\pi \text{ cm}^3$
 19. $A_T = 48\pi \text{ cm}^2$
 20. $A_T = \frac{1}{2} \sqrt[3]{18(1+\sqrt{5})^3} \pi V_T^2$

EJERCICIO 36

1. $A = 64\pi \text{ cm}^2$, $V = \frac{256}{3} \pi \text{ cm}^3$
 2. $V = 180\sqrt{5} \pi \text{ cm}^3$
 3. $V = 6\pi \text{ cm}^3$
 4. $V = 270\pi \text{ cm}^3$
 5. $A = 60\pi \text{ cm}^2$
 6. $A = 96\pi \text{ cm}^2$
 7. $V = \frac{28}{3} \pi \text{ cm}^3$
 8. $V = \frac{52}{3} \pi \text{ cm}^3$
 9. $V = 339\pi \text{ cm}^3$,
 $A = 72\pi \text{ cm}^2$
 10. $V = \frac{211}{216} \pi \text{ cm}^3$
 11. $A = \frac{200}{3} \pi \text{ cm}^2$
 12. $n = 120^\circ$
 13. $V = 72\sqrt{3} \pi \text{ cm}^3$
 14. $r = \frac{9}{2} \text{ cm}$, $A = 9\pi \text{ cm}^2$
 $V = \frac{243}{2} \pi \text{ cm}^3$

CAPÍTULO 11
EJERCICIO 37

Inciso 1)

a)

$$\begin{aligned} \operatorname{sen} A &= \frac{2\sqrt{14}}{9} & \cos A &= \frac{5}{9} & \tan A &= \frac{2\sqrt{14}}{5} \\ \operatorname{ctg} A &= \frac{5\sqrt{14}}{28} & \sec A &= \frac{9}{5} & \csc A &= \frac{9\sqrt{14}}{28} \\ \operatorname{sen} B &= \frac{5}{9} & \cos B &= \frac{2\sqrt{14}}{9} & \tan B &= \frac{5\sqrt{14}}{28} \\ \operatorname{ctg} B &= \frac{2\sqrt{14}}{5} & \sec B &= \frac{9\sqrt{14}}{28} & \csc B &= \frac{9}{5} \end{aligned}$$

b)

$$\begin{aligned} \operatorname{sen} M &= \frac{10\sqrt{149}}{149} & \cos M &= \frac{7\sqrt{149}}{149} & \tan M &= \frac{10}{7} \\ \operatorname{ctg} M &= \frac{7}{10} & \sec M &= \frac{\sqrt{149}}{7} & \csc M &= \frac{\sqrt{149}}{10} \\ \operatorname{sen} N &= \frac{7\sqrt{149}}{149} & \cos N &= \frac{10\sqrt{149}}{149} & \tan N &= \frac{7}{10} \\ \operatorname{ctg} N &= \frac{10}{7} & \sec N &= \frac{\sqrt{149}}{10} & \csc N &= \frac{\sqrt{149}}{7} \end{aligned}$$

c)

$$\begin{aligned} \operatorname{sen} A &= \frac{2}{3} & \cos A &= \frac{\sqrt{5}}{3} & \tan A &= \frac{2\sqrt{5}}{5} \\ \operatorname{ctg} A &= \frac{\sqrt{5}}{2} & \sec A &= \frac{3\sqrt{5}}{5} & \csc A &= \frac{3}{2} \\ \operatorname{sen} B &= \frac{\sqrt{5}}{3} & \cos B &= \frac{2}{3} & \tan B &= \frac{\sqrt{5}}{2} \\ \operatorname{ctg} B &= \frac{2\sqrt{5}}{5} & \sec B &= \frac{3}{2} & \csc B &= \frac{3\sqrt{5}}{5} \end{aligned}$$

d)

$$\begin{aligned} \operatorname{sen} M &= \frac{\sqrt{2}}{2} & \cos M &= \frac{\sqrt{2}}{2} & \tan M &= 1 \\ \operatorname{ctg} M &= 1 & \sec M &= \sqrt{2} & \csc M &= \sqrt{2} \\ \operatorname{sen} N &= \frac{\sqrt{2}}{2} & \cos N &= \frac{\sqrt{2}}{2} & \tan N &= 1 \\ \operatorname{ctg} N &= 1 & \sec N &= \sqrt{2} & \csc N &= \sqrt{2} \end{aligned}$$

Inciso 2)

a)

$$\begin{aligned} \operatorname{sen} \theta &= \frac{2\sqrt{6}}{5} & \cos \theta &= \frac{1}{5} & \tan \theta &= 2\sqrt{6} \\ \operatorname{ctg} \theta &= \frac{\sqrt{6}}{12} & \sec \theta &= 5 & \csc \theta &= \frac{5\sqrt{6}}{12} \\ \operatorname{sen} \alpha &= \frac{1}{5} & \cos \alpha &= \frac{2\sqrt{6}}{5} & \tan \alpha &= \frac{\sqrt{6}}{12} \\ \operatorname{ctg} \alpha &= 2\sqrt{6} & \sec \alpha &= \frac{5\sqrt{6}}{12} & \csc \alpha &= 5 \end{aligned}$$

b)

$$\begin{aligned} \operatorname{sen} A &= \frac{3\sqrt{13}}{13} & \cos A &= \frac{2\sqrt{13}}{13} & \tan A &= \frac{3}{2} \\ \operatorname{ctg} A &= \frac{2}{3} & \sec A &= \frac{\sqrt{13}}{2} & \csc A &= \frac{\sqrt{13}}{3} \\ \operatorname{sen} B &= \frac{2\sqrt{13}}{13} & \cos B &= \frac{3\sqrt{13}}{13} & \tan B &= \frac{2}{3} \\ \operatorname{ctg} B &= \frac{3}{2} & \sec B &= \frac{\sqrt{13}}{3} & \csc B &= \frac{\sqrt{13}}{2} \end{aligned}$$

c)

$$\begin{aligned}
 \operatorname{sen} N &= \frac{1}{2} & \cos N &= \frac{\sqrt{3}}{2} & \tan N &= \frac{\sqrt{3}}{3} \\
 \operatorname{ctg} N &= \sqrt{3} & \sec N &= \frac{2\sqrt{3}}{3} & \csc N &= 2 \\
 \operatorname{sen} M &= \frac{\sqrt{3}}{2} & \cos M &= \frac{1}{2} & \tan M &= \sqrt{3} \\
 \operatorname{ctg} M &= \frac{\sqrt{3}}{3} & \sec M &= 2 & \csc M &= \frac{2\sqrt{3}}{3}
 \end{aligned}$$

d)

$$\begin{aligned}
 \operatorname{sen} \theta &= \frac{\sqrt{33}}{6} & \cos \theta &= \frac{\sqrt{3}}{6} & \tan \theta &= \sqrt{11} \\
 \operatorname{ctg} \theta &= \frac{\sqrt{11}}{11} & \sec \theta &= 2\sqrt{3} & \csc \theta &= \frac{2\sqrt{33}}{11} \\
 \operatorname{sen} \alpha &= \frac{\sqrt{3}}{6} & \cos \alpha &= \frac{\sqrt{33}}{6} & \tan \alpha &= \frac{\sqrt{11}}{11} \\
 \operatorname{ctg} \alpha &= \sqrt{11} & \sec \alpha &= \frac{2\sqrt{33}}{11} & \csc \alpha &= 2\sqrt{3}
 \end{aligned}$$

e)

$$\begin{aligned}
 \operatorname{sen} \beta &= \frac{\sqrt{6}}{4} & \cos \beta &= \frac{\sqrt{10}}{4} & \tan \beta &= \frac{\sqrt{15}}{5} \\
 \operatorname{ctg} \beta &= \frac{\sqrt{15}}{3} & \sec \beta &= \frac{2\sqrt{10}}{5} & \csc \beta &= \frac{2\sqrt{6}}{3} \\
 \operatorname{sen} \alpha &= \frac{\sqrt{10}}{4} & \cos \alpha &= \frac{\sqrt{6}}{4} & \tan \alpha &= \frac{\sqrt{15}}{3} \\
 \operatorname{ctg} \alpha &= \frac{\sqrt{15}}{5} & \sec \alpha &= \frac{2\sqrt{6}}{3} & \csc \alpha &= \frac{2\sqrt{10}}{5}
 \end{aligned}$$

f)

$$\begin{aligned}
 \operatorname{sen} A &= \frac{4\sqrt{29}}{29} & \cos A &= \frac{\sqrt{377}}{29} & \tan A &= \frac{4\sqrt{13}}{13} \\
 \operatorname{ctg} A &= \frac{\sqrt{13}}{4} & \sec A &= \frac{\sqrt{377}}{13} & \csc A &= \frac{\sqrt{29}}{4} \\
 \operatorname{sen} B &= \frac{\sqrt{377}}{29} & \cos B &= \frac{4\sqrt{29}}{29} & \tan B &= \frac{\sqrt{13}}{4} \\
 \operatorname{ctg} B &= \frac{4\sqrt{13}}{13} & \sec B &= \frac{\sqrt{29}}{4} & \csc B &= \frac{\sqrt{377}}{13}
 \end{aligned}$$

EJERCICIO 38

1.

$$\begin{aligned}
 \operatorname{sen} \alpha &= -\frac{5}{13} & \cos \alpha &= \frac{12}{13} & \tan \alpha &= -\frac{5}{12} \\
 \operatorname{ctg} \alpha &= -\frac{12}{5} & \sec \alpha &= \frac{13}{12} & \csc \alpha &= -\frac{13}{5}
 \end{aligned}$$

2.

$$\begin{aligned}
 \operatorname{sen} \alpha &= -\frac{4\sqrt{65}}{65} & \cos \alpha &= -\frac{7\sqrt{65}}{65} & \tan \alpha &= \frac{4}{7} \\
 \operatorname{ctg} \alpha &= \frac{7}{4} & \sec \alpha &= -\frac{\sqrt{65}}{7} & \csc \alpha &= -\frac{\sqrt{65}}{4}
 \end{aligned}$$

3.

$$\begin{aligned}
 \operatorname{sen} \beta &= \frac{3\sqrt{13}}{13} & \cos \beta &= \frac{2\sqrt{13}}{13} & \tan \beta &= \frac{3}{2} \\
 \operatorname{ctg} \beta &= \frac{2}{3} & \sec \beta &= \frac{\sqrt{13}}{2} & \csc \beta &= \frac{\sqrt{13}}{3}
 \end{aligned}$$

4.

$$\begin{aligned}
 \operatorname{sen} \omega &= -\frac{\sqrt{2}}{2} & \cos \omega &= \frac{\sqrt{2}}{2} & \tan \omega &= -1 \\
 \operatorname{ctg} \omega &= -1 & \sec \omega &= \sqrt{2} & \csc \omega &= -\sqrt{2}
 \end{aligned}$$

5.

$$\begin{aligned}
 \operatorname{sen} \alpha &= -\frac{2}{3} & \cos \alpha &= -\frac{\sqrt{5}}{3} & \tan \alpha &= \frac{2\sqrt{5}}{5} \\
 \operatorname{ctg} \alpha &= \frac{\sqrt{5}}{2} & \sec \alpha &= -\frac{3\sqrt{5}}{5} & \csc \alpha &= -\frac{3}{2}
 \end{aligned}$$

6.

$$\begin{aligned}
 \operatorname{sen} \alpha &= -\frac{\sqrt{77}}{11} & \cos \alpha &= \frac{2\sqrt{11}}{11} & \tan \alpha &= -\frac{\sqrt{7}}{2} \\
 \operatorname{ctg} \alpha &= -\frac{2\sqrt{7}}{7} & \sec \alpha &= \frac{\sqrt{11}}{2} & \csc \alpha &= -\frac{\sqrt{77}}{7}
 \end{aligned}$$

7.

$$\operatorname{sen} \beta = \frac{2\sqrt{22}}{13} \quad \cos \beta = -\frac{9}{13} \quad \tan \beta = -\frac{2\sqrt{22}}{9}$$

$$\operatorname{ctg} \beta = -\frac{9\sqrt{22}}{44} \quad \sec \beta = -\frac{13}{9} \quad \csc \beta = \frac{13\sqrt{22}}{44}$$

8.

$$\operatorname{sen} \omega = -\frac{\sqrt{65}}{65} \quad \cos \omega = \frac{8\sqrt{65}}{65} \quad \tan \omega = -\frac{1}{8}$$

$$\operatorname{ctg} \omega = -8 \quad \sec \omega = \frac{\sqrt{65}}{8} \quad \csc \omega = -\sqrt{65}$$

9.

$$\operatorname{sen} \delta = \frac{5}{13} \quad \cos \delta = -\frac{12}{13} \quad \tan \delta = -\frac{5}{12}$$

$$\operatorname{ctg} \delta = -\frac{12}{5} \quad \sec \delta = -\frac{13}{12} \quad \csc \delta = \frac{13}{5}$$

10.

$$\operatorname{sen} \beta = -\frac{\sqrt{6}}{3} \quad \cos \beta = -\frac{\sqrt{3}}{3} \quad \tan \beta = \sqrt{2}$$

$$\operatorname{ctg} \beta = \frac{\sqrt{2}}{2} \quad \sec \beta = -\sqrt{3} \quad \csc \beta = -\frac{\sqrt{6}}{2}$$

11.

$$\operatorname{sen} \alpha = \frac{\sqrt{3}}{2} \quad \cos \alpha = -\frac{1}{2} \quad \tan \alpha = -\sqrt{3}$$

$$\operatorname{ctg} \alpha = -\frac{\sqrt{3}}{3} \quad \sec \alpha = -2 \quad \csc \alpha = \frac{2\sqrt{3}}{3}$$

12.

$$\operatorname{sen} \alpha = -\frac{\sqrt{3}}{2} \quad \cos \alpha = \frac{1}{2} \quad \tan \alpha = -\sqrt{3}$$

$$\operatorname{ctg} \alpha = -\frac{\sqrt{3}}{3} \quad \sec \alpha = 2 \quad \csc \alpha = -\frac{2\sqrt{3}}{3}$$

EJERCICIO 39

Inciso 1)

$$a) -\operatorname{sen} 30^\circ = -\cos 60^\circ$$

$$b) -\tan 15^\circ = -\operatorname{ctg} 75^\circ$$

$$c) \cos 80^\circ = \operatorname{sen} 10^\circ$$

$$d) \csc 60^\circ = \sec 30^\circ$$

$$e) \sec 2^\circ = \csc 88^\circ$$

$$f) -\operatorname{sen} 60^\circ 37' 25'' = -\cos 29^\circ 22' 35''$$

$$g) -\operatorname{ctg} 45^\circ = -\tan 45^\circ$$

$$h) \tan 74^\circ 46' 24'' = \operatorname{ctg} 15^\circ 13' 36''$$

$$i) -\cos 84^\circ 35' = -\operatorname{sen} 5^\circ 25'$$

$$j) \sec 39^\circ 11' 48'' = \csc 50^\circ 48' 12''$$

$$k) \csc 53^\circ = \sec 37^\circ$$

$$l) -\operatorname{ctg} 48^\circ = -\tan 42^\circ$$

$$m) \cos 38^\circ 54' = \operatorname{sen} 51^\circ 6'$$

$$n) -\operatorname{sen} 28^\circ 35' 24'' = -\cos 61^\circ 24' 36''$$

Inciso 2)

$$a) -\operatorname{sen} 160^\circ$$

$$f) -\csc 90^\circ$$

$$b) -\operatorname{ctg} 140^\circ$$

$$g) \cos 225^\circ 15' 46''$$

$$c) \sec 240^\circ$$

$$h) -\operatorname{ctg} 176^\circ 45' 23''$$

$$d) \cos 280^\circ$$

$$i) \sec 108^\circ 32'$$

$$e) -\tan 345^\circ$$

$$j) -\operatorname{sen} 228^\circ 15'$$

Inciso 3)

$$a) -\operatorname{sen} 20^\circ$$

$$g) -\operatorname{sen} 55^\circ$$

$$b) -\operatorname{ctg} 20^\circ$$

$$h) -\tan 76^\circ 34' 42''$$

$$c) \cos 80^\circ$$

$$i) \cos 68^\circ 45' 24''$$

$$d) \tan 45^\circ$$

$$j) \operatorname{ctg} 20^\circ$$

$$e) -\csc 81^\circ 27' 48''$$

$$k) -\sec 40^\circ$$

$$f) -\sec 50^\circ$$

$$l) -\csc 31^\circ 26' 19''$$

Inciso 4)

$$a) 0.3090$$

$$f) 1.0187$$

$$b) 0.9657$$

$$g) 0.9261$$

$$c) 1.1034$$

$$h) 3.8208$$

$$d) 0.1219$$

$$i) 1.0170$$

$$e) 0.7536$$

$$j) 0.4975$$

CAPÍTULO 12

EJERCICIO 40

Grados	Radianes	sen	cos	tan	csc	sec	ctg
0°	0	0	1	0	No existe	1	No existe
30°	$\frac{\pi}{6}$	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{3}}{3}$	2	$\frac{2\sqrt{3}}{3}$	$\sqrt{3}$
45°	$\frac{\pi}{4}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	1	$\sqrt{2}$	$\sqrt{2}$	1
60°	$\frac{\pi}{3}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$	$\frac{2\sqrt{3}}{3}$	2	$\frac{\sqrt{3}}{3}$
90°	$\frac{\pi}{2}$	1	0	No existe	1	No existe	0
120°	$\frac{2\pi}{3}$	$\frac{\sqrt{3}}{2}$	$-\frac{1}{2}$	$-\sqrt{3}$	$\frac{2\sqrt{3}}{3}$	-2	$-\frac{\sqrt{3}}{3}$
135°	$\frac{3\pi}{4}$	$\frac{\sqrt{2}}{2}$	$-\frac{\sqrt{2}}{2}$	-1	$\sqrt{2}$	$-\sqrt{2}$	-1
150°	$\frac{5\pi}{6}$	$\frac{1}{2}$	$-\frac{\sqrt{3}}{2}$	$-\frac{\sqrt{3}}{3}$	2	$-\frac{2\sqrt{3}}{3}$	$-\sqrt{3}$
180°	π	0	-1	0	No existe	-1	No existe
210°	$\frac{7\pi}{6}$	$-\frac{1}{2}$	$-\frac{\sqrt{3}}{2}$	$\frac{\sqrt{3}}{3}$	-2	$-\frac{2\sqrt{3}}{3}$	$\sqrt{3}$
225°	$\frac{5\pi}{4}$	$-\frac{\sqrt{2}}{2}$	$-\frac{\sqrt{2}}{2}$	1	$-\sqrt{2}$	$-\sqrt{2}$	1
240°	$\frac{4\pi}{3}$	$-\frac{\sqrt{3}}{2}$	$-\frac{1}{2}$	$\sqrt{3}$	$-\frac{2\sqrt{3}}{3}$	-2	$\frac{\sqrt{3}}{3}$
270°	$\frac{3\pi}{2}$	-1	0	No existe	-1	No existe	0
300°	$\frac{5\pi}{3}$	$-\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$-\sqrt{3}$	$-\frac{2\sqrt{3}}{3}$	2	$-\frac{\sqrt{3}}{3}$
315°	$\frac{7\pi}{4}$	$-\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	-1	$-\sqrt{2}$	$\sqrt{2}$	-1
330°	$\frac{11\pi}{6}$	$-\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$-\frac{\sqrt{3}}{3}$	-2	$\frac{2\sqrt{3}}{3}$	$-\sqrt{3}$
360°	2π	0	1	0	No existe	1	No existe

- | | | | |
|-------------------------|-------------------|--------------------------|--------|
| 1. $\frac{\sqrt{3}}{2}$ | 5. $\frac{3}{16}$ | 9. 1 | 13. -1 |
| 2. $\frac{\sqrt{3}}{2}$ | 6. $\frac{1}{8}$ | 10. $\frac{\sqrt{3}}{4}$ | 14. 0 |
| 3. $\frac{3}{2}$ | 7. 9 | 11. 2 | 15. 2 |
| 4. 0 | 8. $\sqrt{2}$ | 12. 1 | |

16 a 20. No se incluye la solución por ser demostraciones.

CAPÍTULO 13

EJERCICIO 41

1. Amplitud: 2, Período: $\frac{2}{3}\pi$

Desplazamiento de fase: $\frac{\pi}{6}, \frac{5}{6}\pi$

2. Amplitud: 2, Período: $\frac{1}{2}\pi$

Desplazamiento de fase: 0, $\frac{1}{2}\pi$

3. Amplitud: $\frac{4}{3}$, Período: 3π

Desplazamiento de fase: $-\frac{3}{4}\pi, \frac{9}{4}\pi$

4. Amplitud: 5, Período: 8π

Desplazamiento de fase: $-2\pi, 6\pi$

5. Amplitud: 4

Período: 2π

Desplazamiento de fase: $\frac{3\pi}{4}, \frac{11}{4}\pi$

6. Amplitud: 3

Período: π

Desplazamiento de fase: 0, π

7. Amplitud: $\frac{3}{2}$

Período: $\frac{2}{5}\pi$

Desplazamiento de fase: $-\frac{3\pi}{10}, \frac{\pi}{10}$

8. Amplitud: $\frac{1}{3}$

Período: 8π

Desplazamiento de fase: $-\frac{4\pi}{3}, \frac{20\pi}{3}$

9. Amplitud: 1

Período: 6π

Desplazamiento de fase: 0, 6π

10. Período: $\frac{\pi}{2}$

Asíntotas verticales: $\dots, -\frac{\pi}{4}, \frac{\pi}{4}, \dots$

Desplazamiento de fase: no existe

11. Período: π

Asíntotas verticales: $\dots, -\frac{3}{4}\pi, \frac{\pi}{4}, \dots$

Desplazamiento de fase: $-\frac{\pi}{4}$ a la izq.

12. Período: $\frac{\pi}{3}$

Asíntotas verticales: $\dots, -\frac{\pi}{18}, \frac{5\pi}{18}, \dots$

Desplazamiento de fase: $\frac{\pi}{9}$ a la der.

13. Período: 2π

Asíntotas verticales: $\dots, \pi, 3\pi, \dots$

Desplazamiento de fase: 2π a la der.

14. Período: 4π

Asíntotas verticales: $\dots, 0, 4\pi, \dots$

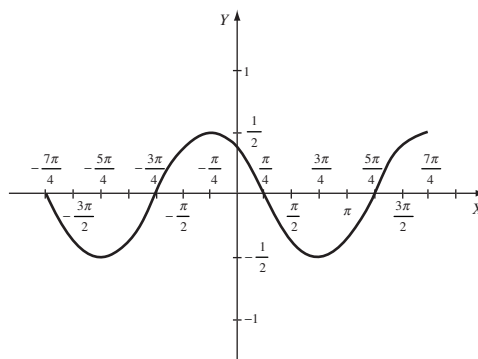
Desplazamiento de fase: 2π a la der.

15. Período: π

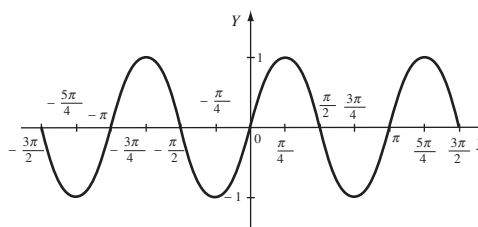
Asíntotas verticales: $\dots, \frac{1}{2}\pi, \frac{3}{2}\pi, \dots$

Desplazamiento de fase: π a la der.

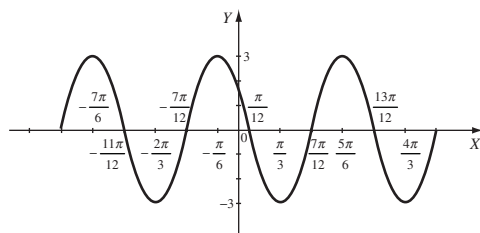
- 16.



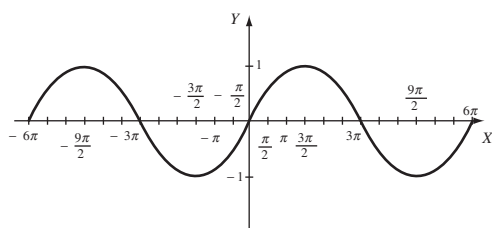
- 17.



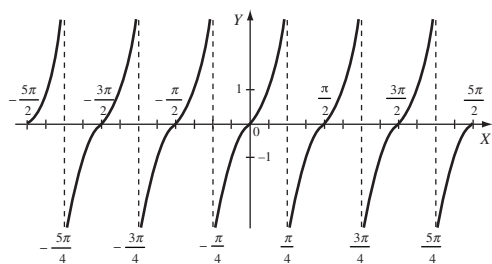
18.



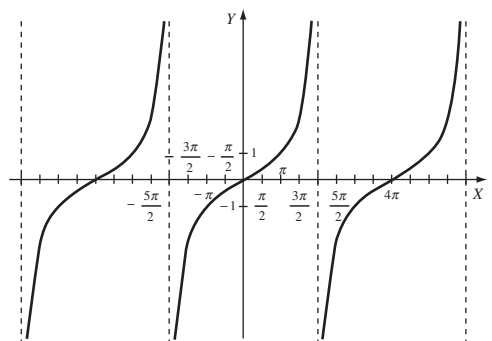
19.



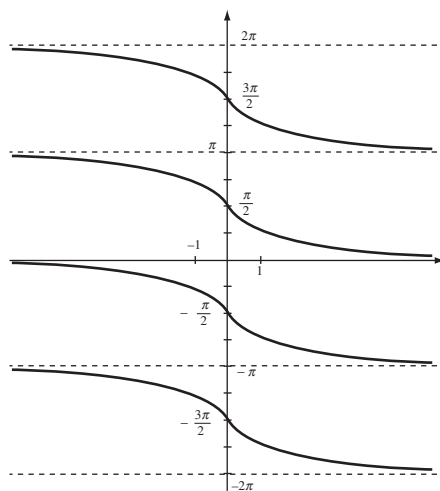
20.



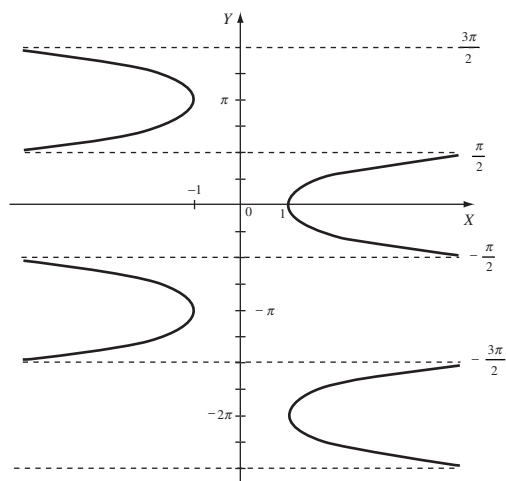
21.



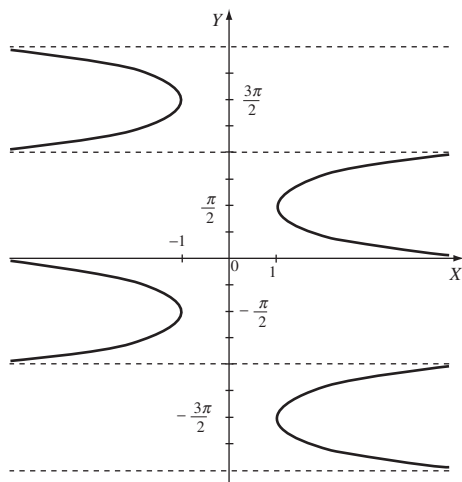
22.



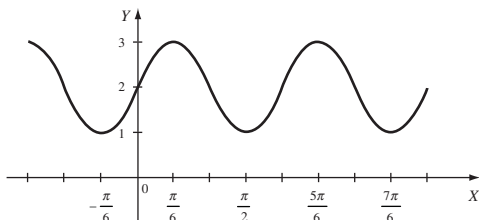
23.



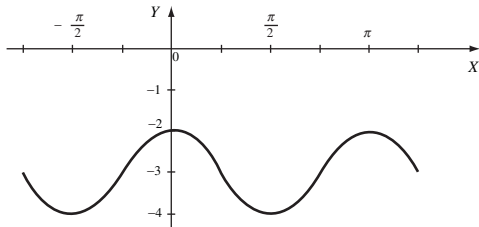
24.



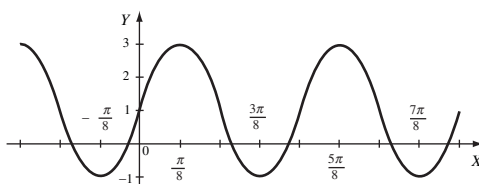
25.



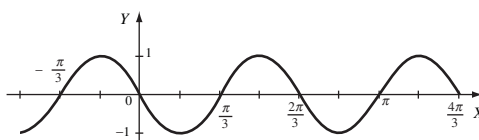
26.



27.



28.



CAPÍTULO 14

EJERCICIO 42

1 a 32. No se incluye la solución por ser demostraciones.

EJERCICIO 43

- | | |
|----------------------------------|--------------------------|
| 1. $\frac{\sqrt{3}}{2}$ | 6. $-\frac{\sqrt{2}}{2}$ |
| 2. $\frac{\sqrt{6}-\sqrt{2}}{4}$ | 7. $-\sqrt{3}$ |
| 3. $\frac{\sqrt{6}+\sqrt{2}}{4}$ | 8. $-\sqrt{2}$ |
| 4. -1 | 9. 1 |
| 5. $-\sqrt{2}$ | 10. 1 |

- | | |
|--|---|
| 11. $\frac{1}{2}(\sqrt{3}\sin\theta + \cos\theta)$ | 16. $\frac{1-\tan\beta}{1+\tan\beta}$ |
| 12. $\frac{\sqrt{2}}{2}(\sin x - \cos x)$ | 17. $\frac{1}{2}(\sqrt{3}\sin x - \cos x)$ |
| 13. $\sin\beta$ | 18. $-\sec 2\omega$ |
| 14. $\frac{1}{\tan x} = \cot x$ | 19. $-\tan\alpha$ |
| 15. $\frac{2}{\sqrt{3}\cos\alpha - \sin\alpha}$ | 20. $\frac{\sqrt{2}}{2}(\sin\theta + \cos\theta)$ |

EJERCICIO 44

- | | |
|----------------------------|-----------------------------------|
| 1. $-(2+\sqrt{3})$ | 6. $\frac{\sqrt{6}-\sqrt{2}}{4}$ |
| 2. $2-\sqrt{3}$ | 7. $\sqrt{3}-2$ |
| 3. $\sqrt{2}(1+\sqrt{3})$ | 8. $-(\sqrt{6}-\sqrt{2})$ |
| 4. $-\sqrt{2}(1+\sqrt{3})$ | 9. $\sqrt{2}-\sqrt{6}$ |
| 5. $2+\sqrt{3}$ | 10. $\frac{\sqrt{2}-\sqrt{3}}{2}$ |

$$11. \sin(\alpha + \beta) = \frac{\sqrt{13}}{65}, \cos(\alpha + \beta) = -\frac{18\sqrt{13}}{65},$$

$$\tan(\alpha + \beta) = -\frac{1}{18}$$

$$12. \sin(\alpha - \beta) = -\frac{\sqrt{2}+\sqrt{6}}{4}, \cos(\alpha - \beta) = \frac{\sqrt{6}-\sqrt{2}}{4},$$

$$\tan(\alpha - \beta) = -(2+\sqrt{3})$$

 13. Funciones del ángulo $(\alpha + \beta)$

$$\sin(\alpha + \beta) = -\frac{\sqrt{3}}{9}(2+\sqrt{10}), \cos(\alpha + \beta) = \frac{\sqrt{3}}{9}(\sqrt{5}-2\sqrt{2})$$

$$\tan(\alpha + \beta) = 3\sqrt{2}+2\sqrt{5}, \operatorname{ctg}(\alpha + \beta) = \frac{2\sqrt{5}-3\sqrt{2}}{2},$$

$$\sec(\alpha + \beta) = -(\sqrt{15}+2\sqrt{6}), \csc(\alpha + \beta) = \sqrt{3}-\frac{1}{2}\sqrt{30}$$

 Funciones del ángulo $(\alpha - \beta)$

$$\sin(\alpha - \beta) = \frac{\sqrt{3}}{9}(2-\sqrt{10}), \cos(\alpha - \beta) = -\frac{\sqrt{3}}{9}(\sqrt{5}+2\sqrt{2})$$

$$\tan(\alpha - \beta) = 2\sqrt{5}-3\sqrt{2}, \operatorname{ctg}(\alpha - \beta) = \frac{2\sqrt{5}+3\sqrt{2}}{2}$$

$$\sec(\alpha - \beta) = \sqrt{15}-2\sqrt{6}, \csc(\alpha - \beta) = -\sqrt{3}-\frac{1}{2}\sqrt{30}$$

14 a 34. No se incluye la solución por ser demostraciones.

EJERCICIO 45

1.

Funciones trigonométricas del ángulo $\frac{\pi}{8}$

$$\operatorname{sen} \frac{\pi}{8} = \frac{\sqrt{2-\sqrt{2}}}{2} \quad \operatorname{ctg} \frac{\pi}{8} = \sqrt{3+2\sqrt{2}}$$

$$\cos \frac{\pi}{8} = \frac{\sqrt{2+\sqrt{2}}}{2} \quad \sec \frac{\pi}{8} = \sqrt{4-2\sqrt{2}}$$

$$\tan \frac{\pi}{8} = \sqrt{3-2\sqrt{2}} \quad \csc \frac{\pi}{8} = \sqrt{4+2\sqrt{2}}$$

Funciones trigonométricas del ángulo $\frac{3}{8}\pi$

$$\operatorname{sen} \frac{3}{8}\pi = \frac{\sqrt{2+\sqrt{2}}}{2} \quad \operatorname{ctg} \frac{3}{8}\pi = \sqrt{3-2\sqrt{2}}$$

$$\cos \frac{3}{8}\pi = \frac{\sqrt{2-\sqrt{2}}}{2} \quad \sec \frac{3}{8}\pi = \sqrt{4+2\sqrt{2}}$$

$$\tan \frac{3}{8}\pi = \sqrt{3+2\sqrt{2}} \quad \csc \frac{3}{8}\pi = \sqrt{4-2\sqrt{2}}$$

Funciones trigonométricas del ángulo $\frac{5}{8}\pi$

$$\operatorname{sen} \frac{5}{8}\pi = \frac{\sqrt{2-\sqrt{2}}}{2} \quad \operatorname{ctg} \frac{5}{8}\pi = -\sqrt{3-2\sqrt{2}}$$

$$\cos \frac{5}{8}\pi = -\frac{\sqrt{2+\sqrt{2}}}{2} \quad \sec \frac{5}{8}\pi = -\sqrt{4+2\sqrt{2}}$$

$$\tan \frac{5}{8}\pi = -\sqrt{3+2\sqrt{2}} \quad \csc \frac{5}{8}\pi = \sqrt{4-2\sqrt{2}}$$

Funciones trigonométricas del ángulo $\frac{7}{8}\pi$

$$\operatorname{sen} \frac{7}{8}\pi = \frac{\sqrt{2-\sqrt{2}}}{2} \quad \operatorname{ctg} \frac{7}{8}\pi = -\sqrt{3+2\sqrt{2}}$$

$$\cos \frac{7}{8}\pi = -\frac{\sqrt{2+\sqrt{2}}}{2} \quad \sec \frac{7}{8}\pi = -\sqrt{4-2\sqrt{2}}$$

$$\tan \frac{7}{8}\pi = -\sqrt{3-2\sqrt{2}} \quad \csc \frac{7}{8}\pi = \sqrt{4+2\sqrt{2}}$$

2.

Funciones trigonométricas del ángulo $\frac{\alpha}{2}$

$$\operatorname{sen} \frac{\alpha}{2} = \frac{\sqrt{8+2\sqrt{15}}}{4} \quad \operatorname{ctg} \frac{\alpha}{2} = \sqrt{31-8\sqrt{15}}$$

$$\cos \frac{\alpha}{2} = \frac{\sqrt{8-2\sqrt{15}}}{4} \quad \sec \frac{\alpha}{2} = 2\sqrt{8+2\sqrt{15}}$$

$$\tan \frac{\alpha}{2} = \sqrt{31+8\sqrt{15}} \quad \csc \frac{\alpha}{2} = 2\sqrt{8-2\sqrt{15}}$$

Funciones trigonométricas del ángulo 2α

$$\operatorname{sen} 2\alpha = -\frac{\sqrt{15}}{8} \quad \operatorname{ctg} 2\alpha = -\frac{7\sqrt{15}}{15}$$

$$\cos 2\alpha = \frac{7}{8} \quad \sec 2\alpha = \frac{8}{7}$$

$$\tan 2\alpha = -\frac{\sqrt{15}}{7} \quad \csc 2\alpha = -\frac{8\sqrt{15}}{15}$$

3.

Funciones trigonométricas del ángulo $\frac{\beta}{2}$

$$\operatorname{sen} \frac{\beta}{2} = \frac{3\sqrt{13}}{13} \quad \operatorname{ctg} \frac{\beta}{2} = -\frac{2}{3}$$

$$\cos \frac{\beta}{2} = -\frac{2\sqrt{13}}{13} \quad \sec \frac{\beta}{2} = -\frac{\sqrt{13}}{2}$$

$$\tan \frac{\beta}{2} = -\frac{3}{2} \quad \csc \frac{\beta}{2} = \frac{\sqrt{13}}{3}$$

Funciones trigonométricas del ángulo 2β

$$\operatorname{sen} 2\beta = \frac{120}{169} \quad \operatorname{ctg} 2\beta = -\frac{119}{120}$$

$$\cos 2\beta = -\frac{119}{169} \quad \sec 2\beta = -\frac{169}{119}$$

$$\tan 2\beta = -\frac{120}{119} \quad \csc 2\beta = \frac{169}{120}$$

4.

Funciones trigonométricas del ángulo $\frac{\omega}{2}$

$$\operatorname{sen} \frac{\omega}{2} = \frac{\sqrt{3}}{4} \quad \operatorname{ctg} \frac{\omega}{2} = -\frac{\sqrt{39}}{3}$$

$$\cos \frac{\omega}{2} = -\frac{\sqrt{13}}{4} \quad \sec \frac{\omega}{2} = -\frac{4\sqrt{13}}{13}$$

$$\tan \frac{\omega}{2} = -\frac{\sqrt{39}}{13} \quad \csc \frac{\omega}{2} = \frac{4\sqrt{3}}{3}$$

Funciones trigonométricas del ángulo 2ω

$$\operatorname{sen} 2\omega = -\frac{5\sqrt{39}}{32} \quad \operatorname{ctg} 2\omega = \frac{7\sqrt{39}}{195}$$

$$\cos 2\omega = -\frac{7}{32} \quad \sec 2\omega = -\frac{32}{7}$$

$$\tan 2\omega = \frac{5\sqrt{39}}{7} \quad \csc 2\omega = -\frac{32\sqrt{39}}{195}$$

5.

Funciones trigonométricas del ángulo $\frac{\alpha}{2}$

$$\operatorname{sen} \frac{\alpha}{2} = \frac{\sqrt{98+28\sqrt{7}}}{14} \quad \operatorname{ctg} \frac{\alpha}{2} = \frac{\sqrt{33-12\sqrt{7}}}{3}$$

$$\cos \frac{\alpha}{2} = \frac{\sqrt{98-28\sqrt{7}}}{14} \quad \sec \frac{\alpha}{2} = \frac{\sqrt{42+12\sqrt{7}}}{3}$$

$$\tan \frac{\alpha}{2} = \frac{\sqrt{33+12\sqrt{7}}}{3} \quad \csc \frac{\alpha}{2} = \frac{\sqrt{42-12\sqrt{7}}}{3}$$

Funciones trigonométricas del ángulo 2α

$$\operatorname{sen} 2\alpha = -\frac{4\sqrt{3}}{7} \quad \operatorname{ctg} 2\alpha = -\frac{\sqrt{3}}{12}$$

$$\cos 2\alpha = \frac{1}{7} \quad \sec 2\alpha = 7$$

$$\tan 2\alpha = -4\sqrt{3} \quad \csc 2\alpha = -\frac{7\sqrt{3}}{12}$$

6.

Funciones trigonométricas del ángulo α

$$\operatorname{sen} \alpha = \frac{2}{3} \quad \cos \alpha = -\frac{\sqrt{5}}{3} \quad \tan \alpha = -\frac{2\sqrt{5}}{5}$$

7.

Funciones trigonométricas del ángulo $\frac{\beta}{2}$

$$\operatorname{sen} \frac{\beta}{2} = \frac{\sqrt{578+136\sqrt{17}}}{34} \quad \operatorname{ctg} \frac{\beta}{2} = -\sqrt{33-8\sqrt{17}}$$

$$\cos \frac{\beta}{2} = -\frac{\sqrt{578-136\sqrt{17}}}{34} \quad \sec \frac{\beta}{2} = -\sqrt{34+8\sqrt{17}}$$

$$\tan \frac{\beta}{2} = -\sqrt{33+8\sqrt{17}} \quad \csc \frac{\beta}{2} = \sqrt{34-8\sqrt{17}}$$

Funciones trigonométricas del ángulo β

$$\operatorname{sen} \beta = -\frac{\sqrt{17}}{17} \quad \operatorname{ctg} \beta = 4$$

$$\cos \beta = -\frac{4\sqrt{17}}{17} \quad \sec \beta = -\frac{\sqrt{17}}{4}$$

$$\tan \beta = \frac{1}{4} \quad \csc \beta = -\sqrt{17}$$

8.

Funciones trigonométricas del ángulo α

$$\operatorname{sen} \alpha = \frac{2\sqrt{5}}{5} \quad \tan \alpha = 2 \quad \sec \alpha = \sqrt{5}$$

$$\cos \alpha = \frac{\sqrt{5}}{5} \quad \operatorname{ctg} \alpha = \frac{1}{2} \quad \csc \alpha = \frac{\sqrt{5}}{2}$$

9.

Funciones trigonométricas del ángulo $\frac{\beta}{2}$

$$\operatorname{sen} \frac{\beta}{2} = \frac{\sqrt{3}}{3} \quad \tan \frac{\beta}{2} = \frac{\sqrt{2}}{2} \quad \sec \frac{\beta}{2} = \frac{\sqrt{6}}{2}$$

$$\cos \frac{\beta}{2} = \frac{\sqrt{6}}{3} \quad \operatorname{ctg} \frac{\beta}{2} = \sqrt{2} \quad \csc \frac{\beta}{2} = \sqrt{3}$$

Funciones trigonométricas del ángulo β

$$\operatorname{sen} \beta = \frac{2\sqrt{2}}{3} \quad \tan \beta = 2\sqrt{2} \quad \sec \beta = 3$$

$$\cos \beta = \frac{1}{3} \quad \operatorname{ctg} \beta = \frac{\sqrt{2}}{4} \quad \csc \beta = \frac{3\sqrt{2}}{4}$$

10.

Funciones trigonométricas del ángulo ω

$$\operatorname{sen} \omega = -\frac{3}{5} \quad \operatorname{ctg} \omega = -\frac{4}{3}$$

$$\cos \omega = \frac{4}{5} \quad \sec \omega = \frac{5}{4}$$

$$\tan \omega = -\frac{3}{4} \quad \csc \omega = -\frac{5}{3}$$

Funciones trigonométricas del ángulo 2ω

$$\operatorname{sen} 2\omega = -\frac{24}{25} \quad \operatorname{ctg} 2\omega = -\frac{7}{24}$$

$$\cos 2\omega = \frac{7}{25} \quad \sec 2\omega = \frac{25}{7}$$

$$\tan 2\omega = -\frac{24}{7} \quad \csc 2\omega = -\frac{25}{24}$$

Funciones trigonométricas del ángulo 4ω

$$\operatorname{sen} 4\omega = -\frac{336}{625} \quad \operatorname{ctg} 4\omega = \frac{527}{336}$$

$$\cos 4\omega = -\frac{527}{625} \quad \sec 4\omega = -\frac{625}{527}$$

$$\tan 4\omega = \frac{336}{527} \quad \csc 4\omega = -\frac{625}{336}$$

11 a 25. No se incluye la solución por ser demostraciones.

EJERCICIO 46

1. $\frac{1}{2}[\operatorname{sen}(2\alpha) + \operatorname{sen}(2\beta)]$
2. $\frac{1}{2}[\operatorname{sen}(105^\circ) + \operatorname{sen}(15^\circ)]$
3. $-\frac{1}{2}[\cos(2\gamma) - \cos(2\beta)]$
4. $\frac{1}{2}\left[\cos\left(\frac{2}{3}\pi\right) + \cos\left(\frac{1}{6}\pi\right)\right]$
5. $\frac{1}{2}[\operatorname{sen}(120^\circ) + \operatorname{sen}(45^\circ)]$
6. $-\frac{1}{2}[\cos(45^\circ) - \cos(30^\circ)]$
7. $\frac{1}{2}[\operatorname{sen}(2x) - \operatorname{sen}(2\alpha)]$
8. $\frac{1}{2}\left[\cos(\pi) + \cos\left(\frac{1}{6}\pi\right)\right]$
9. $\frac{1}{2}[\operatorname{sen}(45^\circ) - \operatorname{sen}(30^\circ)]$
10. $\frac{1}{2}\left[\cos\left(\frac{11}{6}\pi\right) + \cos\left(\frac{5}{3}\pi\right)\right]$
11. $-2[\cos(4\alpha) - \cos(2\alpha)]$
12. $\frac{5}{2}[\operatorname{sen}(8\alpha) + \operatorname{sen}(4\alpha)]$
13. $\frac{1}{2}[\operatorname{sen}(90^\circ) - \operatorname{sen}(4^\circ)]$
14. $\frac{1}{2}\left[\cos\frac{1}{3}(2\alpha+5\beta) + \cos\frac{1}{3}(2\alpha-5\beta)\right]$
15. $\frac{3}{2}\left[\operatorname{sen}\left(\frac{19}{2}\alpha\right) + \operatorname{sen}\left(\frac{17}{2}\alpha\right)\right]$
16. $\frac{2}{\cos\frac{\pi}{2} + \cos\frac{\pi}{6}}$
17. $\frac{\operatorname{sen}3\alpha + \operatorname{sen}\alpha}{\operatorname{sen}3\alpha - \operatorname{sen}\alpha}$
18. $\frac{2}{\operatorname{sen}\pi - \operatorname{sen}\frac{\pi}{2}}$
19. $\frac{\cos2\alpha - \cos2x}{\cos2\alpha + \cos2x}$
20. $\frac{1}{2}[\operatorname{sen}(4\alpha) + \operatorname{sen}(2\beta)]$

EJERCICIO 47

1 a 14. No se incluye la solución por ser demostraciones.

EJERCICIO 48

1. $2[\operatorname{sen}(120^\circ) \cdot \cos(45^\circ)]$
2. $2\left[\cos\left(\frac{5}{2}\beta\right)\cos\left(\frac{9}{2}\beta\right)\right]$
3. $2[\operatorname{sen}(180^\circ)\cos(60^\circ)]$
4. $-2[\operatorname{sen}(4\theta)\operatorname{sen}(\theta)]$
5. $2[\cos(45^\circ)\cos(7^\circ31')]$
6. $2\left[\cos\left(\frac{1}{3}\pi\right)\operatorname{sen}\left(\frac{1}{4}\pi\right)\right]$
7. $2\left[\cos\left(\frac{1}{4}\pi\right)\cos\left(\frac{1}{36}\pi\right)\right]$
8. $2[\cos(30^\circ)\operatorname{sen}(5^\circ)]$
9. $-2\left[\operatorname{sen}\left(\frac{5}{12}\pi\right)\operatorname{sen}\left(\frac{1}{3}\pi\right)\right]$
10. $2\left[\cos(\beta)\cos\left(\frac{1}{6}\pi\right)\right]$
11. $2\left[\operatorname{sen}\left(\frac{7}{24}\pi\right)\cos\left(\frac{1}{24}\pi\right)\right]$
12. $2\left[\operatorname{sen}\left(\frac{3}{4}(\alpha+\beta)\right)\cos\left(\frac{1}{4}(\alpha-\beta)\right)\right]$
13. $-2\left[\operatorname{sen}(\alpha)\operatorname{sen}\left(\frac{\pi}{4}\right)\right]$
14. $2\left[\operatorname{sen}(\beta)\cos\left(\frac{\pi}{8}\right)\right]$
15. $2\left[\operatorname{sen}\left(\frac{3}{4}\pi\right)\cos\left(\alpha - \frac{\pi}{8}\right)\right]$
16. $-2\left[\operatorname{sen}\left(\frac{\alpha}{2}\right)\operatorname{sen}\left(\frac{\beta}{2}\right)\right]$

EJERCICIO 49

1 a 12. No se incluye la solución por ser demostraciones.

EJERCICIO 50

1. $\frac{\pi}{4}, \frac{5}{4}\pi$
2. $\frac{\pi}{2}, 36^\circ 52' 11''$
3. $\frac{7}{12}\pi, \frac{23}{12}\pi$
4. $\frac{1}{4}\pi, \frac{5}{4}\pi$
5. $\frac{1}{6}\pi, \frac{5}{6}\pi$
6. $\frac{1}{3}\pi, \frac{5}{3}\pi$
7. $0, \pi, 2\pi$
8. $0, \pi, 2\pi, \frac{7}{6}\pi, \frac{11}{6}\pi$
9. $0, 2\pi, 152^\circ 44', 207^\circ 15'$
10. $\frac{1}{4}\pi, \frac{7}{4}\pi, \frac{3}{4}\pi, \frac{5}{4}\pi$
11. $\frac{1}{2}\pi, \frac{1}{4}\pi, \frac{5}{4}\pi$
12. $\frac{7}{6}\pi, \frac{11}{6}\pi$
13. $\frac{1}{4}\pi, \frac{5}{4}\pi$
14. $\frac{1}{3}\pi, \frac{2}{3}\pi, \frac{4}{3}\pi, \frac{5}{3}\pi$
15. $\frac{1}{6}\pi, \frac{7}{6}\pi$
16. $\frac{1}{2}\pi, \frac{1}{6}\pi, \frac{5}{6}\pi$
17. $\frac{1}{4}\pi, \frac{5}{4}\pi$
18. $0, \pi, 2\pi, \frac{2}{3}\pi, \frac{4}{3}\pi$
19. $\frac{1}{3}\pi, \frac{5}{3}\pi$
20. $0, \frac{1}{3}\pi, \frac{2}{3}\pi, \frac{4}{3}\pi, \frac{5}{3}\pi, \pi, 2\pi$

$$21. 0, \frac{2}{3}\pi, \frac{5}{3}\pi, \pi, 2\pi$$

$$22. \frac{1}{2}\pi, \frac{3}{4}\pi, \frac{5}{4}\pi$$

$$23. \frac{3}{2}\pi, \frac{1}{3}\pi, \frac{2}{3}\pi$$

$$24. \frac{1}{3}\pi, \frac{5}{3}\pi$$

$$25. \frac{7}{6}\pi, \frac{11}{6}\pi, \frac{1}{3}\pi, \frac{5}{3}\pi$$

$$26. \frac{7}{6}\pi, \frac{1}{6}\pi$$

$$27. \frac{7}{12}\pi, \frac{11}{12}\pi$$

$$28. \frac{7}{6}\pi, \frac{11}{6}\pi, \frac{1}{6}\pi, \frac{5}{6}\pi$$

$$29. \frac{1}{3}\pi, \frac{4}{3}\pi$$

$$30. \frac{1}{3}\pi, \pi, \frac{5}{3}\pi$$

CAPÍTULO 15**EJERCICIO 51**

1. $c = \sqrt{145}, A = 44^\circ 54', \angle C = 45^\circ 6'$
2. $a = 2.11, c = 3.39, \angle C = 58^\circ$
3. $c = 5.23, b = 7.24, \angle A = 43^\circ 40'$
4. $b = 52.55, \angle A = 38^\circ 11' 40'', \angle C = 51^\circ 48' 20''$
5. $c = 13, b = 13\sqrt{2}, \angle C = 45^\circ$
6. $a = 13.28, c = 18.28, \angle A = 36^\circ$
7. $a = 12.51, \angle A = 33^\circ 46' 46'', \angle C = 56^\circ 13' 14''$
8. $a = 25.71, c = 22.9, \angle C = 41^\circ 48'$
9. $a = 82.68, b = 100.36, \angle A = 55^\circ 28'$
10. $c = 7.87, \angle A = 66^\circ 39' 17'', \angle C = 23^\circ 20' 43''$
11. $b = 22.36, c = 18.86, \angle C = 57^\circ 33'$
12. $c = \sqrt{13}, \angle A = 29^\circ 1' 1'', \angle C = 60^\circ 58' 59''$
13. $a = 15.27, c = 17.19, \angle A = 41^\circ 37'$
14. $b = 7.9, \angle A = 71^\circ 33' 54'', \angle C = 18^\circ 26' 5''$
15. $a = 6.28, b = 14.44, \angle C = 64^\circ 11'$
16. $\angle A = 26^\circ 33' 54'', \angle C = 63^\circ 26' 6''$

17. $a = 5, b = 13, c = 12, \angle A = 22^\circ 37' 11'', \angle C = 67^\circ 22' 48''$

18. $a = 4, b = 5, c = 3, \angle A = 53^\circ 7' 49'', \angle C = 36^\circ 52' 11''$

19. $\angle A = \angle C = 45^\circ$

20. $\angle A = 19^\circ 28' 16'', \angle C = 70^\circ 31' 44''$

EJERCICIO 52

1. 288.4 m

2. 4.2 m

3. $38^\circ 44' 4'', 1.65 \text{ m}$

4. $(10\sqrt{2} + 1) \text{ m}$

5. $54^\circ 8'$

6. 52.07 m

7. 11.25 m

8. a) 53.6 m , b) 59.1 m , c) 22.6 m

9. $53^\circ 7', 3 \text{ m}$

10. $21^\circ 47', 14 \text{ dm}$

11.
$$L = \frac{\pi R}{90} \left[180 - \cos^{-1} \left(\frac{R-r}{l} \right) \right] + \frac{\pi r}{90} \cos^{-1} \left(\frac{R-r}{l} \right) + 2\sqrt{l^2 - (R-r)^2}$$

12. sí

CAPÍTULO 16**EJERCICIO 53**

1. $a = 20.9, c = 14.7, \angle A = 79^\circ 1'$

2. $b = 52.4, a = 47.7, \angle B = 79^\circ 16'$

3. $b = 21.03, a = 46.9, \angle C = 67^\circ 44'$

4. $b = 86.21, c = 66.87, \angle B = 76^\circ 39'$

5. $a = 23.35, c = 25.23, \angle A = 67^\circ$

6. $b = 17.09, c = 22.3, \angle C = 99^\circ$

7. $c = 9.43, \angle B = 57^\circ 58' 51'', \angle C = 90^\circ 1' 8''$

8. $a = 19.8, \angle A = 118^\circ 23' 35'', \angle B = 26^\circ 21' 24''$

9. $c = 15.11, \angle A = 40^\circ 5' 50'', \angle C = 83^\circ 19' 9''$

10. $b = 11.4, \angle A = 46^\circ 14' 25'', \angle B = 66^\circ 24' 34''$

11. $\angle A = 31^\circ 48' 52'', \angle B = 34^\circ 12' 58'', \angle C = 113^\circ 58' 10''$

12. $\angle A = 27^\circ 25' 16'', \angle B = 44^\circ 1' 54'', \angle C = 108^\circ 32' 50''$

13. $\angle A = 52^\circ 17' 24'', \angle B = 44^\circ 33' 55'', \angle C = 83^\circ 8' 41''$

14. $\angle A = 48^\circ 20' 58'', \angle B = 36^\circ 42' 37'', \angle C = 94^\circ 56' 23''$

15. $c = 15.3, \angle A = 46^\circ 39' 8'', \angle B = 65^\circ 20' 52''$

16. $b = 37.07, \angle A = 47^\circ 7' 45'', \angle C = 56^\circ 52' 15''$

17. $a = 46.05, \angle B = 34^\circ 5' 24'', \angle C = 110^\circ 54' 36''$

18. $c = 15.65, \angle A = 41^\circ 52' 18'', \angle B = 82^\circ 7' 42''$

EJERCICIO 54

1. $\overline{AB} = 369.95 \text{ m}$

7. 322.92 km

2. 1.76 cm

8. 307.4 m

3. 30.34 km

9. 29.07 km

4. 19.4 km

10. 180.37 m

5. 8.03 m

11. 29.7 cm

6. 4.7 cm

12 a 13. No se incluye la solución por ser demostraciones.

CAPÍTULO 17**EJERCICIO 55**

1. $z = \sqrt{17} \text{cis} 345^\circ 37' 49''$

2. $z = 2 \text{cis} 30^\circ$

3. $z = 2\sqrt{2} \text{cis} 135^\circ$

4. $z = 5 \text{cis} 0^\circ$

5. $z = 3 \text{cis} 270^\circ$

6. $z = \frac{5}{6} \text{cis} 53^\circ 7' 48''$

7. $z = \text{cis} 315^\circ$

8. $z = \text{cis} 150^\circ$

9. $z_1 \cdot z_2 = \sqrt{26} \text{cis} 75^\circ$

10. $z_2 \cdot z_4 = \sqrt{26} \text{cis} 165^\circ$

11. $z_1 \cdot z_3 = 2\sqrt{2} \text{cis} 105^\circ$

12. $z_1 \cdot z_2 \cdot z_3 = 2\sqrt{26} \text{cis} 135^\circ$

13. $z_1 \cdot z_3 \cdot z_4 = 4 \text{cis} 240^\circ$

14. $\frac{z_1}{z_4} = \text{cis} 270^\circ$

15. $\frac{z_2}{z_4} = \frac{\sqrt{26}}{2} \text{cis} 255^\circ$

16. $\frac{z_1}{z_3} = \frac{\sqrt{2}}{2} \text{cis} 345^\circ$

17. $\frac{z_1 \cdot z_2}{z_3} = \frac{\sqrt{26}}{2} \text{cis} 15^\circ$

18. $\frac{z_2}{z_1 \cdot z_4} = \frac{\sqrt{13}}{2} \text{cis} 210^\circ$

$$19. \frac{z_2 \cdot z_3}{z_1 \cdot z_4} = \sqrt{13} \operatorname{cis} 270^\circ$$

$$20. \frac{z_1 \cdot z_2 \cdot z_3}{z_4} = 2\sqrt{13} \operatorname{cis} 0^\circ$$

$$21. z^2 = 9 \operatorname{cis} 240^\circ$$

$$22. z^4 = 81 \operatorname{cis} 100^\circ$$

$$23. z^3 = 125 \operatorname{cis} 45^\circ$$

$$24. z_1 = 4 \operatorname{cis} 30^\circ, z_2 = 4 \operatorname{cis} 210^\circ$$

$$25. z_1 = 2 \operatorname{cis} 20^\circ, z_2 = 2 \operatorname{cis} 80^\circ, z_3 = 2 \operatorname{cis} 140^\circ, z_4 = 2 \operatorname{cis} 200^\circ$$

$$z_5 = 2 \operatorname{cis} 260^\circ, z_6 = 2 \operatorname{cis} 320^\circ$$

$$26. z_1 = \operatorname{cis} 60^\circ, z_2 = \operatorname{cis} 180^\circ, z_3 = \operatorname{cis} 300^\circ$$

$$27. (z \cdot z_1)^2 = 36 \operatorname{cis} 120^\circ$$

$$28. z_1 = 2 \operatorname{cis} 30^\circ, z_2 = 2 \operatorname{cis} 150^\circ, z_3 = 2 \operatorname{cis} 270^\circ$$

$$29. 28 \operatorname{cis} 100^\circ$$

$$30. z_1 = 4 \operatorname{cis} 10^\circ, z_2 = 4 \operatorname{cis} 130^\circ, z_3 = 4 \operatorname{cis} 250^\circ$$

Solución a los ejercicios de geometría analítica



CAPÍTULO 1

EJERCICIO 1

1. $3u$
2. $4u$
3. $5\sqrt{3}u$
4. $\frac{5}{2}u$
5. $\frac{5}{4}u$
6. $1u$
7. $\frac{13}{24}u$
8. $\frac{17}{5}u$
9. $5a$
10. $\frac{13}{12}a$
11. $\overline{AB} = 7u$
12. $\overline{DC} = \frac{5}{4}u$
13. $\overline{AD} = \frac{5}{2}u$
14. $\overline{BA} = -7u$
15. $\overline{CB} = \frac{13}{4}u$
16. $\overline{DA} = -\frac{5}{2}u$
17. $\overline{DB} = \frac{9}{2}u$
18. $\overline{CA} = -\frac{15}{4}u$
19. $\overline{BC} = -\frac{13}{4}u$
20. $\overline{CD} = -\frac{5}{4}u$

EJERCICIO 2

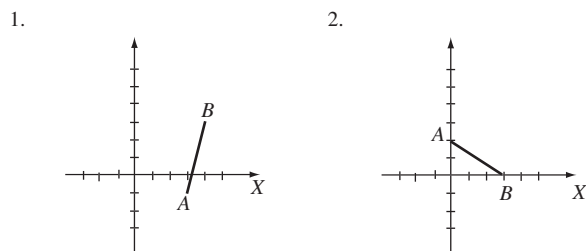
1. $r = \frac{1}{3}$
2. $r = \frac{1}{4}$
3. $r = 1$
4. $r = -\frac{7}{2}$
5. $r = -\frac{4}{35}$
6. $r = \frac{3}{5}a$
7. $r = \frac{7}{3}$
8. $r = -\frac{7}{9}$
9. $x = -\frac{14}{3}$
10. $x = \frac{27}{4}$
11. $x = -\frac{5}{2}$
12. $x = \frac{5}{2}$
13. $x = -\frac{17}{20}$
14. $x = -\frac{23}{40}$

EJERCICIO 3

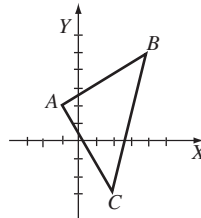
1. 1
2. $\frac{11}{2}$
3. -5
4. $-\frac{7}{12}$
5. $\frac{5}{12}$
6. $-\frac{7}{24}$
7. $\frac{3}{2}\sqrt{3}$
8. $-\frac{5}{24}a$
9. $-\frac{7}{4}$
10. $-\frac{3}{2}$

CAPÍTULO 2

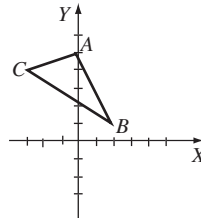
EJERCICIO 4



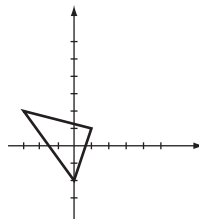
3.



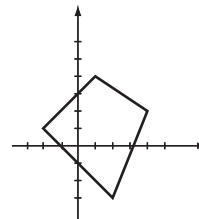
4.



5.



6.



EJERCICIO 5

1. $10u$
2. $\sqrt{5}u$
3. $5\sqrt{2}u$
4. $4\sqrt{2}u$
5. $8u$
6. $\frac{\sqrt{181}}{6}u$
7. $\frac{5}{4}u$
8. $\frac{\sqrt{5}}{4}u$
9. $\frac{\sqrt{17}}{2}u$
10. $1u$
11. $28.72u$
12. $15.64u$
13. $25.10u$
14. $12u$
15. Es triángulo isósceles, debido a que $d_{AC} = d_{BC}$.
16. Perímetro = $40.96 u$; Área = $133.517 u^2$
17. Ordenada ($y = 11$, $y = 1$), puntos $(-4, 11)$ y $(-4, 1)$
18. $P(4, 6)$ y $P\left(-\frac{84}{13}, -\frac{126}{13}\right)$
19. sí
20. no
21. sí
22. no

EJERCICIO 6

1. $r = -\frac{1}{2}$
2. $r = -\frac{4}{3}$
3. $r = \frac{1}{2}$
4. $r = -2$
5. $r = -\frac{1}{10}$
6. $r = \frac{2}{7}$
7. $P(6, -5)$
8. $P(5, 0)$
9. $P\left(\frac{2}{5}, \frac{19}{5}\right)$
10. $P\left(-\frac{4}{9}, \frac{4}{3}\right)$
11. $P\left(4, -\frac{9}{2}\right)$
12. $P(-a, 3b)$
13. $r = -\frac{5}{2}$
14. $P\left(\frac{19}{5}, \frac{37}{5}\right)$
15. $P(3, 7)$
16. $r = 2$
17. $P(2, 1)$
18. $P\left(5, -\frac{11}{2}\right)$
19. $P_2\left(\frac{11}{3}, \frac{2}{3}\right)$
20. $P\left(\frac{2}{5}, \frac{7}{5}\right)$

$$21. r=1; x=\frac{x_1+x_2}{2}; y=\frac{y_1+y_2}{2}$$

22. Los puntos de trisección son:

$$A\left(\frac{x_1+2x_2}{3}, \frac{y_1+2y_2}{3}\right)$$

$$B\left(\frac{2x_1+x_2}{3}, \frac{2y_1+y_2}{3}\right)$$

EJERCICIO 7

$$1. Pm\left(\frac{5}{2}, 2\right); P_i: \left(\frac{8}{3}, 3\right) y \left(\frac{7}{3}, 1\right)$$

$$2. Pm\left(\frac{3}{2}, \frac{11}{2}\right); P_i: (1, 5) y (2, 6)$$

$$3. Pm(4, 7); P_i: \left(\frac{7}{3}, \frac{17}{3}\right) y \left(\frac{17}{3}, \frac{25}{3}\right)$$

$$4. Pm\left(8, -\frac{11}{2}\right); P_i: (7, -6) y (9, -5)$$

$$5. Pm\left(\frac{5}{12}, \frac{3}{2}\right); P_i: \left(\frac{4}{9}, \frac{4}{3}\right) y \left(\frac{7}{18}, \frac{5}{3}\right)$$

$$6. Pm\left(\frac{11}{24}, -\frac{1}{2}\right); P_i: \left(\frac{7}{18}, 0\right) y \left(\frac{19}{36}, -1\right)$$

$$7. P_2(-5, -5)$$

$$8. (-1, 1), (7, 9) y (-3, 5)$$

EJERCICIO 8

$$1. 3u^2$$

$$4. 10u^2$$

$$7. 31u^2$$

$$10. 50.5u^2$$

$$2. 15u^2$$

$$5. 6u^2$$

$$8. 17u^2$$

$$3. 28.5u^2$$

$$6. a^2u^2$$

$$9. 19u^2$$

CAPÍTULO 3

EJERCICIO 9

$$1. \frac{2}{5}$$

7. No existe

$$13. 30^\circ$$

$$2. -\frac{7}{5}$$

$$8. -\frac{17}{5}$$

$$14. 35^\circ 15'$$

$$3. -\frac{1}{8}$$

$$9. -\frac{5}{72}$$

$$15. 90^\circ$$

$$4. \frac{4}{3}$$

$$10. \frac{b}{a}$$

17. Son colineales

$$5. -\frac{2}{3}$$

$$11. 45^\circ$$

18. Son colineales

$$6. 0$$

$$12. 135^\circ$$

19. Son colineales

20. No son colineales

21. Son colineales 24. Son colineales

$$22. \text{No son colineales} \quad 25. x = \frac{2}{3}$$

$$27. y = -3 + \sqrt{3}$$

$$23. \text{Son colineales} \quad 26. x = 4$$

$$28. y = -6$$

EJERCICIO 10

1. Son perpendiculares

2. Los lados opuestos son paralelos y de igual pendiente

$$m_{BC} = m_{AD} = \frac{2}{5} \text{ y } m_{CD} = m_{AB} = 3$$

3. Las rectas son paralelas

$$m_{AB} = m_{CD} = -\frac{5}{3}$$

4. En un triángulo rectángulo los catetos son perpendiculares.

$$m_{AB} \cdot m_{AC} = \frac{1}{2} \cdot (-2) = -1$$

5. En un cuadrado los lados opuestos son paralelos y los lados adyacentes son perpendiculares.

$$m_{AB} = m_{CD} = 4, m_{BC} = m_{AD} = -\frac{1}{4} \text{ y } m_{AC} \cdot m_{BD} = -1$$

$$6. x = 5$$

$$7. m_{AB} = m_{CD} = -2; m_{BC} = m_{AD} = \frac{2}{7}$$

EJERCICIO 11

$$1. \alpha = 57^\circ 5' 41''$$

$$2. 26^\circ 33' 54'', 116^\circ 33' 54''$$

$$36^\circ 52' 11''$$

$$3. 63^\circ 26' 5'', 63^\circ 26' 5'' y$$

$$53^\circ 7' 48''$$

4. Un ángulo de 90°

los restantes de 45°

$$5. 63^\circ 26' 5'' y 26^\circ 33' 54''$$

$$6. \alpha = 131^\circ 49' 12''$$

$$7. 130^\circ 14' 10''; 49^\circ 45' 50''$$

$$8. 117^\circ 16' 36''$$

$$9. m_1 = \frac{2}{\sqrt{3}} = \frac{2\sqrt{3}}{3}$$

$$10. x = 8$$

$$11. m_1 = -\frac{1}{7}, m_2 = 7$$

$$12. -2 + \sqrt{3}; 2 + \sqrt{3}$$

CAPÍTULO 4

EJERCICIO 12

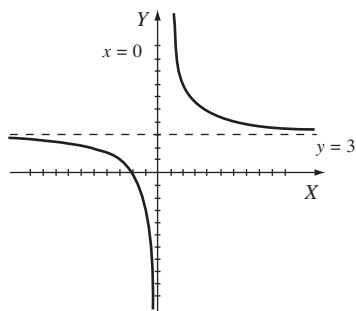
1. Intersecciones con los ejes:
- $(-2, 0)$

Simetría: No existe

 Extensión: $\{x \in \mathbb{R} \mid x \neq 0\}; \{y \in \mathbb{R} \mid y \neq 3\}$

 Asíntotas: Horizontal: $y = 3$, Vertical: $x = 0$

Gráfica



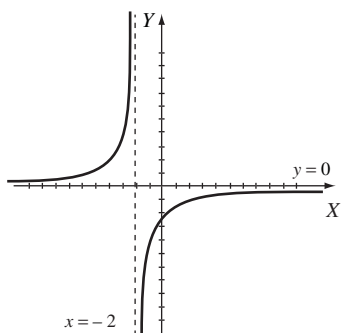
2. Intersecciones con los ejes:
- $(0, -2)$

Simetría: No existe

 Extensión: $\{x \in \mathbb{R} \mid x \neq -2\}; \{y \in \mathbb{R} \mid y \neq 0\}$

 Asíntotas: Horizontal: $y = 0$, Vertical: $x = -2$

Gráfica



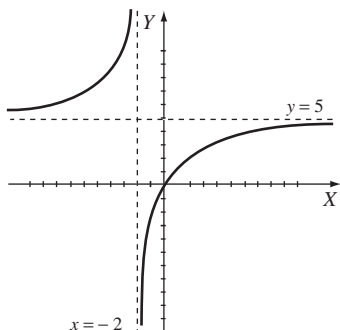
3. Intersecciones con los ejes:
- $(0, 0)$

Simetría: No existe

 Extensión: $\{x \in \mathbb{R} \mid x \neq -2\}; \{y \in \mathbb{R} \mid y \neq 5\}$

 Asíntotas: Horizontal: $y = 5$, Vertical: $x = -2$

Gráfica



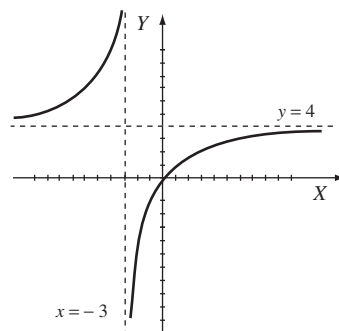
4. Intersecciones con los ejes:
- $(0, 0)$

Simetría: No existe

 Extensión: $\{x \in \mathbb{R} \mid x \neq -3\}; \{y \in \mathbb{R} \mid y \neq 4\}$

 Asíntotas: Horizontal: $y = 4$, Vertical: $x = -3$

Gráfica



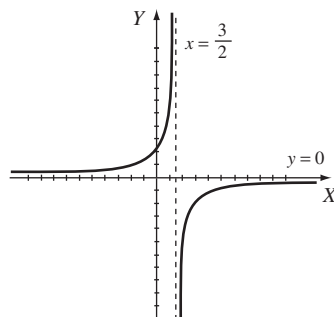
5. Intersecciones con los ejes:
- $(0, 2)$

Simetría: No existe

 Extensión: $\left\{x \in \mathbb{R} \mid x \neq \frac{3}{2}\right\}; \{y \in \mathbb{R} \mid y \neq 0\}$

 Asíntotas: Horizontal: $y = 0$, Vertical: $x = \frac{3}{2}$

Gráfica



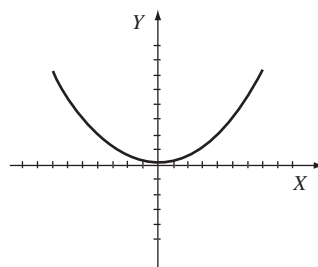
6. Intersecciones con los ejes:
- $(0, 0)$

Simetría: Sólo con el eje Y

 Extensión: $\{x \in \mathbb{R}\}; \{y \in \mathbb{R} \mid y \geq 0\}$

Asíntotas: No existen

Gráfica



7. Intersecciones con los ejes: Eje $X \rightarrow (-4, 0), (4, 0)$

Eje $Y \rightarrow (0, 2), (0, -2)$

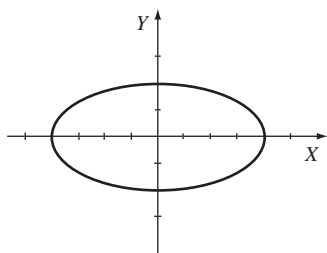
Simetría: Es simétrica con ambos ejes y con el origen

Extensión: $\{x \in \mathbb{R} \mid -4 \leq x \leq 4\}$

$\{y \in \mathbb{R} \mid -2 \leq y \leq 2\}$

Asíntotas: No existen

Gráfica



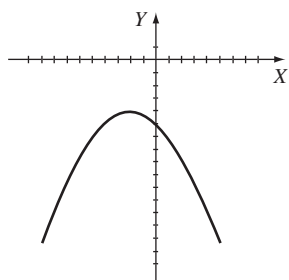
8. Intersecciones con los ejes: $(0, -5)$

Simetría: No existe

Extensión: $\{x \in \mathbb{R}\}; \{y \in \mathbb{R} \mid y \leq -4\}$

Asíntotas: No existen

Gráfica



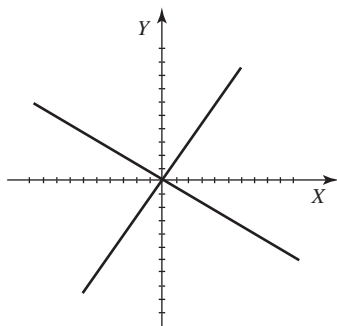
9. Intersecciones con los ejes: $(0, 0)$

Simetría: Sólo respecto al origen

Extensión: $\{x \in \mathbb{R}\}; \{y \in \mathbb{R}\}$

Asíntotas: No existen

Gráfica



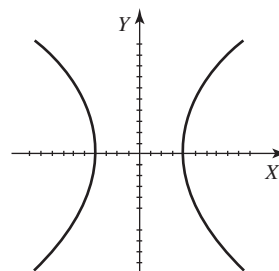
10. Intersecciones con los ejes: Eje $X \rightarrow (-4, 0), (4, 0)$

Simetría: Es simétrica con los ejes y con el origen

Extensión: $\{x \in \mathbb{R} \mid x \leq -4 \text{ o } x \geq 4\}; \{y \in \mathbb{R}\}$

Asíntotas: No hay horizontales o verticales

Gráfica



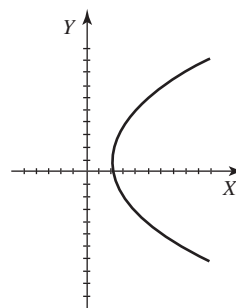
11. Intersecciones con los ejes: $\left(\frac{17}{8}, 0\right)$

Simetría: No existe

Extensión: $\{x \in \mathbb{R} \mid x \geq 2\}; \{y \in \mathbb{R}\}$

Asíntotas: No existen

Gráfica



12. Intersecciones con los ejes: $(0, 0)$ y $(6, 0)$

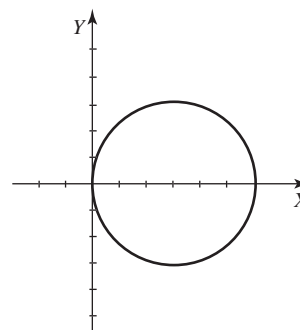
Simetría: Sólo con el eje X

Extensión: $\{x \in \mathbb{R} \mid 0 \leq x \leq 6\}$

$\{y \in \mathbb{R} \mid -3 \leq y \leq 3\}$

Asíntotas: No existen

Gráfica



EJERCICIO 13

1. $y - x = 2$
2. $xy = 1$
3. $x - 2y = 0$
4. $x^2 + y^2 = 25$
5. $7x - 3y + 4 = 0$
6. $x^2 + y^2 - 8x + 6y = 0$
7. $x^2 + 10y + 25 = 0$
8. $2x + y + 5 = 0$
9. $x^2 + y^2 + 6x + 4y - 51 = 0$
10. $x^2 + y^2 - 6x - 6y + 9 = 0$
11. $21x^2 - 12xy + 16y^2 + 60x + 60y - 600 = 0$
12. $9x^2 + 25y^2 - 225 = 0$
13. $28y^2 - 36x^2 - 63 = 0$; $36x^2 - 28y^2 + 63 = 0$
14. $16x^2 + 7y^2 - 112 = 0$
15. $7x^2 - 9y^2 - 28x + 90y - 260 = 0$

CAPÍTULO 5
EJERCICIO 14

- | | |
|-------------------------------------|----------------------------------|
| 1. $2x + 5y - 14 = 0$ | 19. $2x + y - 4 = 0$ |
| 2. $2x - y + 3 = 0$ | 20. $6x + 5y - 82 = 0$ |
| 3. $2y - 1 = 0$ | 21. $2x - 3y + 8 = 0$ |
| 4. $2x + 2y + 1 = 0$ | 22. $4x - 7y + 10 = 0$ |
| 5. $3x - 5y + 11 = 0$ | $7x - 5y - 26 = 0$ |
| 6. $4x - 3y + 6 = 0$ | $3x + 2y + 22 = 0$ |
| 7. $x - 3 = 0$ | 23. $4x - 6y - 5 = 0$ |
| 8. $8x - 4y - 7 = 0$ | 24. $10x - 3y - 4 = 0$ |
| 9. $3x + 2y - 2 = 0$ | 25. $2x - 3y - 7 = 0$ |
| 10. $3x + 5y - 12 = 0$ | 26. $2x + 11y + 5 = 0$ |
| 11. $3x + 2y - 5 = 0$ | $5x - 16y - 31 = 0$ |
| 12. $\sqrt{3}x + y - 3\sqrt{3} = 0$ | 27. $(8, 6), (-2, -8), (-6, -2)$ |
| 13. $x + y - 4 = 0$ | 28. $(-3, 0), (3, 2), (4, -2)$ |
| 14. $x - 3 = 0$ | 29. $(-3, 2), (-1, -2)$ |
| 15. $x + 4 = 0$ | $(7, 0), (5, 4)$ |
| 16. $3x - 2y - 12 = 0$ | 30. $120x - y = 0$ |
| 17. $x + 4y + 2 = 0$ | 31. $3t - 2v + 4 = 0$ |
| 18. $2x + y = 0$ | 32. $x + 40y - 600 = 0$ |
| $3x - 4y + 11 = 0$ | 33. $120x - y + 1200 = 0$ |
| $5x + 2y - 25 = 0$ | 34. $9T_C - 5T_F + 160 = 0$ |
| $y = 0$ | |

EJERCICIO 15

1. $y = -x + 4$; $\frac{x}{4} + \frac{y}{4} = 1$
2. $y = \frac{2}{5}x + 1$; $\frac{x}{-5/2} + \frac{y}{1} = 1$
3. $y = \frac{1}{3}x + \frac{8}{3}$; $\frac{x}{-8} + \frac{y}{8/3} = 1$

4. $y = 2x$

5. $y = -\frac{1}{8}x + \frac{1}{2}$; $\frac{x}{4} + \frac{y}{1/2} = 1$

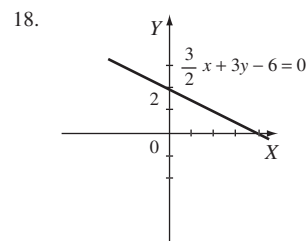
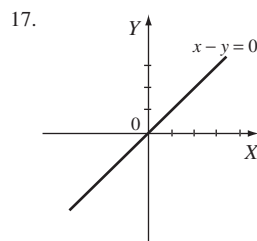
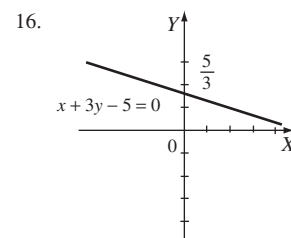
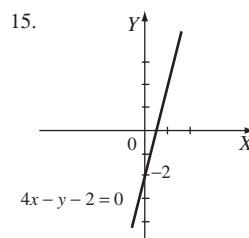
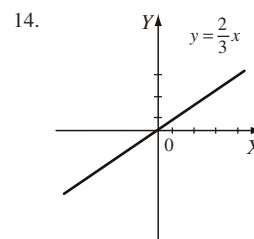
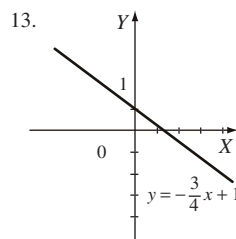
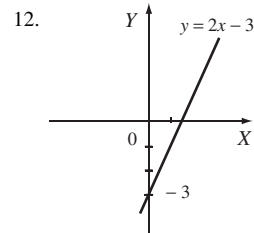
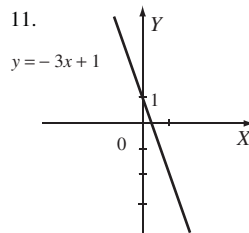
6. $y = \frac{3}{4}x - 3$; $\frac{x}{4} + \frac{y}{-3} = 1$

7. $y = -\frac{3}{5}x + 2$; $\frac{x}{10/3} + \frac{y}{2} = 1$

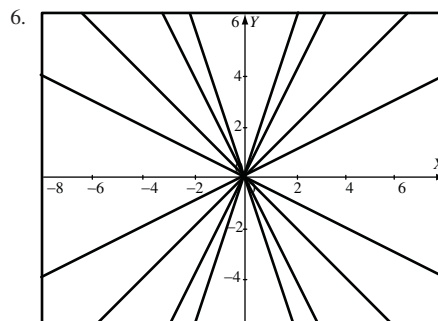
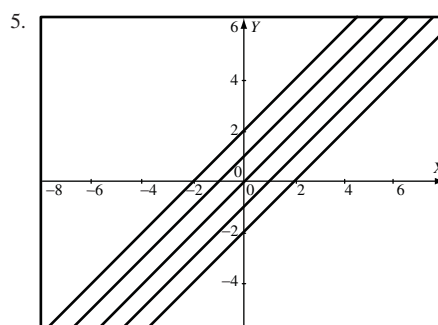
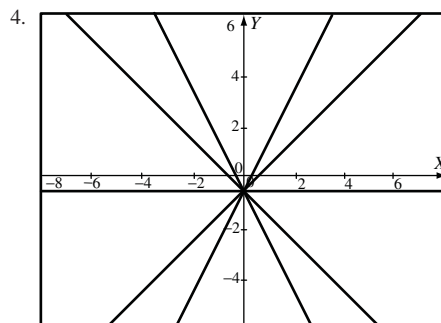
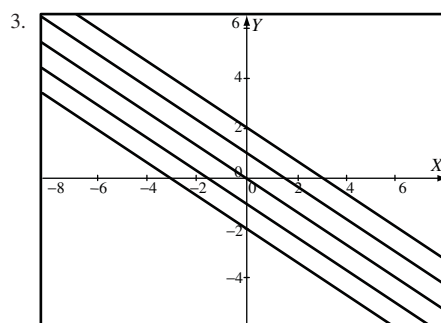
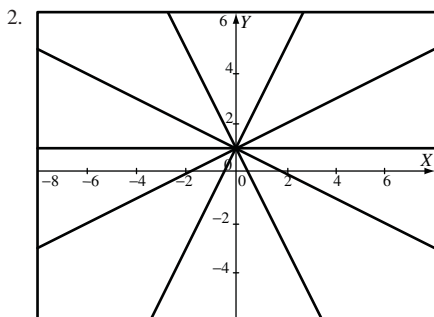
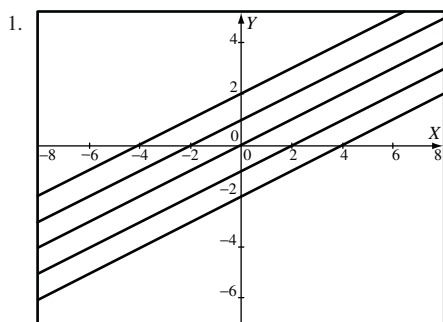
8. $y = -\frac{1}{6}x - \frac{5}{3}$; $\frac{x}{-10} + \frac{y}{-5/3} = 1$

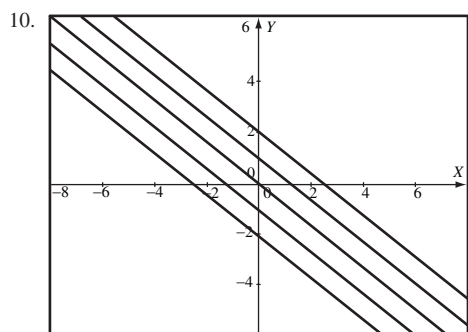
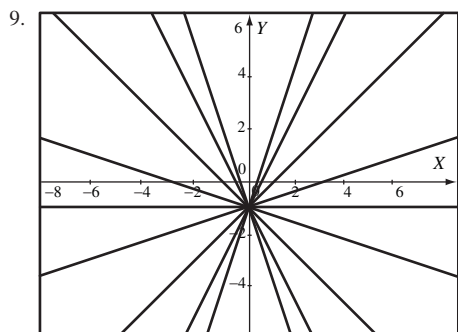
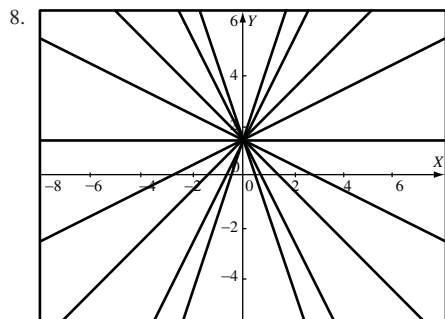
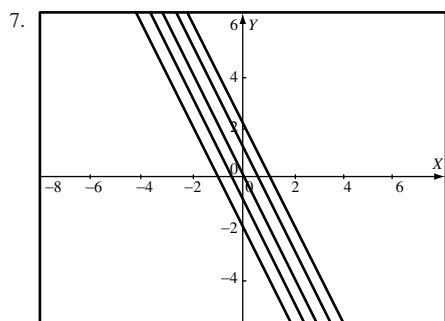
9. $y = \frac{6}{5}x + 12$; $\frac{x}{-10} + \frac{y}{12} = 1$

10. $y = -x \operatorname{ctg} w + p \operatorname{csc} w$; $\frac{x}{p \sec w} + \frac{y}{p \csc w} = 1$



19. $x + y + 5 = 0$
 20. $y = 2x + 4$
 21. $y = -3x + 11$
 22. $y = -\frac{1}{6}x + \frac{13}{6}$
 23. $y = 5x + 5$
 24. $5x - 3y + 20 = 0$
 25. $4x + y - 15 = 0$
 26. $x - y + 1 = 0$
 27. $6x + 4y + 5 = 0$
 28. $3x - 2y - 6 = 0$
 29. $161^\circ 33' 54''$
 30. $36^\circ 1' 38''$
 $102^\circ 20' 20''$
 $41^\circ 38'$
 31. $(20, 13)$
 32. a) $y = 50x + 30\,000$
 b) \$70 000.00
 33. a) $4m/s^2$
 b) $31m/s$
34. a) $C = 30x + 6\,000$
 b) \$1 000.00
 c) no
 d) 120 platillos
35. $\frac{x}{5} + \frac{y}{-5} = 1$
 36. $\frac{x}{8} + \frac{y}{-4} = 1$
 37. $\frac{x}{\frac{4}{-3}} + \frac{y}{\frac{4}{-5}} = 1$
 38. $\frac{x}{\frac{7}{-4}} + \frac{y}{\frac{7}{2}} = 1$
 39. $\frac{x}{\frac{6}{5}} + \frac{y}{\frac{3}{-2}} = 1$

EJERCICIO 16


EJERCICIO 17

1. $\frac{2x}{\sqrt{13}} + \frac{3y}{\sqrt{13}} - \frac{5}{\sqrt{13}} = 0$
2. $-\frac{x}{\sqrt{2}} + \frac{y}{\sqrt{2}} - \frac{5}{\sqrt{2}} = 0$
3. $\frac{5x}{\sqrt{34}} + \frac{3y}{\sqrt{34}} = 0$
4. $\frac{-x}{\sqrt{10}} + \frac{3y}{\sqrt{10}} - \frac{7}{\sqrt{10}} = 0$
5. $-\frac{12x}{13} - \frac{5y}{13} - 1 = 0$
6. $\frac{\sqrt{5}x}{3} + \frac{2y}{3} - \frac{1}{3} = 0$
7. $\sqrt{3}x + y - 8 = 0$
8. $x + y - 2 = 0$
9. $x + \sqrt{3}y - 6 = 0$
10. $x - \sqrt{3}y + 2 = 0$
11. $x + \sqrt{3} = 0$
12. $\sqrt{3}x - y + 10 = 0$
13. $x + y + 6 = 0$
14. $x - \sqrt{3}y - 8\sqrt{3} = 0$
15. $\frac{x}{2} - \frac{\sqrt{3}y}{2} - 2 = 0$
16. $\frac{2x}{\sqrt{29}} + \frac{5y}{\sqrt{29}} - \frac{12}{\sqrt{29}} = 0$
17. $-\frac{3x}{\sqrt{10}} - \frac{y}{\sqrt{10}} - \frac{7}{\sqrt{10}} = 0$
18. $\theta = 60^\circ, \theta' = 120^\circ$
19. $3x - 4y - 12 = 0$
 $3x + 4y + 12 = 0$

EJERCICIO 18

1. $\frac{23\sqrt{53}}{53}$
2. $\frac{9}{5}$
3. $5\sqrt{2}$
4. $\frac{49}{13}$
5. $\frac{7\sqrt{2}}{2}$
6. 1
7. $\frac{32\sqrt{17}}{17}$
8. 10
9. $\frac{9\sqrt{10}}{5}$
10. $5u^2$
11. $r = \frac{7}{5}$
12. $K_1 = 2$
 $K_2 = \frac{1}{2}$
13. $\frac{2\sqrt{13}}{13}$
14. $-\frac{6}{5}$
15. $\frac{14}{5}$
16. $\frac{11\sqrt{29}}{29}$
17. $-\frac{7}{10}$
18. $-\frac{27\sqrt{2}}{8}$
19. $3x - 4y - 10 = 0$
 $3x - 4y + 20 = 0$
20. $y = -\sqrt{17} + \frac{7}{4}$
 $y = \sqrt{17} + \frac{7}{4}$
21. $\frac{7\sqrt{5}}{5}$
22. $\frac{11\sqrt{5}}{15}$
23. $x - y - 3 - 3\sqrt{2} = 0$
 $x - y - 3 + 3\sqrt{2} = 0$
24. $2x - y - 1 = 0$
 $2x + y - 7 = 0$

EJERCICIO 19

1. $2x - y + 3 = 0$
2. $4x - 6y + 13 = 0$
3. $x + 3y - 9 = 0$
4. $y + 6 = 0$
5. $2x - 6y - 11 = 0; 6x + 2y - 1 = 0$
6. $99x - 27y + 50 = 0; 21x + 77y - 80 = 0$
7. $y - 2 = 0; 7x - y - 5 = 0; x + y - 3 = 0; (1, 2)$
8. $2x + y - 6 = 0; x - 6y - 10 = 0; 3x - 5y - 16 = 0;$
 $\left(\frac{46}{13}, -\frac{14}{13}\right)$
9. $2x + y - 3 = 0; 3x - 5y + 16 = 0; x - 6y + 19 = 0;$
 $\left(-\frac{1}{13}, \frac{41}{13}\right)$
10. $11x + 4y - 27 = 0; 7x - y - 16 = 0; 4x + 5y - 11 = 0;$
 $\left(\frac{7}{3}, \frac{1}{3}\right)$
11. $(5\sqrt{5} - \sqrt{34})x + (3\sqrt{5} + 2\sqrt{34})y - (4\sqrt{5} + 7\sqrt{34}) = 0$
 $(\sqrt{37} + 6\sqrt{5})x + (\sqrt{5} - 2\sqrt{37})y + (7\sqrt{37} - 23\sqrt{5}) = 0$
 $(5\sqrt{37} + 6\sqrt{34})x + (3\sqrt{37} + \sqrt{34})y - (4\sqrt{37} + 23\sqrt{34}) = 0$
 $(1.5965, 2.2438)$
12. $55x + 47y - 144 = 0$

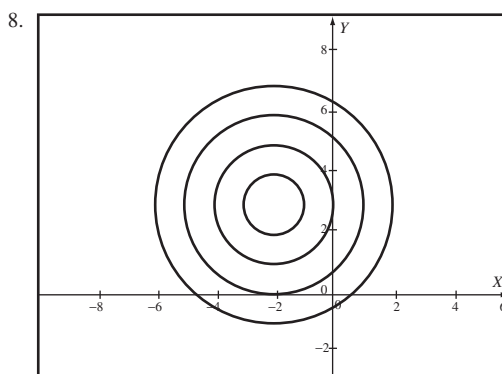
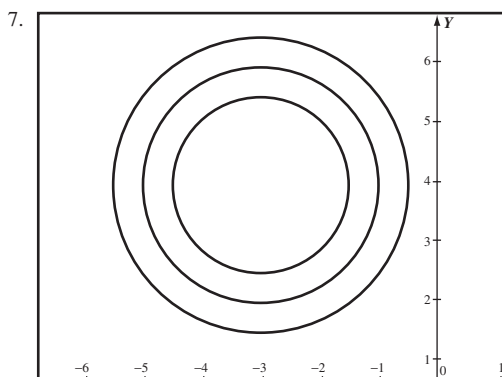
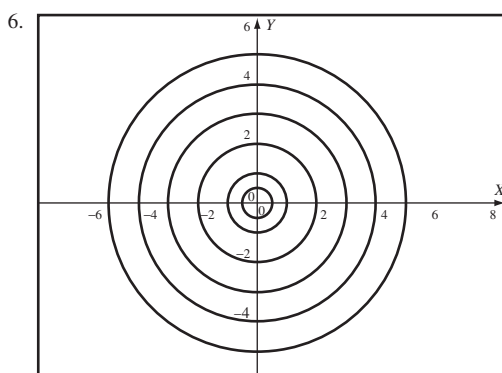
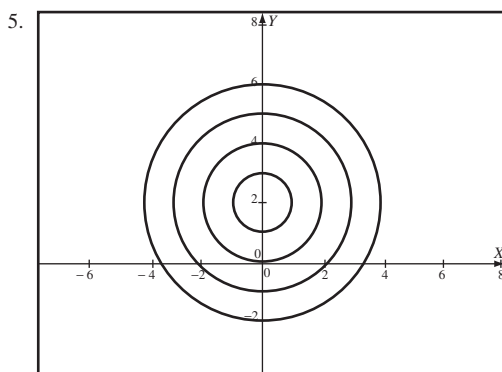
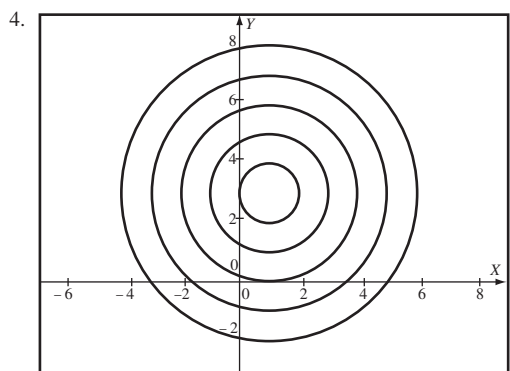
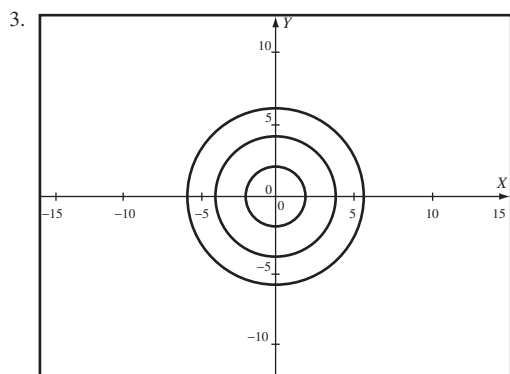
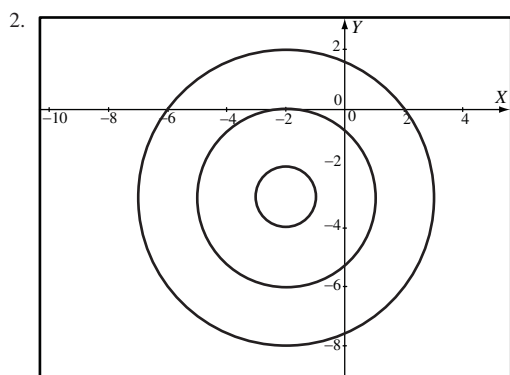
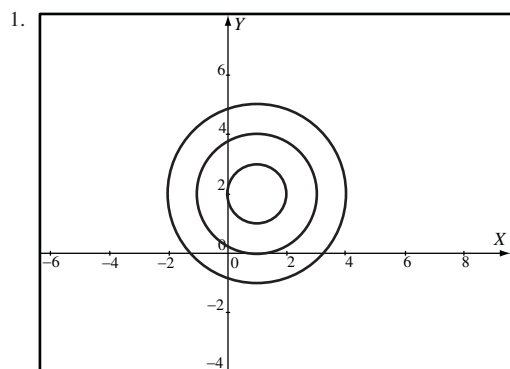
CAPÍTULO 6**EJERCICIO 20**

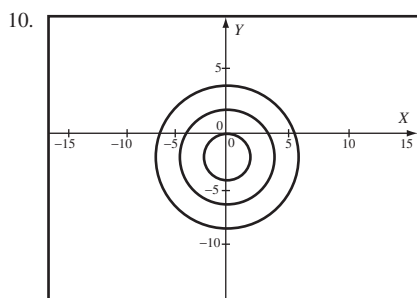
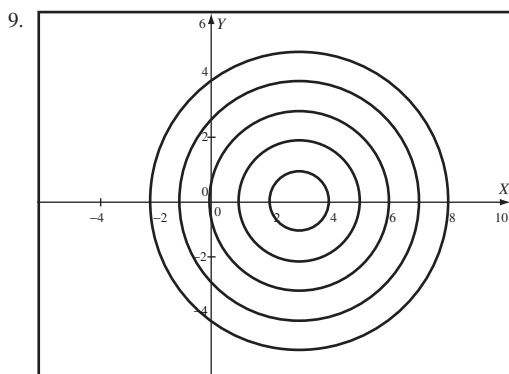
1. $x^2 + y^2 - 16 = 0$
2. $4x^2 + 4y^2 - 3 = 0$
3. $x^2 + y^2 - 2x + 6y + 6 = 0$
4. $3x^2 + 3y^2 + 3x + 4y = 0$
5. $x^2 + y^2 - 13 = 0$
6. $x^2 + y^2 - 2x - 6y - 31 = 0$
7. $x^2 + y^2 - 2x + 6y - 35 = 0$
8. $x^2 + y^2 + 2x + 10y + 25 = 0$
9. $x^2 + y^2 - 10x + 4y = 0$
10. $x^2 + y^2 + 8x - 4y + 4 = 0$
11. $x^2 + y^2 - 10x + 10y + 25 = 0$
12. $3x^2 + 3y^2 + 13x - 65 = 0$
13. $4x^2 + 4y^2 + 41y + 66 = 0$
14. $x^2 + y^2 + 4y = 0$

15. $x^2 + y^2 - 8x + 6y + 21 = 0$
16. $x^2 + y^2 - 2x - 4y - 11 = 0$
17. $x^2 + y^2 + 6x + 8y - 144 = 0$
18. $x^2 + y^2 - 2x - 4y - 59 = 0$
19. $2x^2 + 2y^2 + 15x - 11y - 51 = 0$
20. $17x^2 + 17y^2 - 88x + 58y - 544 = 0$
21. $9x^2 + 9y^2 - 43x + 9y - 140 = 0$
22. $x^2 + y^2 - 4x + 4y - 2 = 0$
23. $x^2 + y^2 - 2x - 4y - 4 = 0$
24. $3x^2 + 3y^2 + 8x + 10y - 43 = 0$
25. $3x^2 + 3y^2 - x + 7y - 10 = 0$
26. $x^2 + y^2 - 6x + 4y - 5 = 0$
27. $x^2 + y^2 + 4x - 2y - 29 = 0$

EJERCICIO 21

1. $C(-1, -1), r = 2$
2. $C(3, -4), r = \sqrt{5}$
3. Punto, $C(-3, -1), r = 0$
4. Imaginaria con $C(2, -1)$ y $r = \sqrt{-9}$
5. $C(-7, 4), r = 5$
6. $C(0, 4), r = 3$
7. $C(-2, 0), r = 1$
8. $C\left(-\frac{5}{2}, \frac{3}{2}\right), r = \sqrt{7}$
9. $C\left(\frac{1}{2}, -\frac{3}{2}\right), r = \frac{1}{2}$
10. $C\left(\frac{1}{5}, 3\right), r = \frac{4}{5}$
11. $C\left(\frac{3}{4}, -\frac{1}{6}\right), r = \frac{5}{12}$
12. $C\left(-\frac{2}{3}, \frac{1}{2}\right), r = 3$
13. $x - 3y + 8 = 0$
 $79x + 3y - 568 = 0$
14. $x^2 + y^2 - 8x - 4y + 10 = 0$
15. $x^2 + y^2 - 8x - 8y + 19 = 0$
 $x^2 + y^2 + 4y - 9 = 0$
16. $x^2 + y^2 + 10x - 4 = 0$
17. $k = 5, k = -\frac{19}{17}$
18. $(-5, 3)$ y $(1, 9)$
19. $(2, 2)$
20. No existe intersección
21. $(3, 5)$ y $(4, 4)$
22. $(4, -1)$ y $(6, -3)$

EJERCICIO 22



11. $(x-1)^2 + (y-1)^2 = p^2$ 13. $(x+2)^2 + (y-3)^2 = p^2$
 12. $(x+1)^2 + (y+3)^2 = p^2$ 14. $(x-1)^2 + (y-0)^2 = p^2$

CAPÍTULO 7

EJERCICIO 23

- | | |
|---------------------------|--------------------------------|
| 1. $(2, 2)$ | 10. $y'^2 - 12x' = 0$ |
| 2. $(9, 1)$ | 11. $9x'^2 + 16y'^2 - 144 = 0$ |
| 3. $(-1, 4)$ | 12. $4x'^2 + 5y'^2 - 20 = 0$ |
| 4. $(-10, -4)$ | 13. $9x'^2 + 4y'^2 - 72 = 0$ |
| 5. $(-8, 7)$ | 14. $x'^2 - 2y'^2 - 2 = 0$ |
| 6. $y'^2 - 8x' = 0$ | 15. $4x'^2 - 9y'^2 - 36 = 0$ |
| 7. $x'^2 - 4y' = 0$ | 16. $y' = x'^3$ |
| 8. $x'^2 + y'^2 - 9 = 0$ | 17. $y'^2 = x'^3 - 1$ |
| 9. $x'^2 + y'^2 - 25 = 0$ | |

EJERCICIO 24

- | | |
|--------------------------|--------------------------------|
| 1. $x'^2 - 8y' = 0$ | 5. $9x'^2 + 4y'^2 - 36 = 0$ |
| 2. $y'^2 - 16x' = 0$ | 6. $16x'^2 + 16y'^2 - 9 = 0$ |
| 3. $x'^2 + y'^2 - 8 = 0$ | 7. $25x'^2 - 16y'^2 + 400 = 0$ |
| 4. $x'^2 + y'^2 - 4 = 0$ | 8. $y'^3 - x' = 0$ |

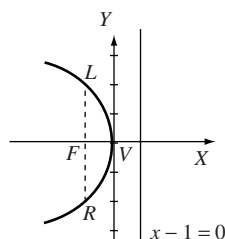
CAPÍTULO 8

EJERCICIO 25

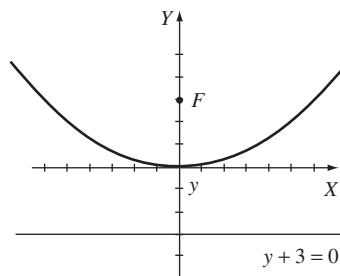
- $y^2 + 8x = 0$
- $x^2 + 4y = 0$
- $y^2 - 12x + 2y + 1 = 0$
- $x^2 - 16y = 0$
- $x^2 - 4x + 4y - 16 = 0$
- $y^2 + 12x - 4y + 52 = 0$
- $x^2 - 2xy + y^2 + 8x - 4y + 2 = 0$
- $9x^2 - 12xy + 4y^2 - 48x + 32y + 64 = 0$

EJERCICIO 26

- Foco: $F(-1, 0)$, Directriz: $x - 1 = 0$, $LR = 4$,
Eje: $y = 0$



- Foco: $F(0, 3)$, Directriz: $y + 3 = 0$, $LR = 12$,
Eje: $x = 0$



- Foco: $F(5, 0)$, Directriz: $x + 5 = 0$, $LR = 20$,
Eje: $y = 0$

